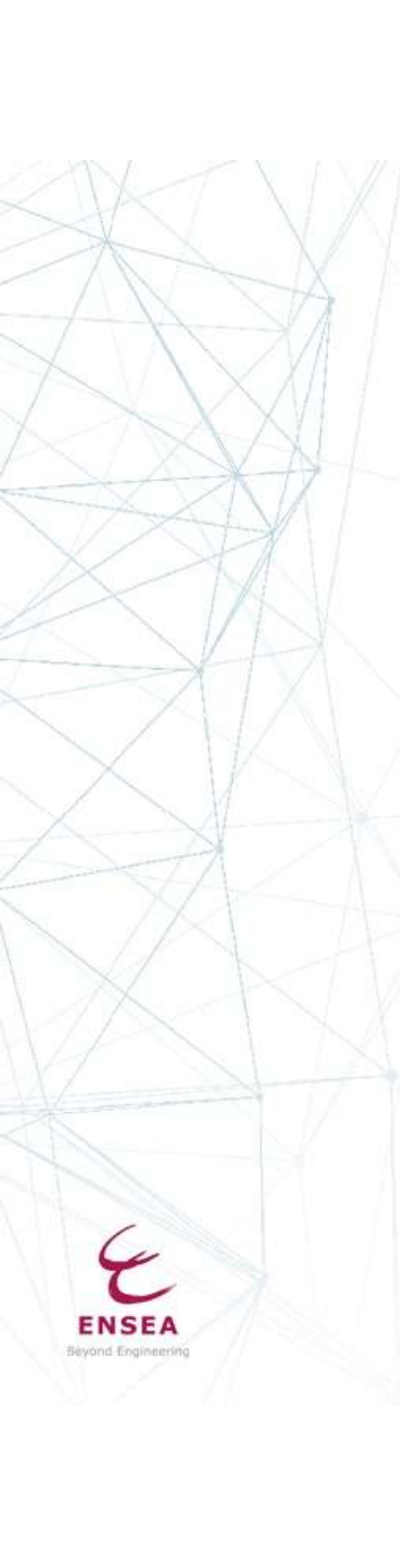




Power Electronics

alexis.martin@ensea.fr
Desk D216





Layout

- Non isolated choppers
 - Introduction
 - Direct conversion
 - Buck
 - Boost
 - Reversible choppers
 - Half bridge
 - H – bridge
 - Inverter
 - Energy storage based chopper
 - Buck-boost
 - Cuck
 - SEPIC
- Isolated power supplies
- Power components

Progress

- Lesson : 8 x 2h
- Tutorials : 9 x 2h
- Labs : 4 x 4h
 - Buck chopper
 - Reversible chopper for DC motor
 - Switch mode power supply Flyback
 - Switch mode power supply Forward
- Marks : Final exam - Labs



Prerequisites

- **Kirchhoff's circuit laws**
- **Average value of a signal, integral calculus, RMS value**
- **AC and DC real power, power factor, complex representation, phasor**
- **Transformers (magnetic field, flux)**
- **Linear control (Laplace transform, Bode diagram, gain/phase margin, PI controller)**

Introduction

Direct conversion

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Introduction

Introduction

Direct conversion

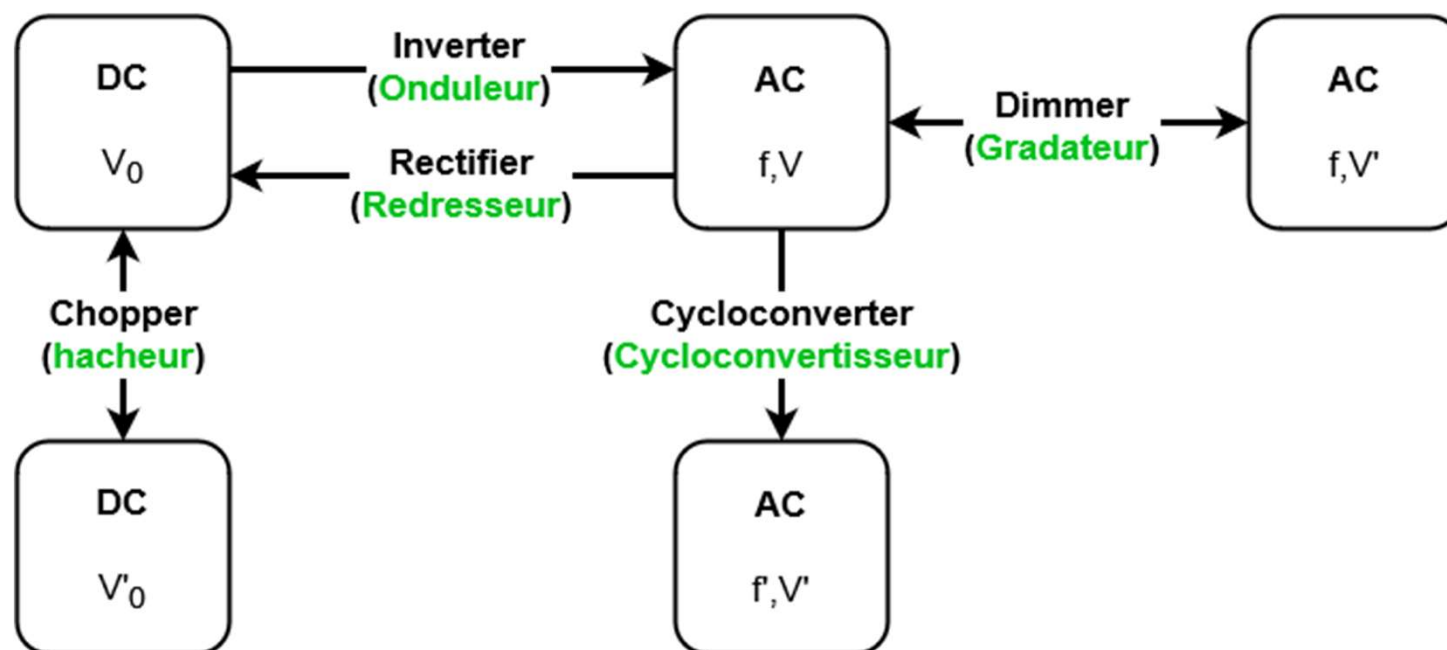
Reversible
choppers

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chopper control

Energy storage
based chopper

Energy conversion: principles



Introduction

Direct conversion

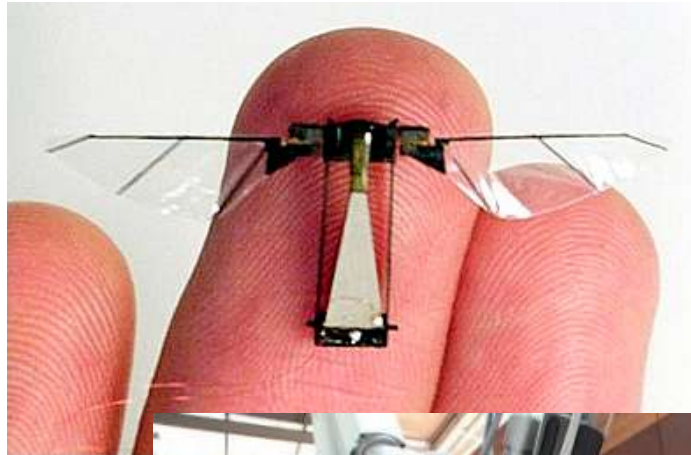
Reversible
choppers

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based chopper

Micro-mechanisms



1-100 W

Introduction

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Use : automatisms



0,1 à 10 kW

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Use : rail traction



100 kW

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based chopper

High speed rail traction



10 MW

Introduction

Direct conversion

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choppers

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chopper control

Energy storage
based chopper

Photovoltaic plant



300 MWc

Introduction

Direct conversion

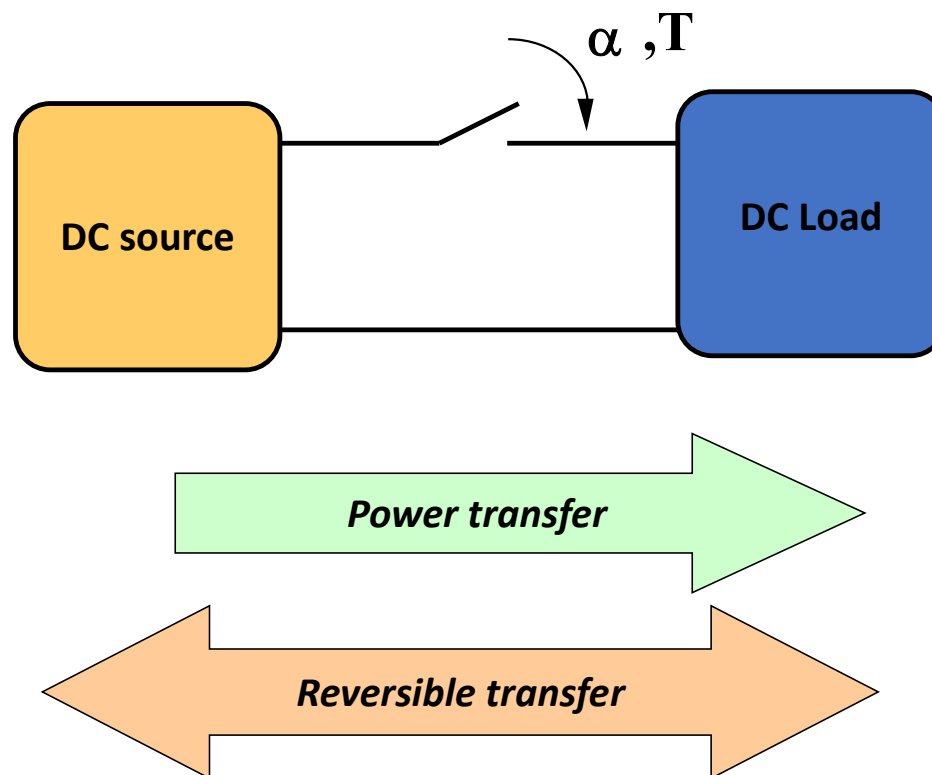
Reversible
choppers

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Energy storage
based chopper

«Chopper » function : a switching electronic



Introduction

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Link power sources

Voltage Source

Current Source

Active
(Energy generator)

Passive
(Energy receiver)

Theorem

*Only two sources with different category
can be directly connected.*

Introduction

Direct conversion

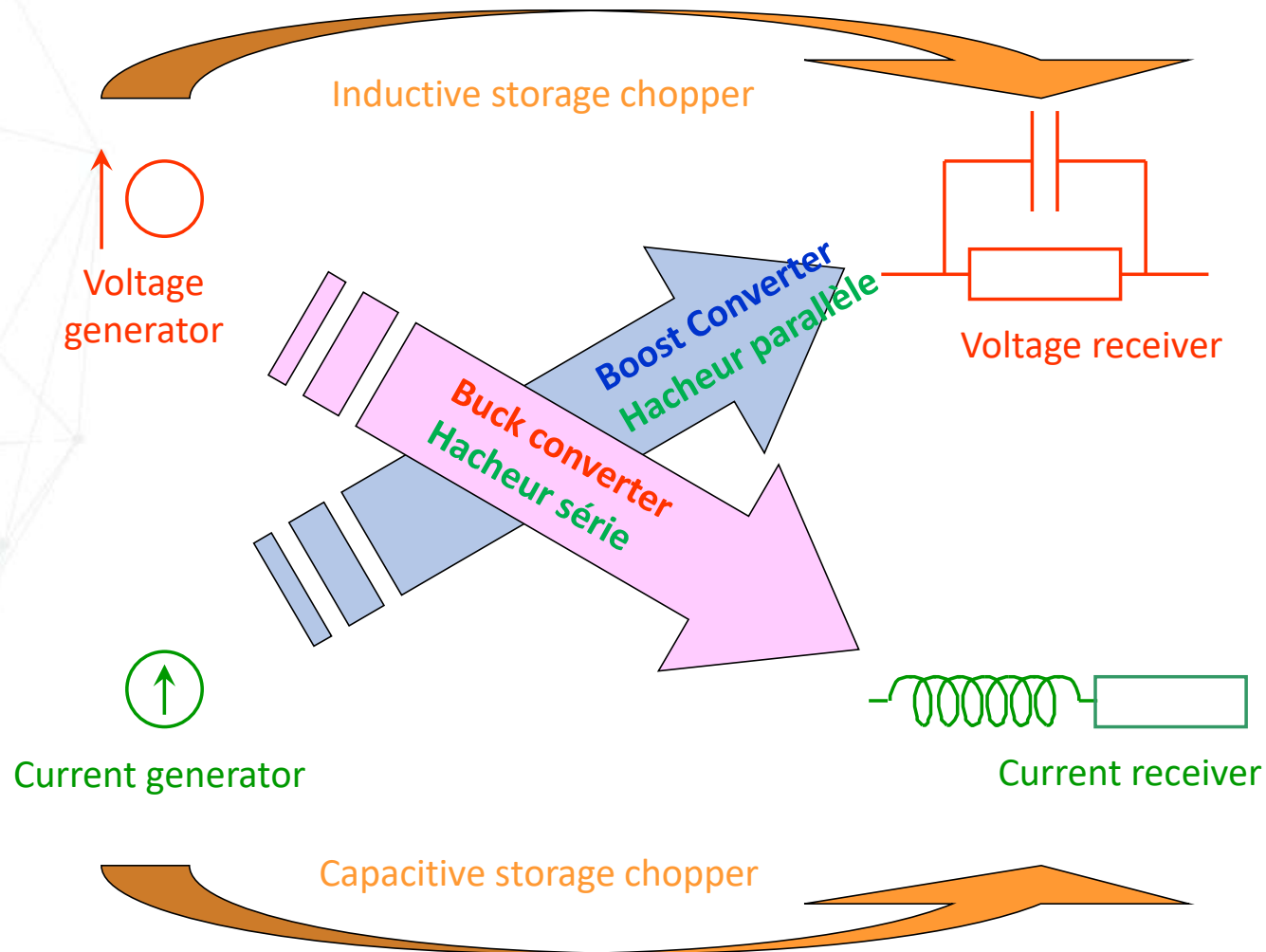
Reversible
choppers

Inverters

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Energy storage
based chopper

Non reversible chopper



Introduction

Direct conversion

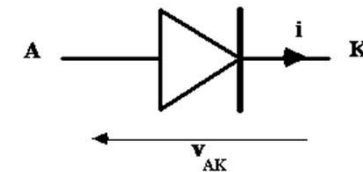
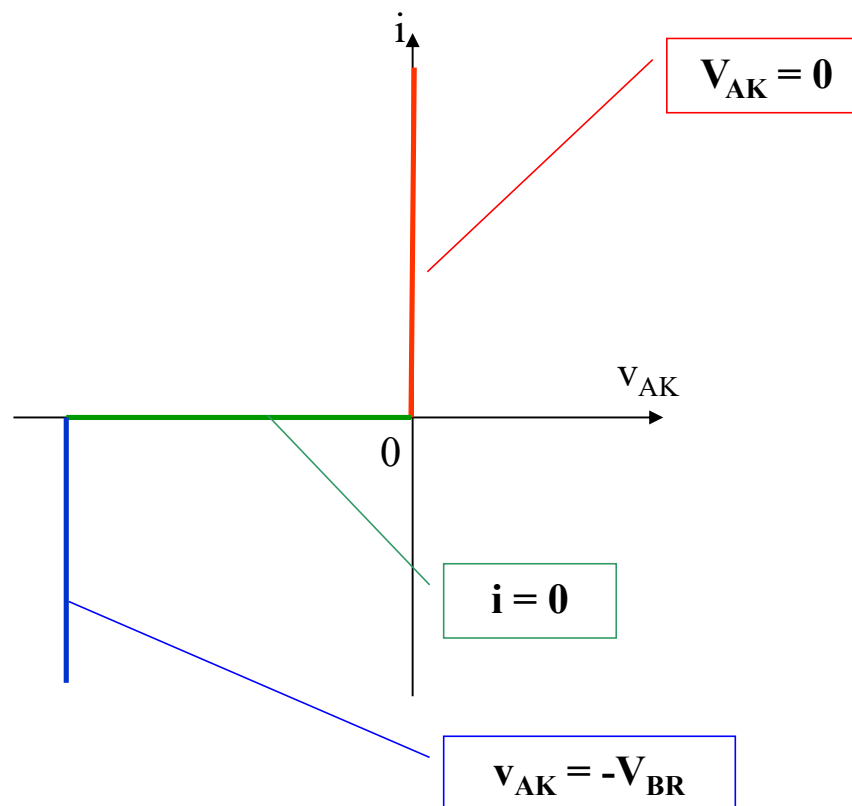
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

« Perfect » diode



Introduction

Direct conversion

Reversible
choppers

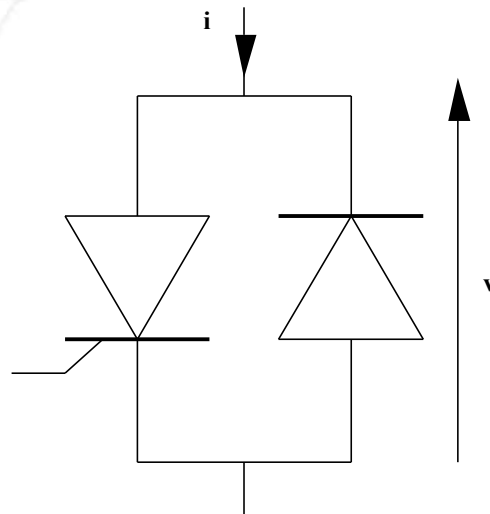
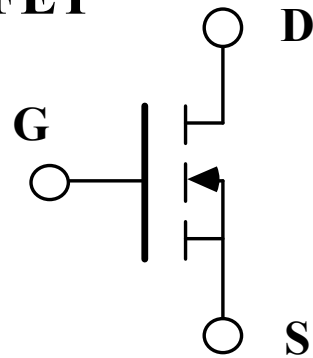
Inverters

Closed loop
chopper control

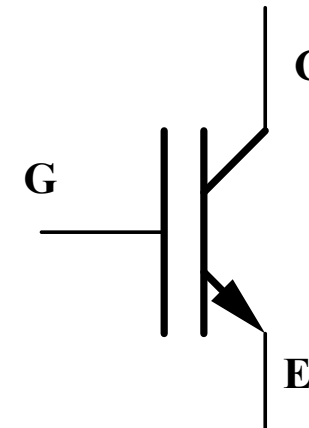
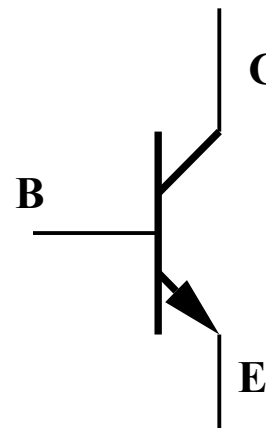
Energy storage
based chopper

Controlled switch

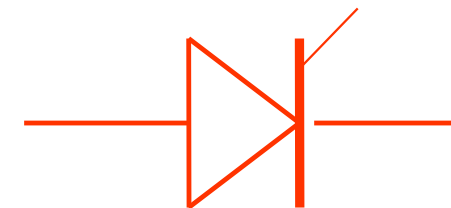
MOSFET



**Bipolar
transistor**



IGBT



Thyristor

Introduction

Direct conversion

Reversible
choppers

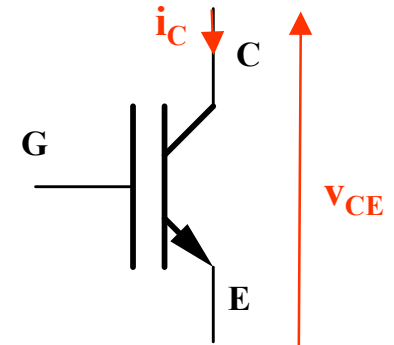
Inverters

Closed loop
chopper control

Energy storage
based chopper

« Perfect » controlled switch ex : IGBT

- On state: « short circuit »
 - $v_{CE} = 0$, i_C fixed by circuit
- Off state: « open circuit »
 - $i_C = 0$, v_{CE} fixed by circuit



Introduction

Direct conversion

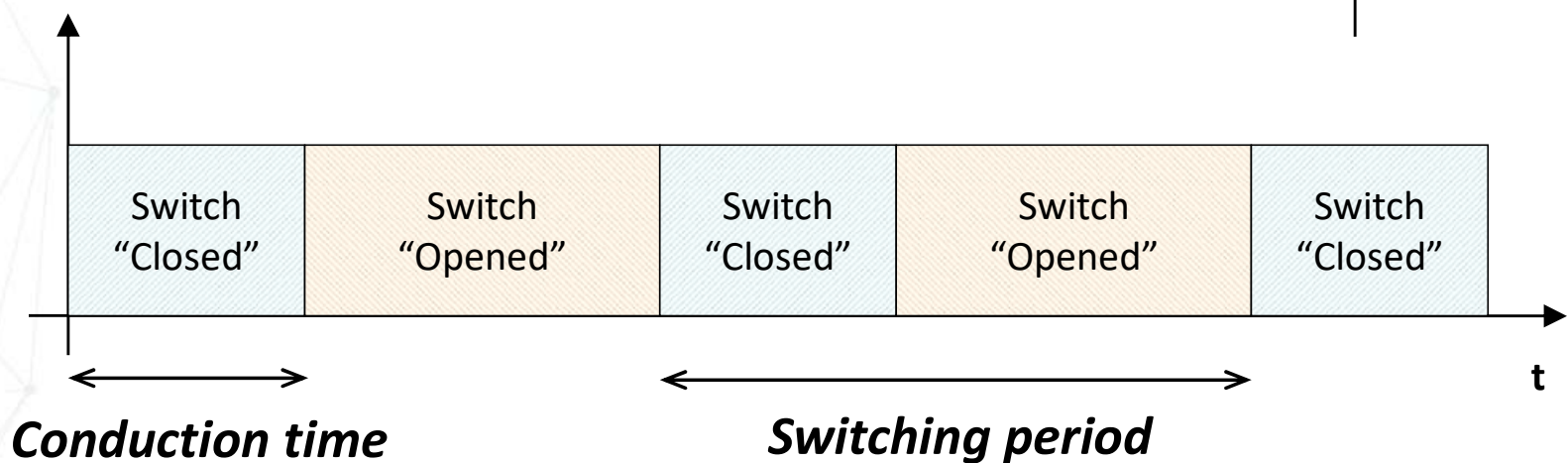
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Duty cycle



$$\alpha = \frac{\text{Conduction time}}{\text{Switching period}}$$

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Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

Inverters

Closed loop
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Energy storage
based chopper

Buck Chopper

Introduction

Direct conversion

Buck chopper

Boost chopper

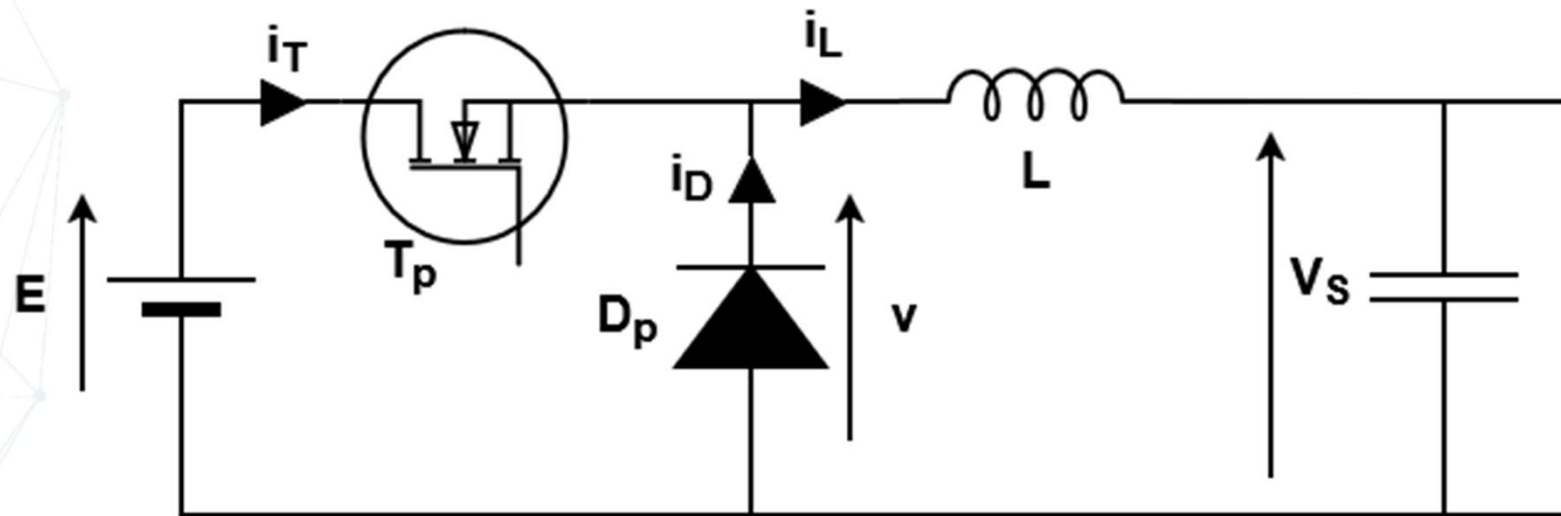
Reversible
choppers

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Energy storage
based chopper

Buck chopper : principle



Hypothesis : $V_s = \text{constant}$

Introduction

Direct conversion

Buck chopper

Boost chopper

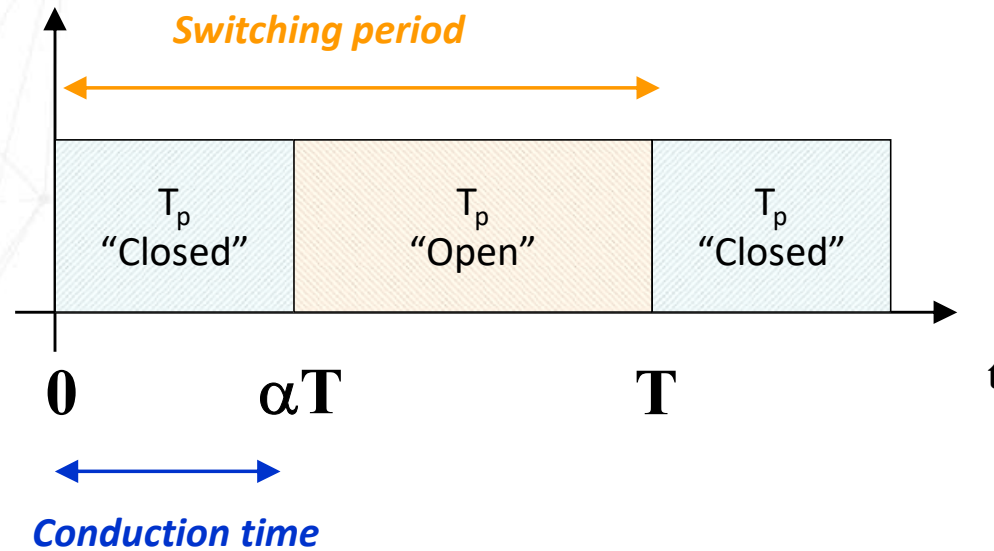
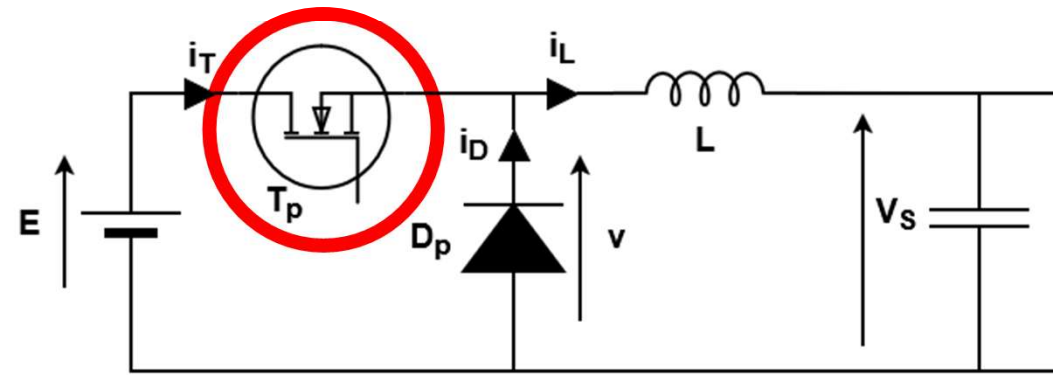
Reversible
choppers

Inverters

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Energy storage
based chopper

Control



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Buck chopper

Boost chopper

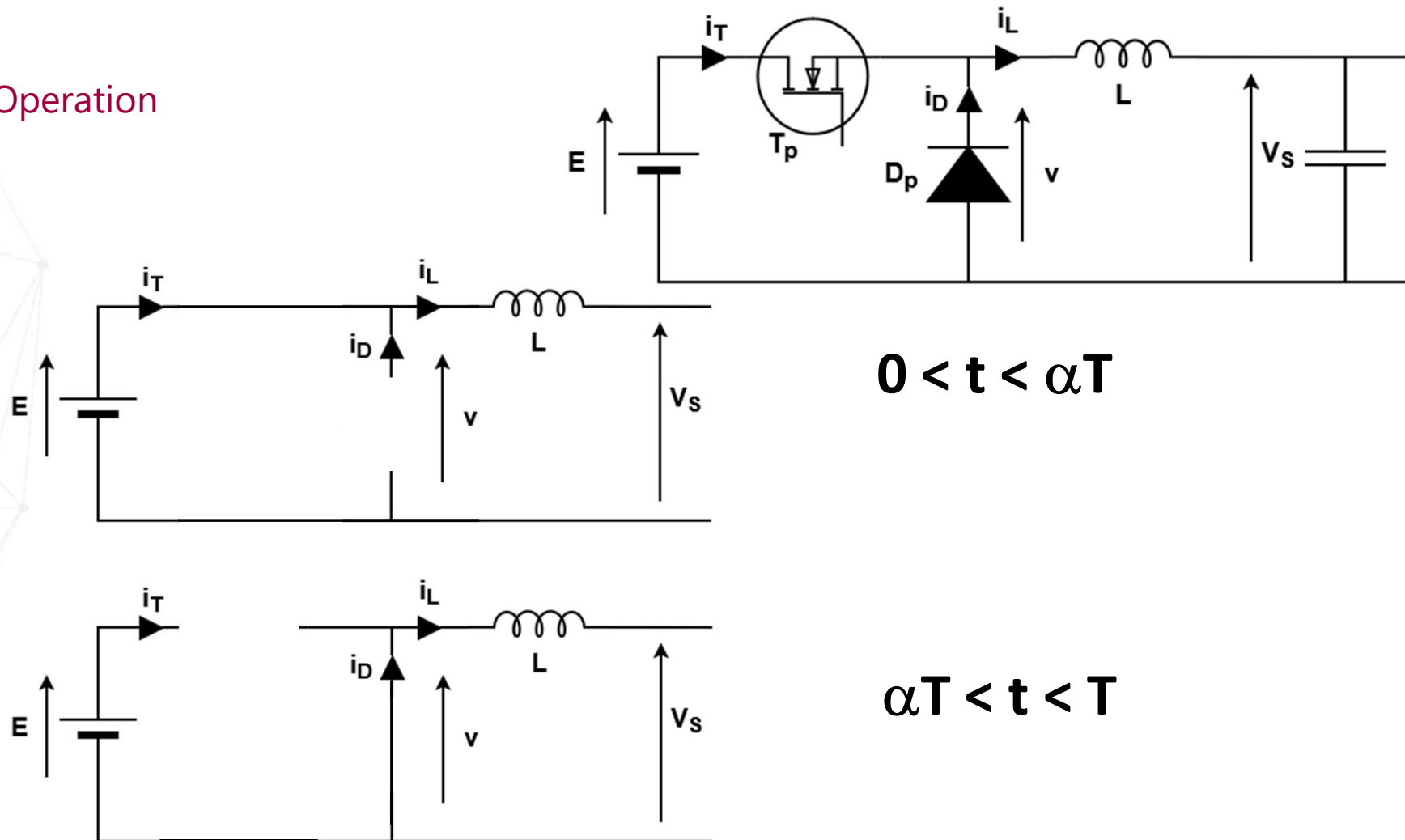
Reversible
choppers

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Operation



Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

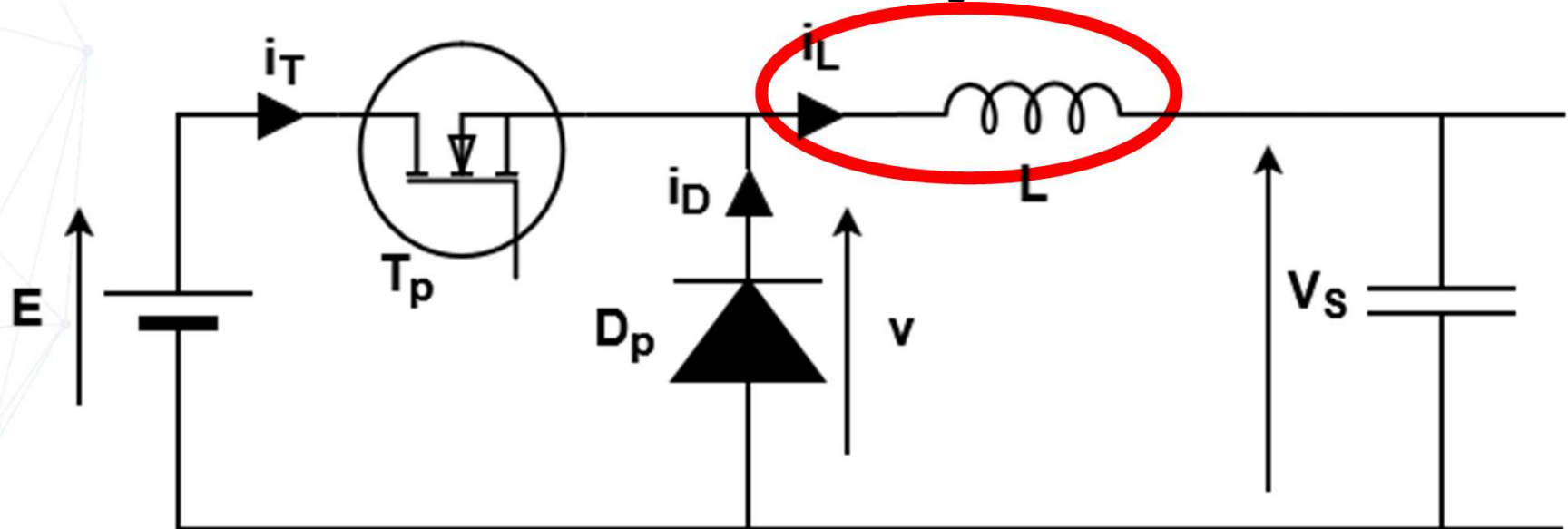
Inverters

Closed loop
chopper control

Energy storage
based chopper

Modes

- ***Continuous conduction mode: $i_L > 0$***



- ***Discontinuous conduction mode: $i_L(t_0) = 0 ; t_0 < T$***

Introduction

Direct conversion

Buck chopper

Boost chopper

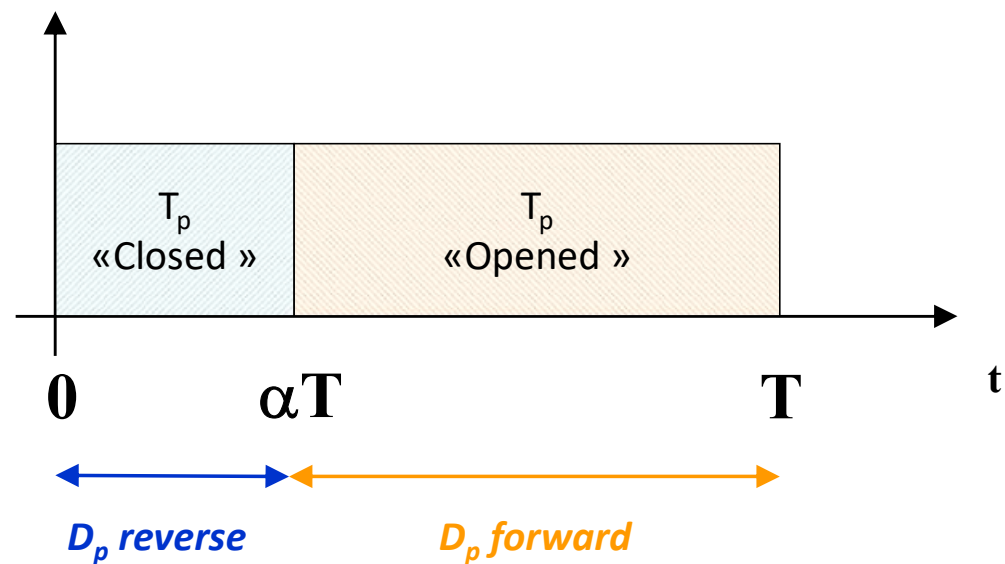
Reversible
choppers

Inverters

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Energy storage
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Continuous mode



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Direct conversion

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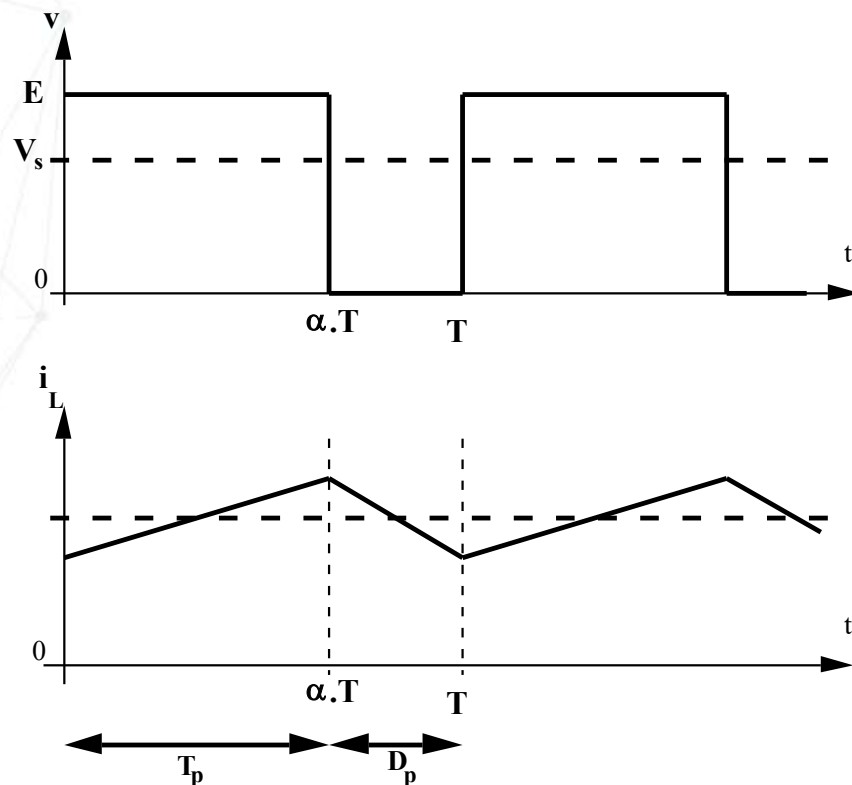
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Chronograms



- $0 < t < \alpha T$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$

- $\alpha T < t < T$

$$\frac{di}{dt} = -\frac{V_s}{L}$$

$$E > V_s$$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible choppers

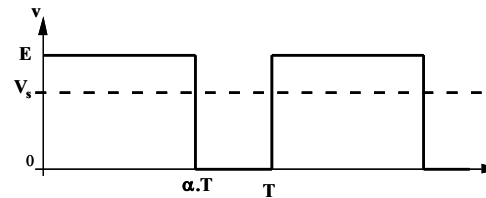
Inverters

Closed loop chopper control

Energy storage based chopper

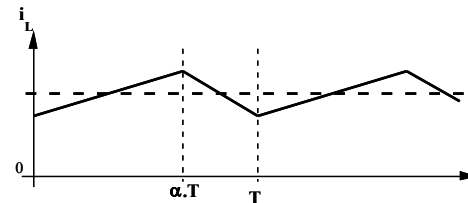
Continuous mode (Calculation)

Output voltage



$$\langle v \rangle = \alpha \cdot E = V_s$$

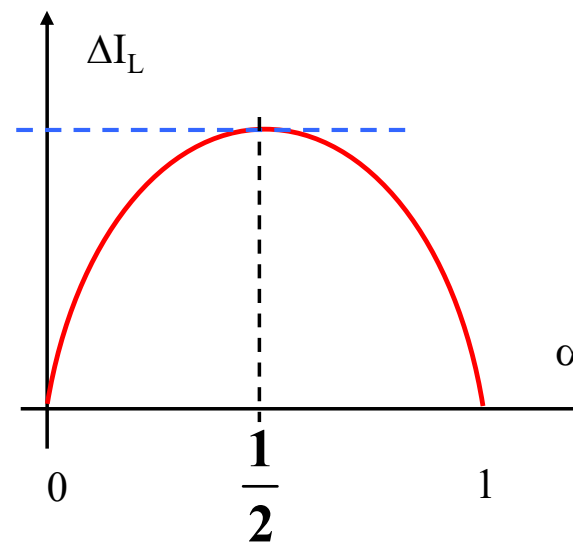
Current ripple i_L



$$\Delta I_L = \alpha \cdot (1 - \alpha) \cdot \frac{E \cdot T}{L}$$

Maximum ripple

$$\Delta I_{L \max} = \frac{E \cdot T}{4 L}$$



Introduction

Direct conversion

Buck chopper

Boost chopper

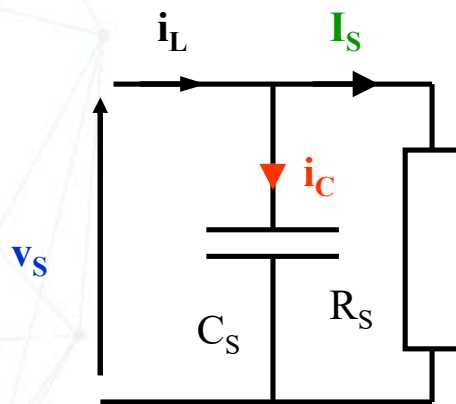
Reversible
choppers

Inverters

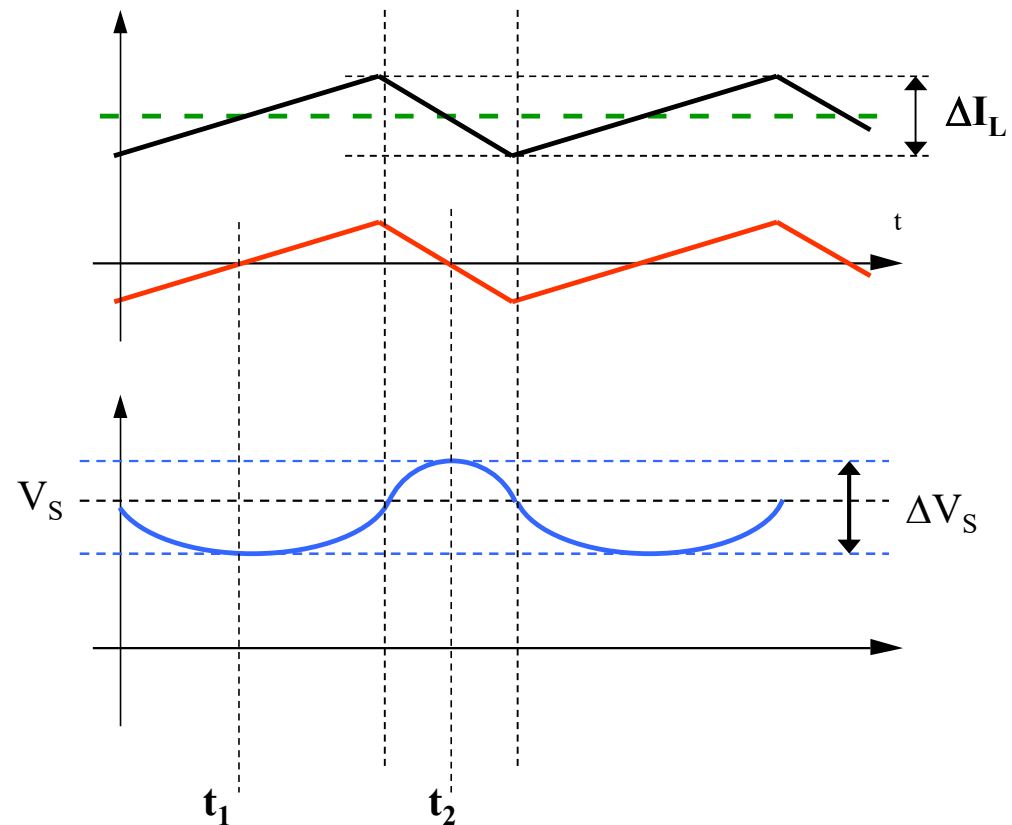
Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Output filtering principle



$$i_c = C_s \cdot \frac{dv_s}{dt}$$



Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
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Energy storage
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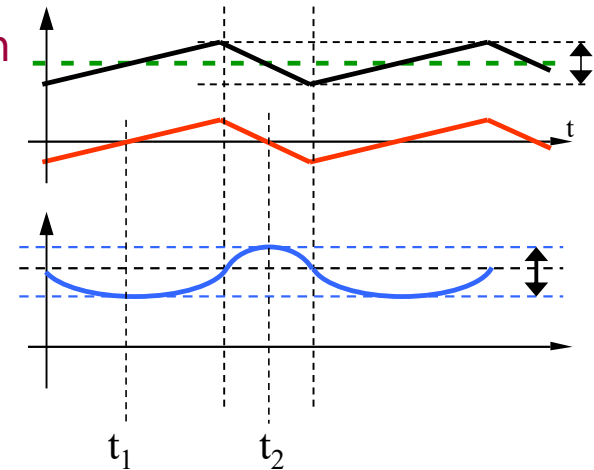
Continuous mode: Output filtering calculation

$$t_1 = \alpha \frac{T}{2}$$

$$t_2 = \frac{(1+\alpha).T}{2}$$

$$\Delta I_L = \alpha \cdot (1 - \alpha) \cdot \frac{E \cdot T}{L}$$

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{1}{C_S} \cdot \frac{1}{2} \cdot \frac{\Delta I_L}{2} \cdot (t_2 - t_1) = \frac{\Delta I_L \cdot T}{8 \cdot C_S}$$



$$C_S = \frac{\alpha \cdot (1 - \alpha) \cdot E \cdot T^2}{8 \cdot L \cdot \Delta V_S}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

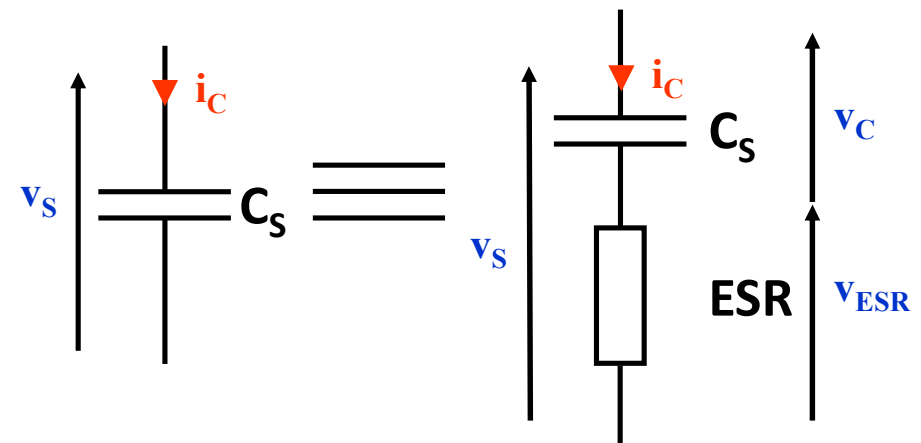
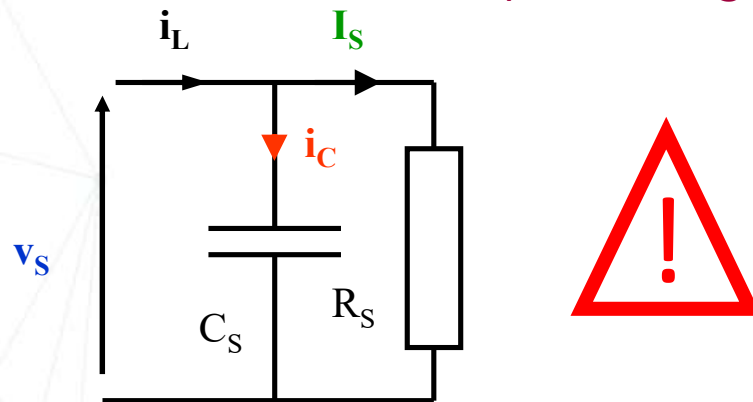
Reversible
choppers

Inverters

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Energy storage
based chopper

Continuous mode: Output filtering calculation



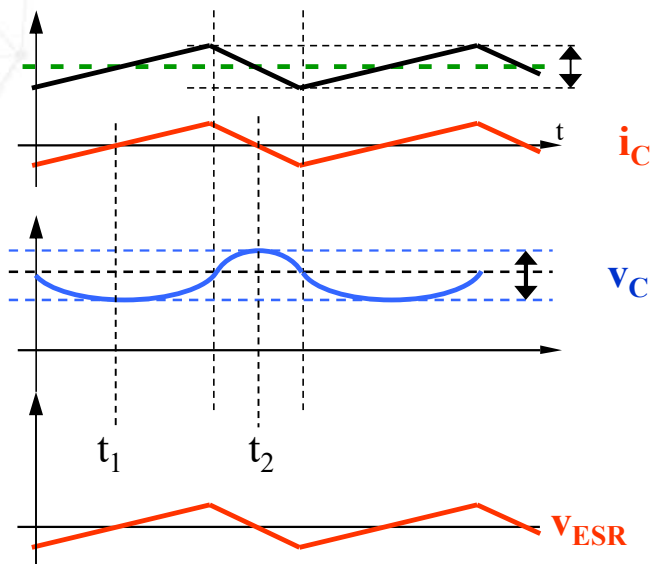
$$v_s = v_C + v_{ESR}$$

$$\Delta v_s < \Delta v_C + \Delta v_{ESR}$$

For high value of C_s ,

$$\Delta v_s \approx \Delta v_{ESR} = ESR \cdot \Delta i_C = ESR \cdot \Delta i_L$$

$$ESR < \frac{\Delta v_s}{\Delta i_L}$$



Introduction

Direct conversion

Buck chopper

Boost chopper

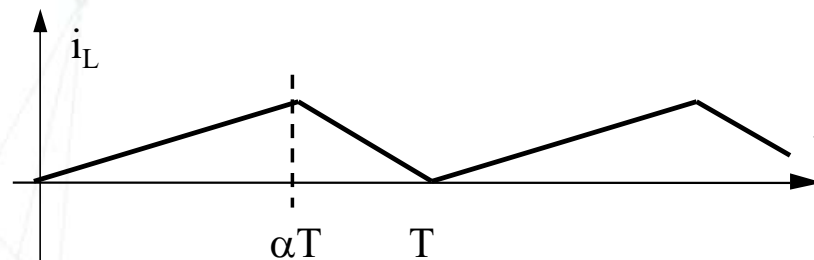
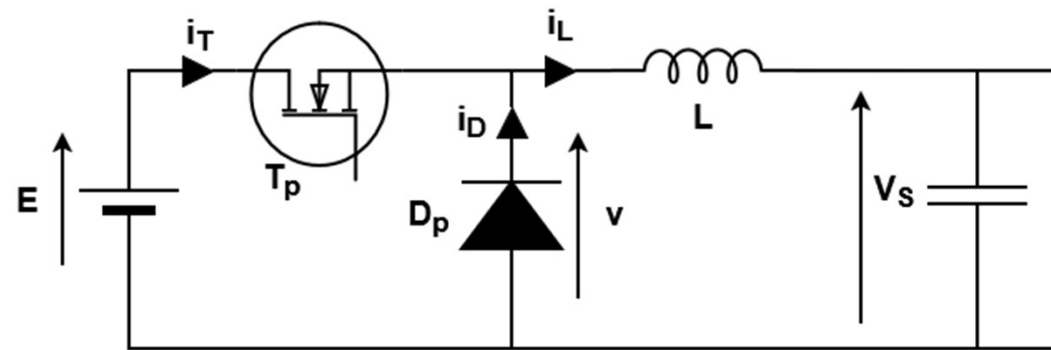
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

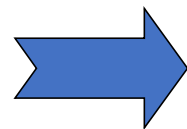
Boundary mode



$$i_L(T) = 0$$

$$\langle I_L \rangle = \frac{1}{2} \cdot \left(\frac{E - V_S}{L} \cdot \alpha T \right) = \frac{1}{2} \cdot \frac{\alpha(1-\alpha) \cdot E \cdot T}{L}$$

$$V_S = \alpha \cdot E$$



$$\langle I_L \rangle = \frac{T}{2L} \cdot V_S \left(1 - \frac{V_S}{E} \right)$$

*Curve of mode
separation (V_S, I_S)*

Introduction

Direct conversion

Buck chopper

Boost chopper

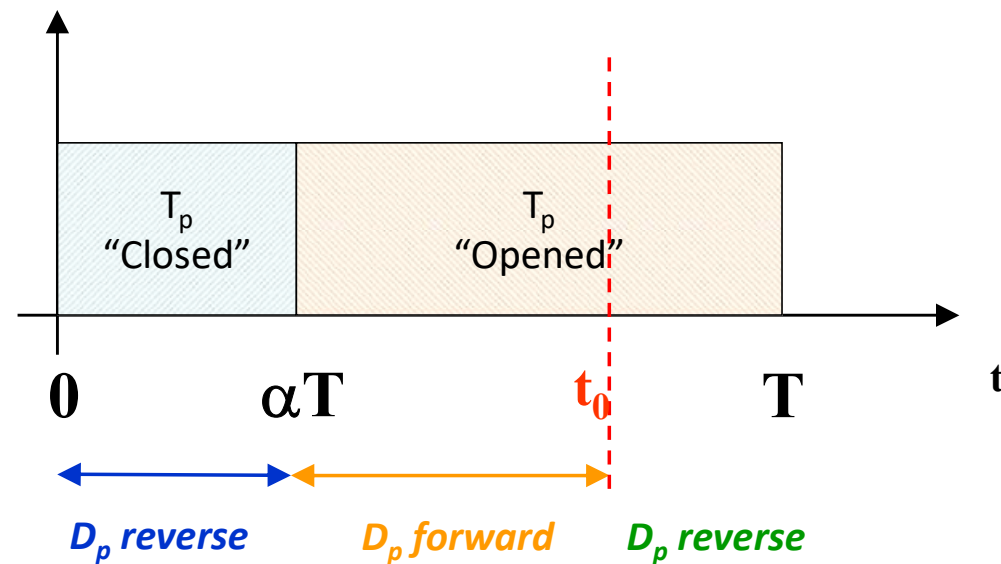
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode



At $t_0 = \alpha' T$: diode switches to reverse mode

Introduction

Direct conversion

Buck chopper

Boost chopper

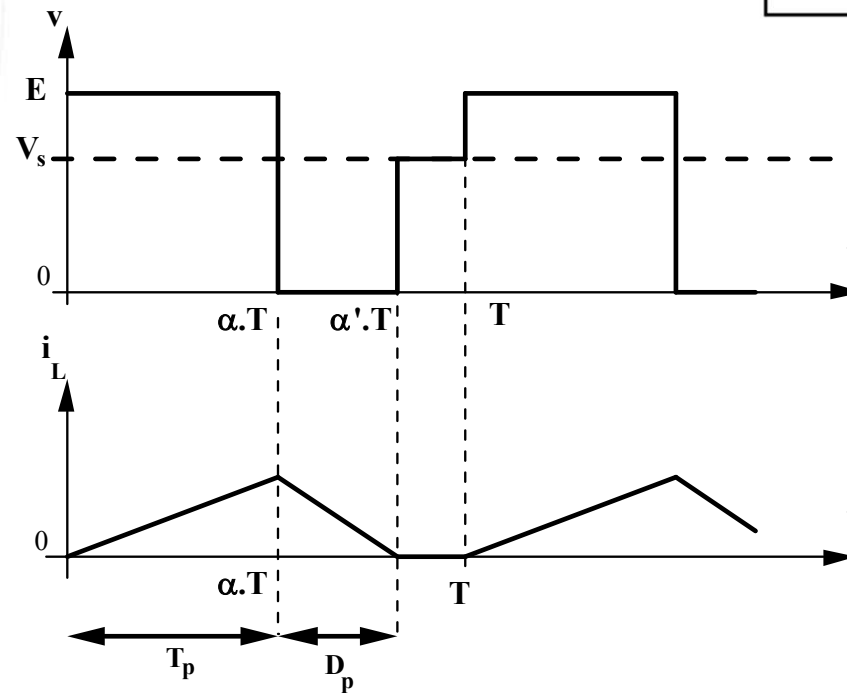
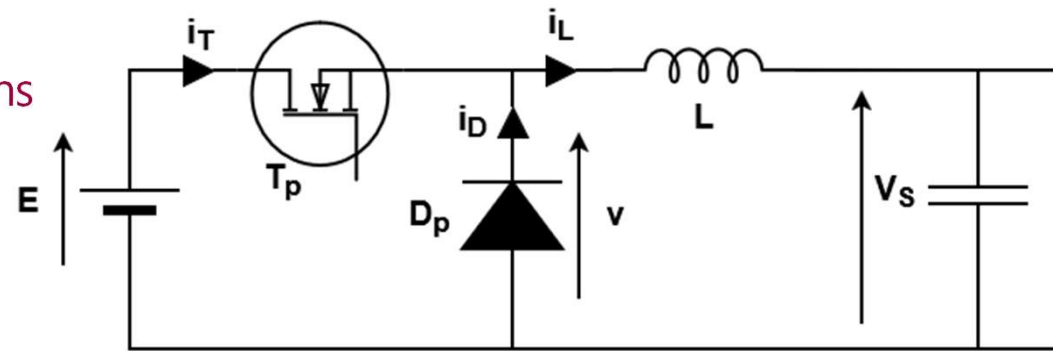
Reversible
choppers

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Energy storage
based chopper

Discontinuous mode: Chronograms



Introduction

Direct conversion

Buck chopper

Boost chopper

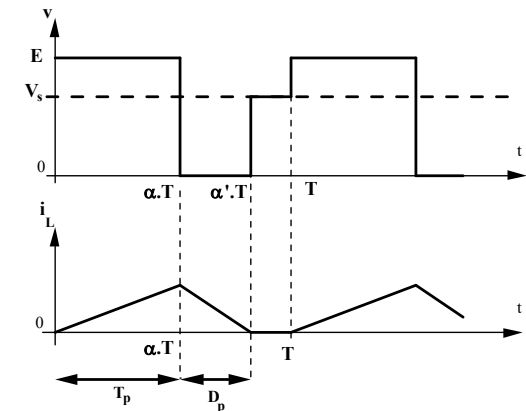
Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper

Discontinuous mode: calculation



$$I_{max} = \frac{E - V_s}{L} \cdot \alpha T = \frac{V_s}{L} \cdot (\alpha' T - \alpha T) \quad \Rightarrow \quad \frac{\alpha}{\alpha'} = \frac{V_s}{E}$$

$$\langle I_L \rangle = \frac{1}{2} \cdot I_{max} \cdot \alpha' = \frac{E - V_s}{2L} \cdot \alpha \cdot \alpha' \cdot T = \frac{\alpha \cdot E \cdot T}{2L} (\alpha' - \alpha)$$
$$\Rightarrow \alpha' = \alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}$$

$$\Rightarrow V_s = \frac{\alpha \cdot E}{\alpha + \frac{2L \cdot \langle I_L \rangle}{\alpha \cdot E \cdot T}}$$

Output current: $\langle I_L \rangle = I_s$

Introduction

Direct conversion

Buck chopper

Boost chopper

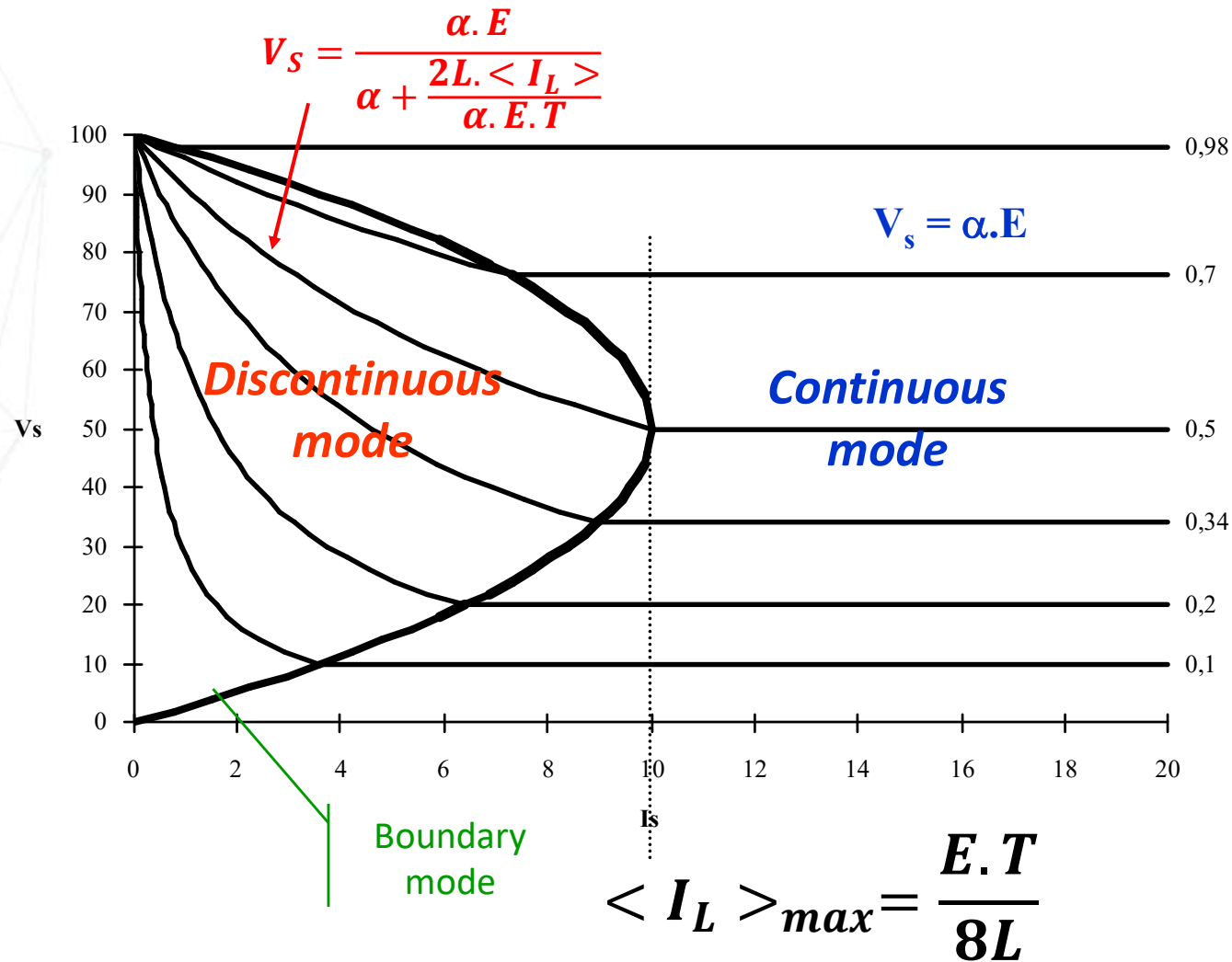
Reversible
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Energy storage
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Output characteristics



Introduction

Direct conversion

Buck chopper

Boost chopper

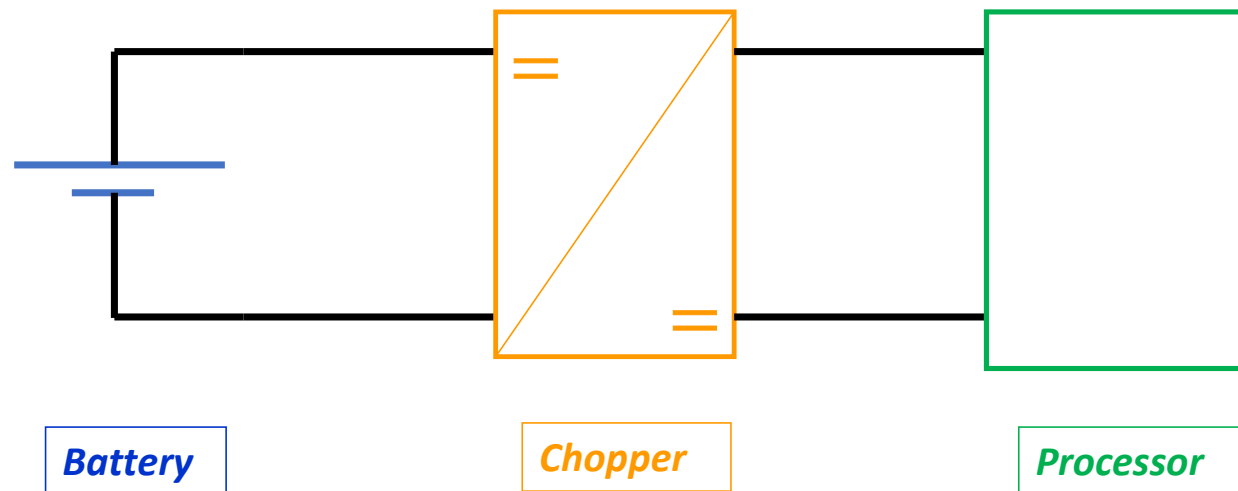
Reversible
choppers

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Energy storage
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Example



=> DC supply from battery

Introduction

Direct conversion

Buck chopper

Boost chopper

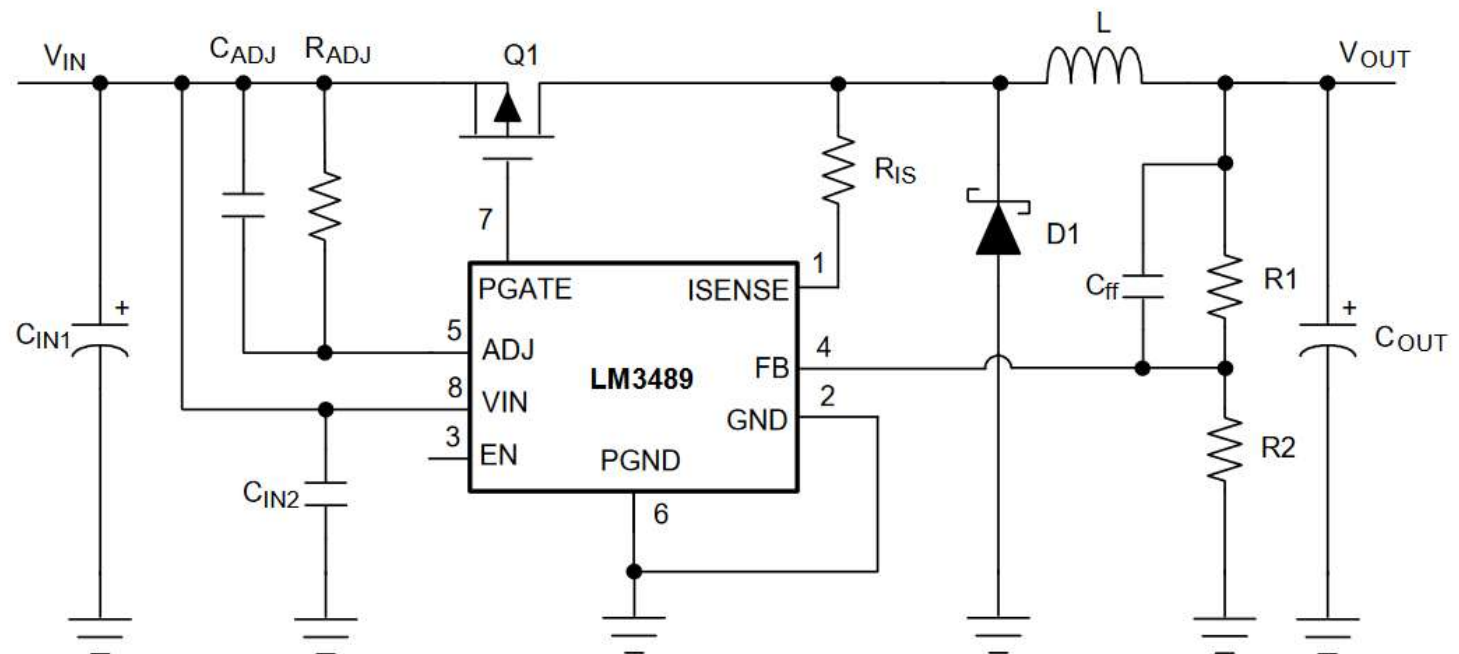
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Example : LM3489



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Introduction

Direct conversion

Buck chopper

Boost chopper

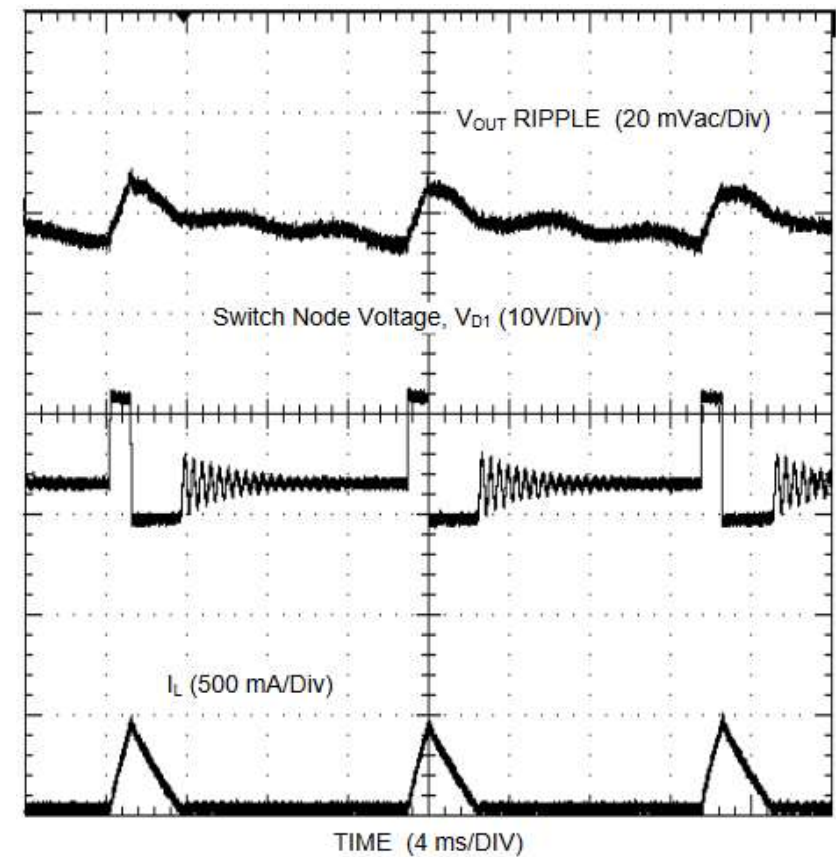
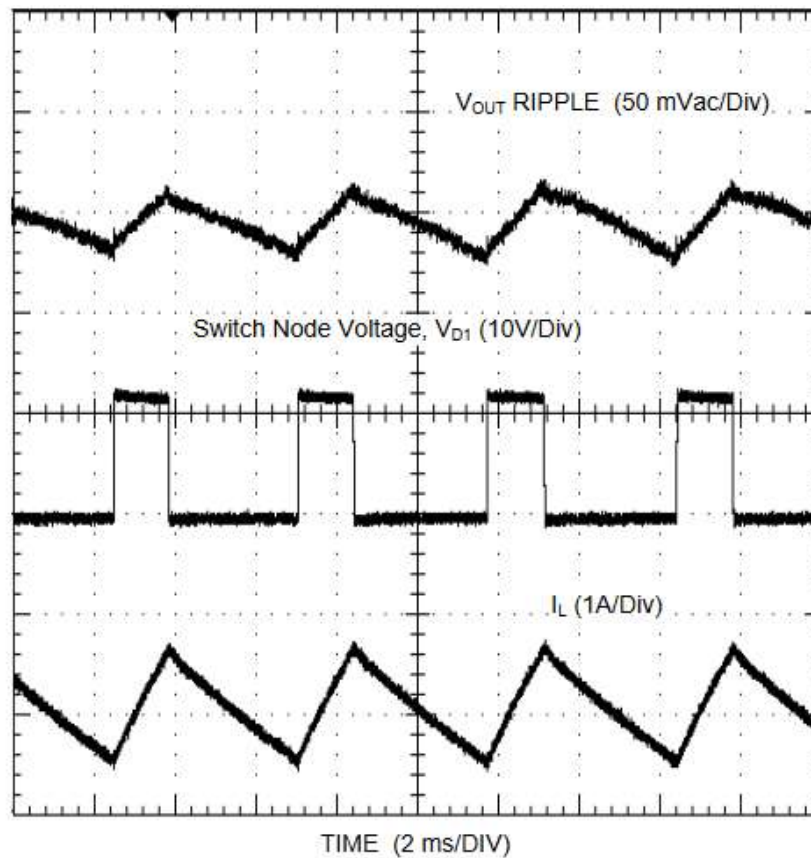
Reversible
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Example : LM3489



Introduction

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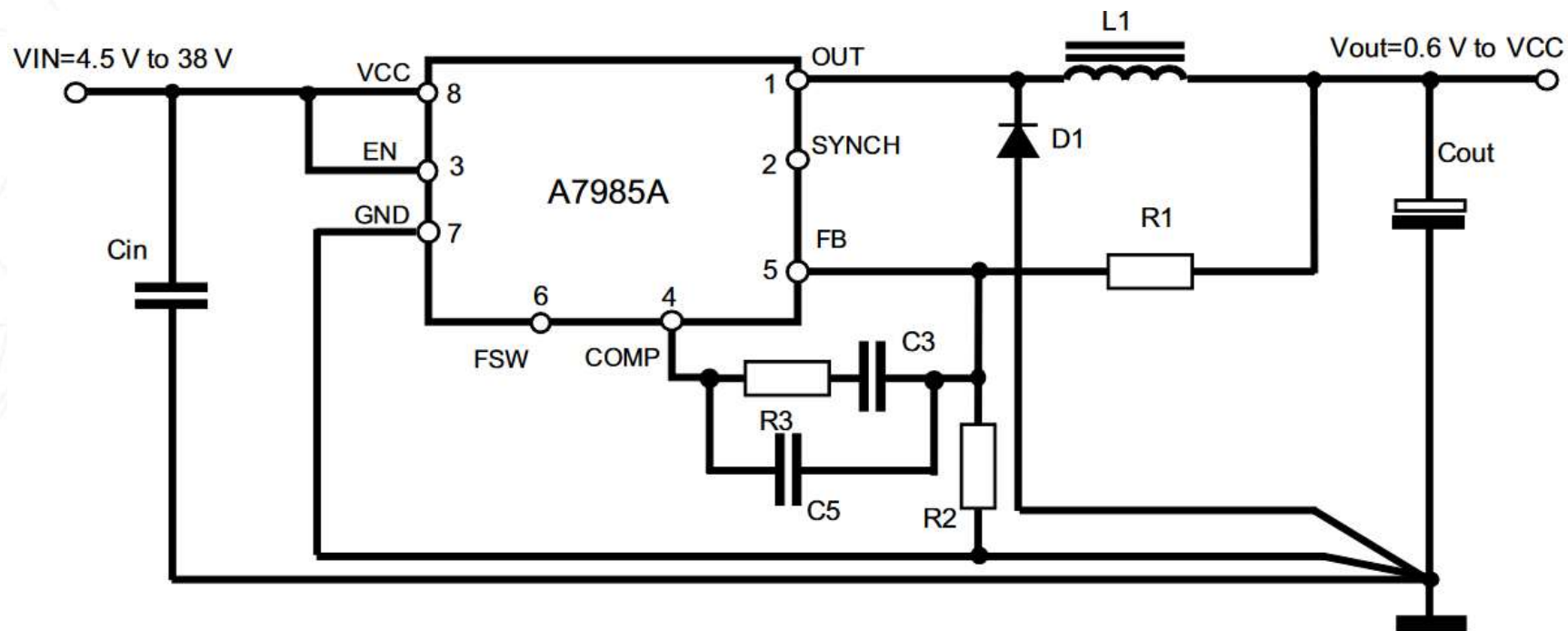
Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper

Example : A7985A



Introduction

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Boost chopper

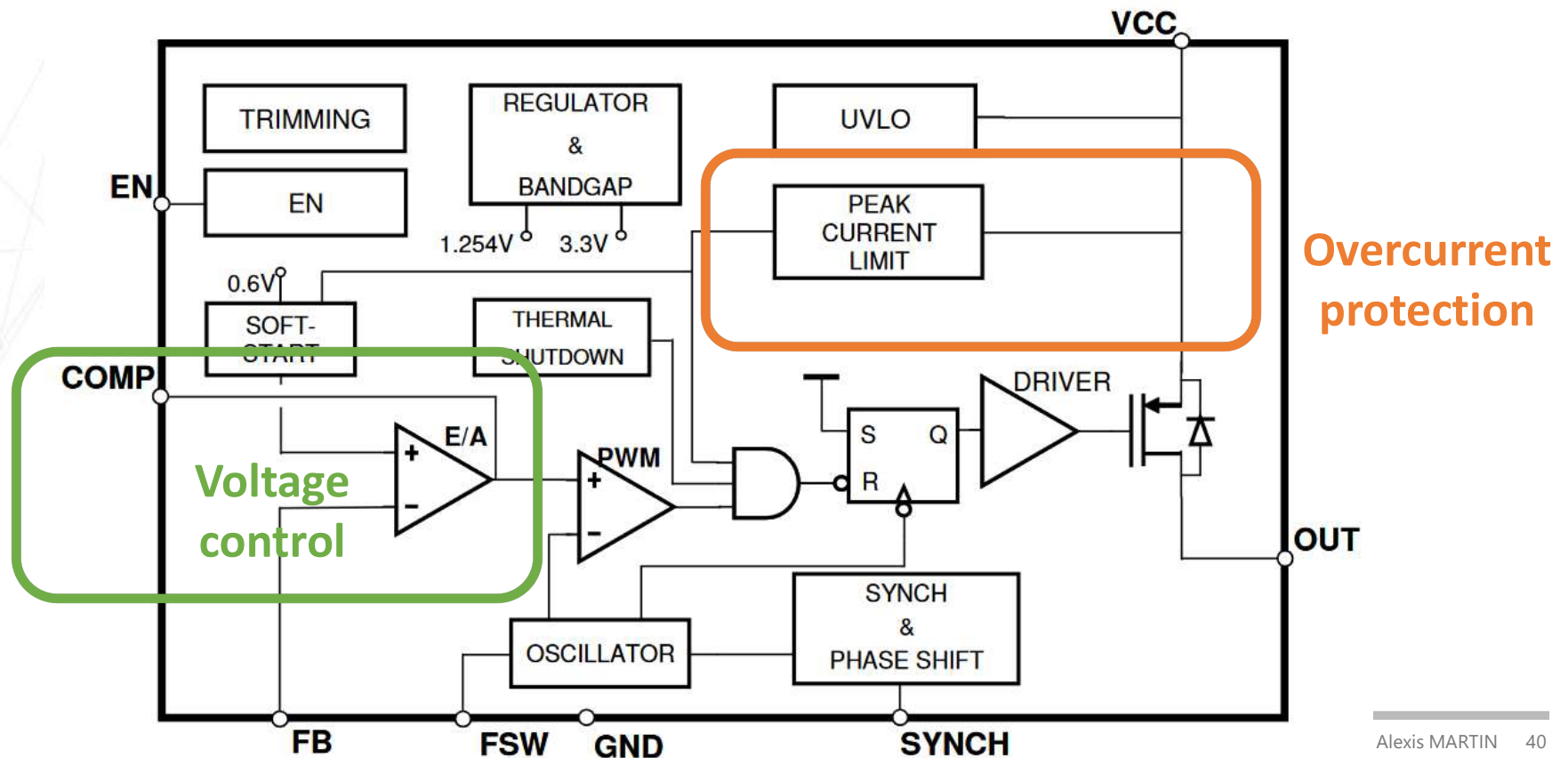
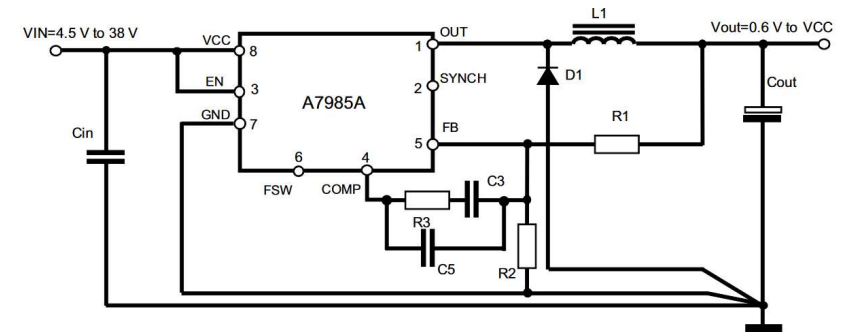
Reversible choppers

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Energy storage based chopper

Example : A7985A



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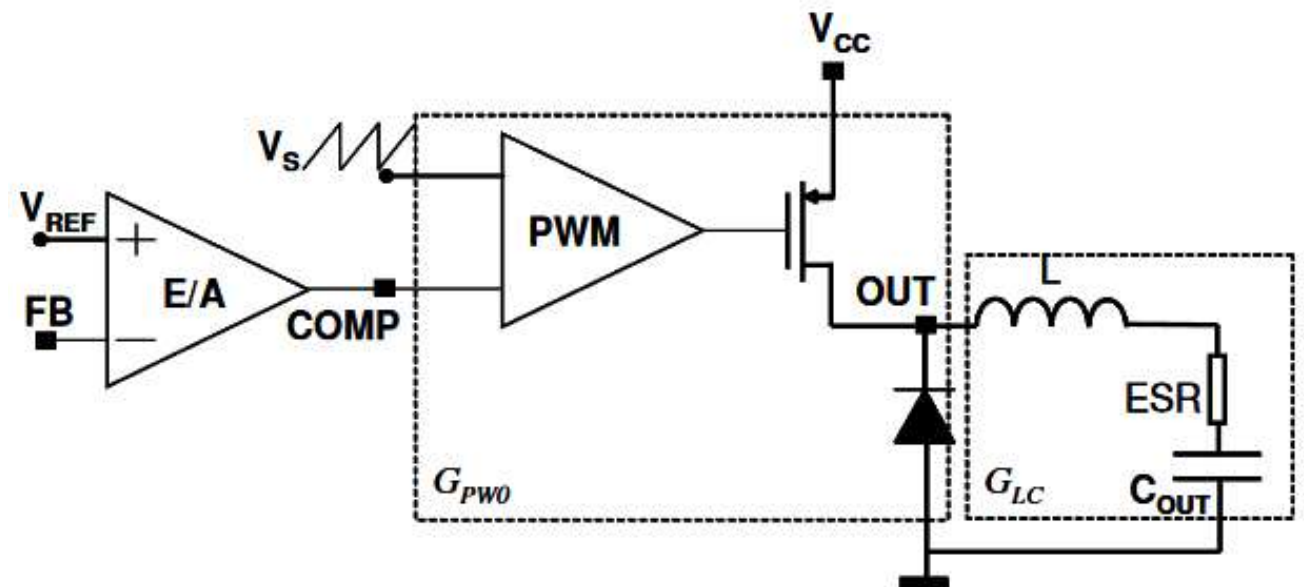
Reversible choppers

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Closed loop chopper control

Energy storage based chopper

Example : A7985A



$$G_{LC}(s) = \frac{1 + \frac{s}{2\pi \cdot f_{zESR}}}{1 + \frac{s}{2\pi \cdot Q \cdot f_{LC}} + \left(\frac{s}{2\pi \cdot f_{LC}}\right)^2}$$

$$f_{LC} = \frac{1}{2\pi \cdot \sqrt{L \cdot C_{OUT}} \cdot \sqrt{1 + \frac{ESR}{R_{OUT}}}}, \quad f_{zESR} = \frac{1}{2\pi \cdot ESR \cdot C_{OUT}}$$

$$Q = \frac{\sqrt{R_{OUT} \cdot L \cdot C_{OUT} \cdot (R_{OUT} + ESR)}}{L + C_{OUT} \cdot R_{OUT} \cdot ESR}$$

Usually : $f_{LC} < f_{zESR}$

→ Slightly depends on the load

Introduction

Direct conversion

Buck chopper

Boost chopper

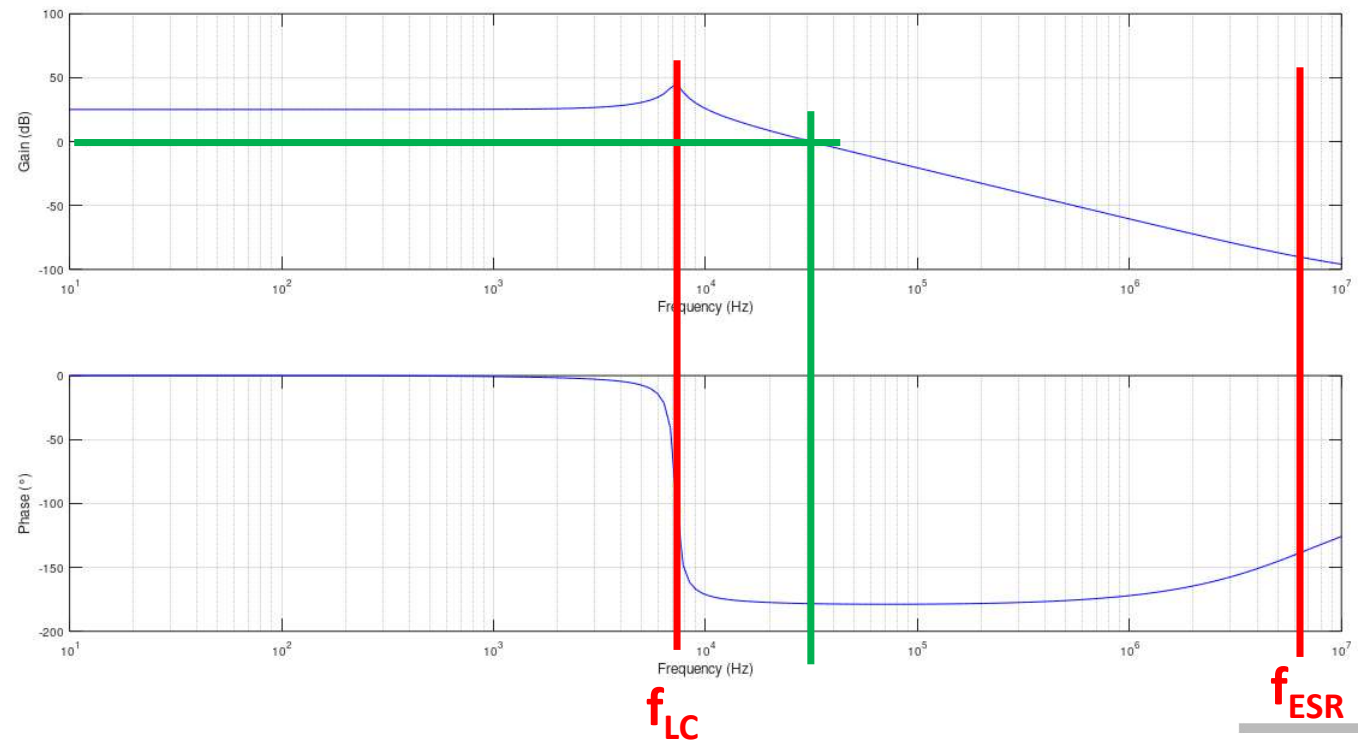
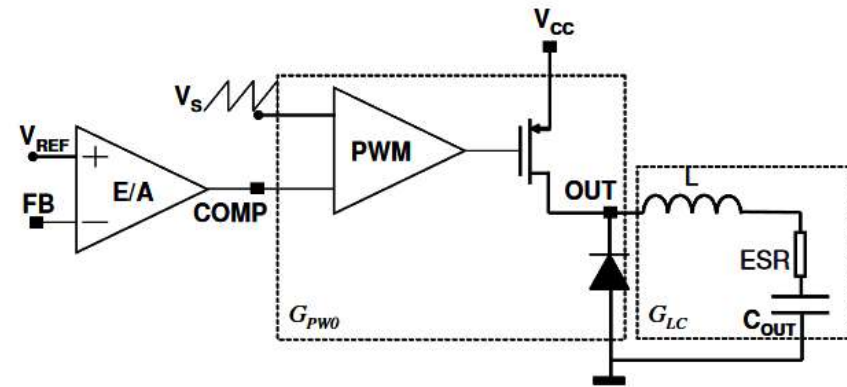
Reversible choppers

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Example : A7985A



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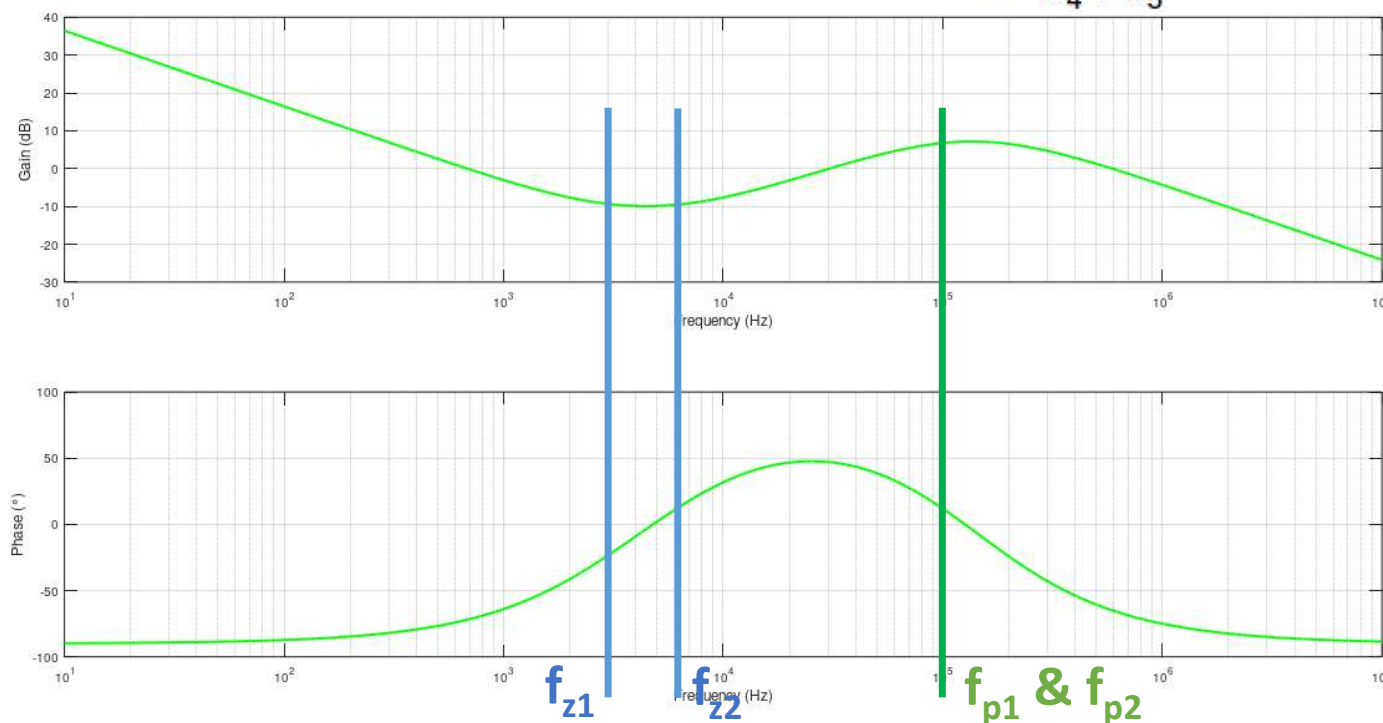
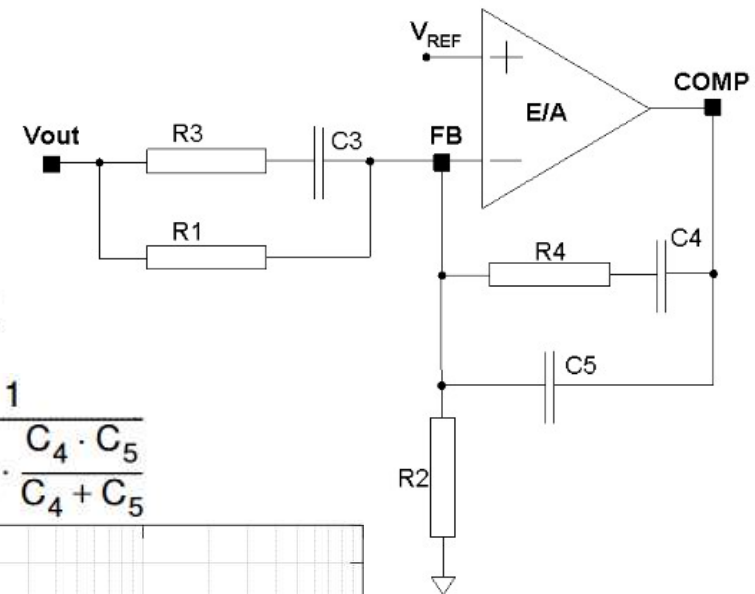
Energy storage based chopper

Example : A7985A

If $f_{zESR} \gg BW$ (desired)
(ESR < 1mΩ)

Zeroes : $f_{z1} = \frac{1}{2\pi \cdot C_3 \cdot (R_1 + R_3)}$, $f_{z2} = \frac{1}{2\pi \cdot R_4 \cdot C_4}$

Poles : $f_{p0} = 0$, $f_{p1} = \frac{1}{2\pi \cdot R_3 \cdot C_3}$, $f_{p2} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$



Introduction

Direct conversion

Buck chopper

Boost chopper

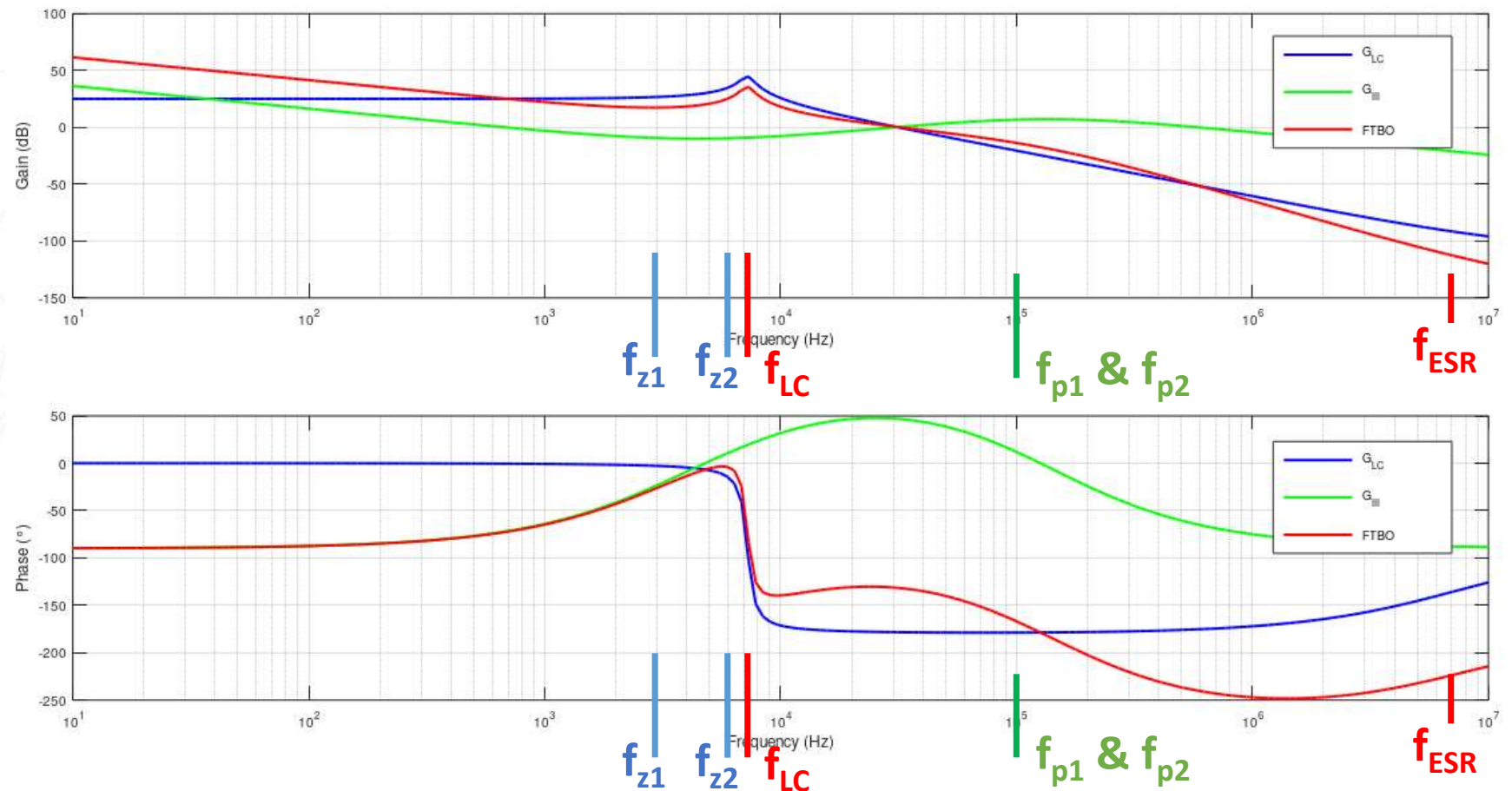
Reversible
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Inverters

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Energy storage
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Example : A7985A



Introduction

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Reversible choppers

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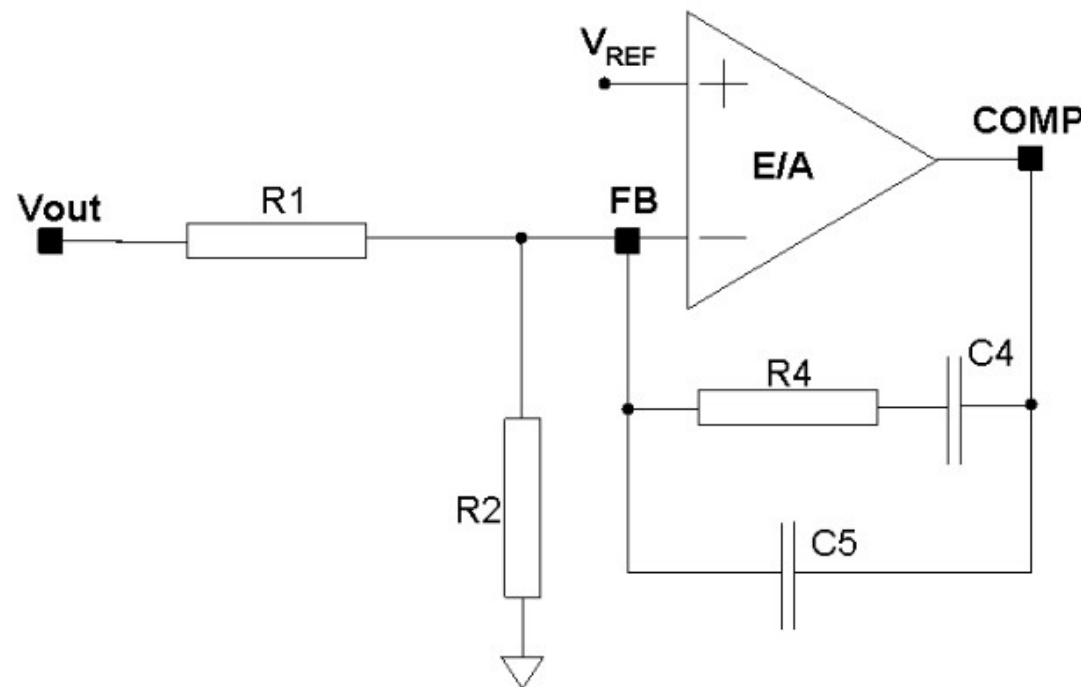
Closed loop chopper control

Energy storage based chopper

Example : A7985A

If $f_{zESR} \approx BW$ (desired)
 $ESR > 30 \text{ m}\Omega$

→ Use of the zero of ESR to stabilize the loop



Zero : $f_{z1} = \frac{1}{2\pi \cdot R_4 \cdot C_4},$

Poles : $f_{p0} = 0,$

$$f_{p1} = \frac{1}{2\pi \cdot R_4 \cdot \frac{C_4 \cdot C_5}{C_4 + C_5}}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Boost Chopper

Introduction

Direct conversion

Buck chopper

Boost chopper

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based chopper

Boost chopper

Buck chopper : Voltage source



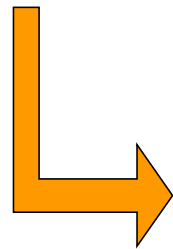
Current receiver

Boost chopper : Current source



Voltage receiver

Current source $\Leftrightarrow$ Voltage source + inductor



Voltage receiver $\Leftrightarrow$ Battery or capacitive receiver

Introduction

Direct conversion

Buck chopper

Boost chopper

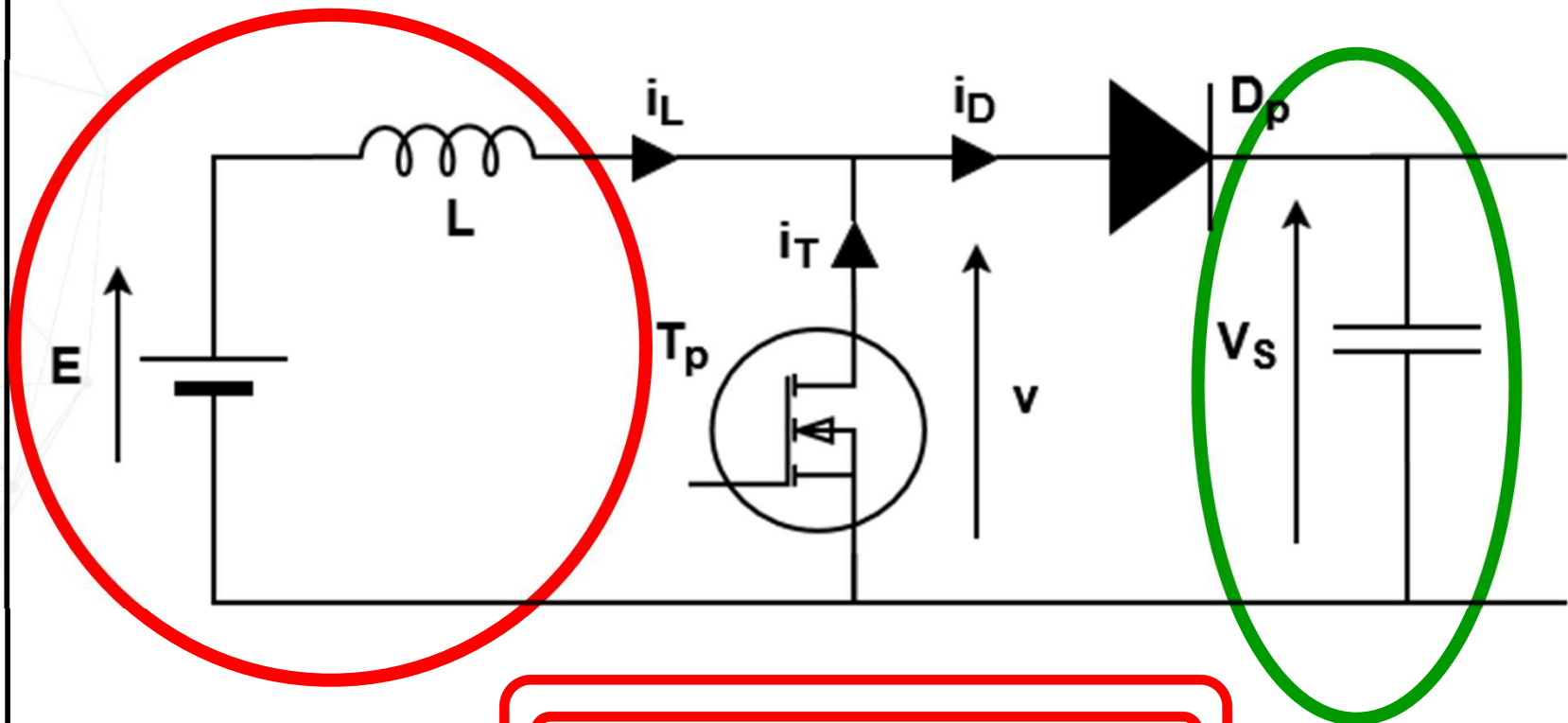
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Boost chopper: principle



Hypothesis : $V_s = \text{constant}$

Introduction

Direct conversion

Buck chopper

Boost chopper

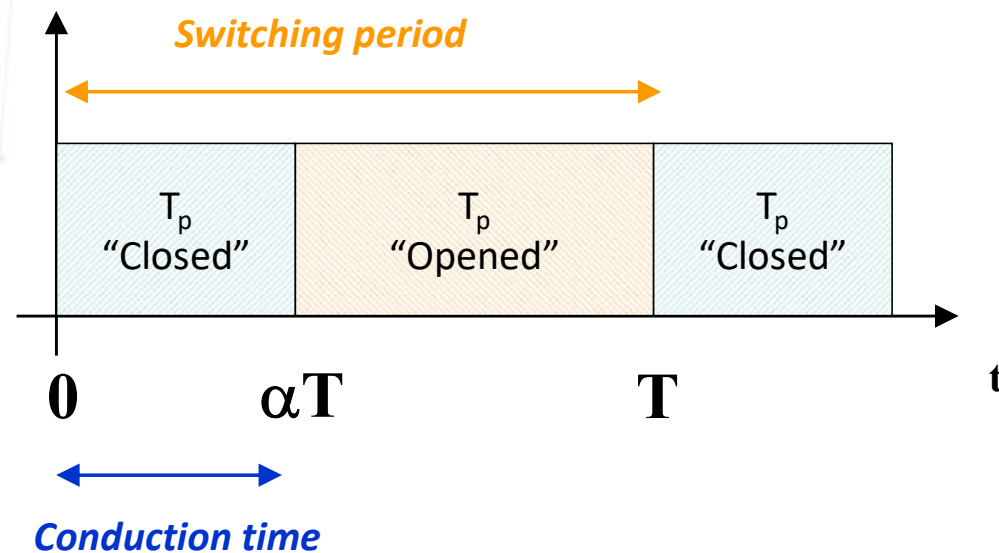
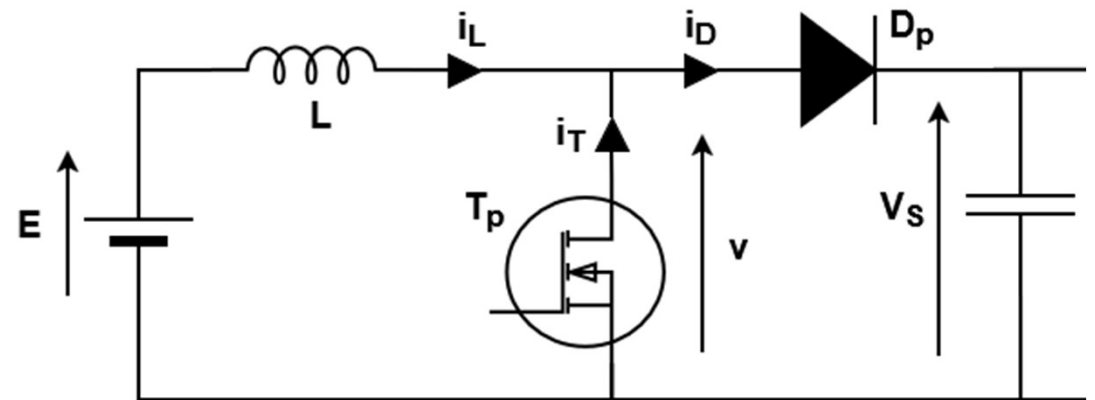
Reversible
choppers

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based chopper

Control



Introduction

Direct conversion

Buck chopper

Boost chopper

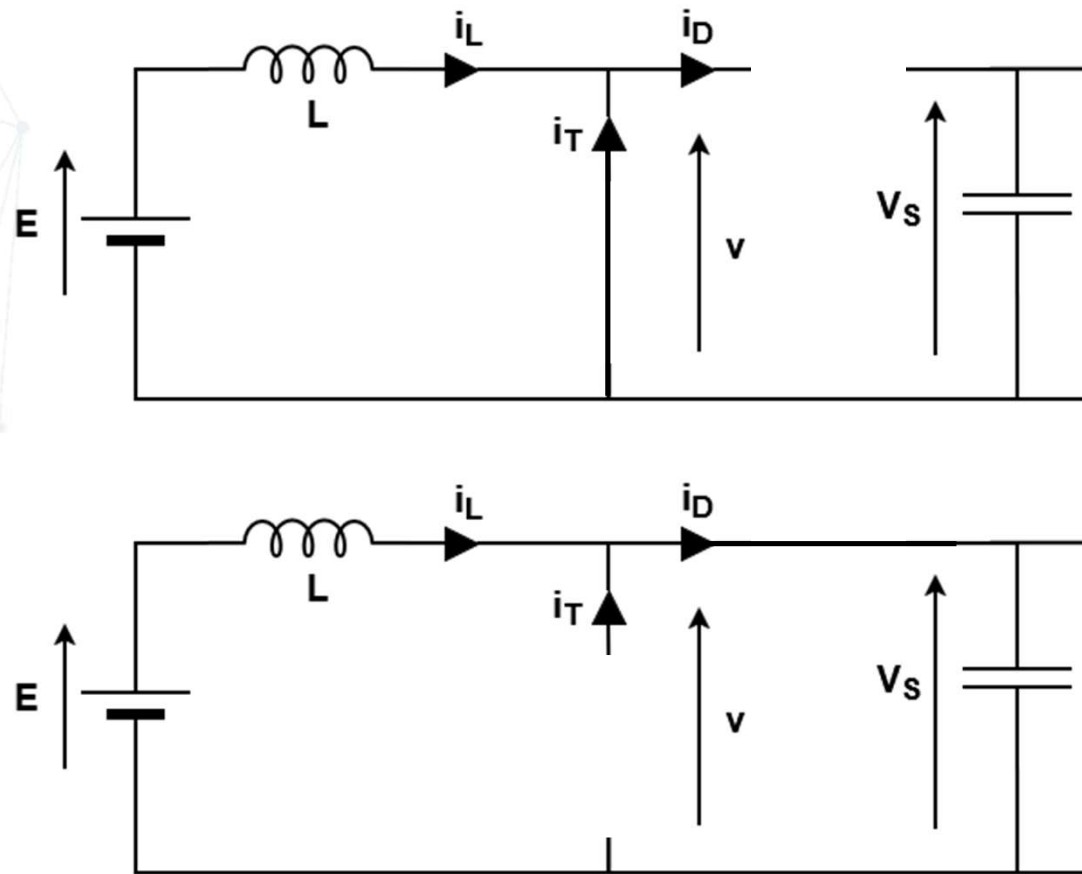
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Principle



$$0 < t < \alpha T$$

$$\alpha T < t < T$$

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

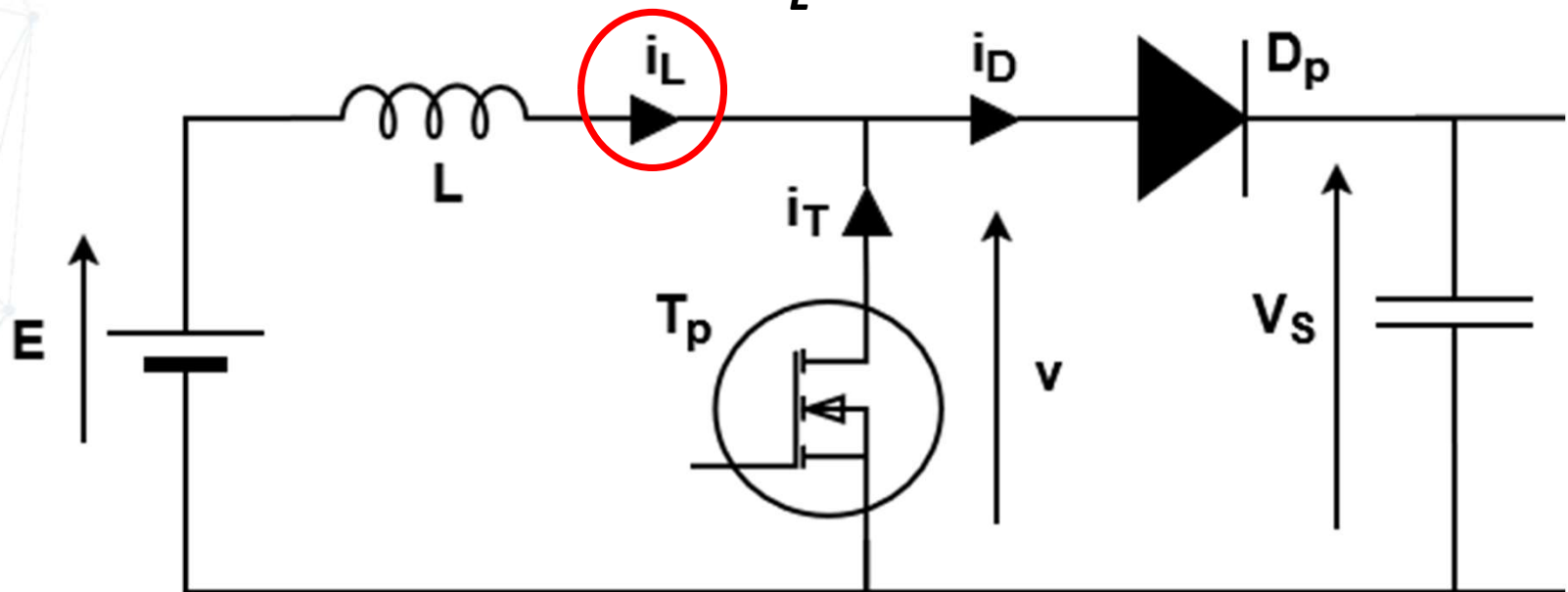
Inverters

Closed loop
chopper control

Energy storage
based chopper

Modes

- **Continuous conduction: $i_L > 0$**



- **Discontinuous conduction: $i_L(t_0) = 0 ; t_0 < T$**

Introduction

Direct conversion

Buck chopper

Boost chopper

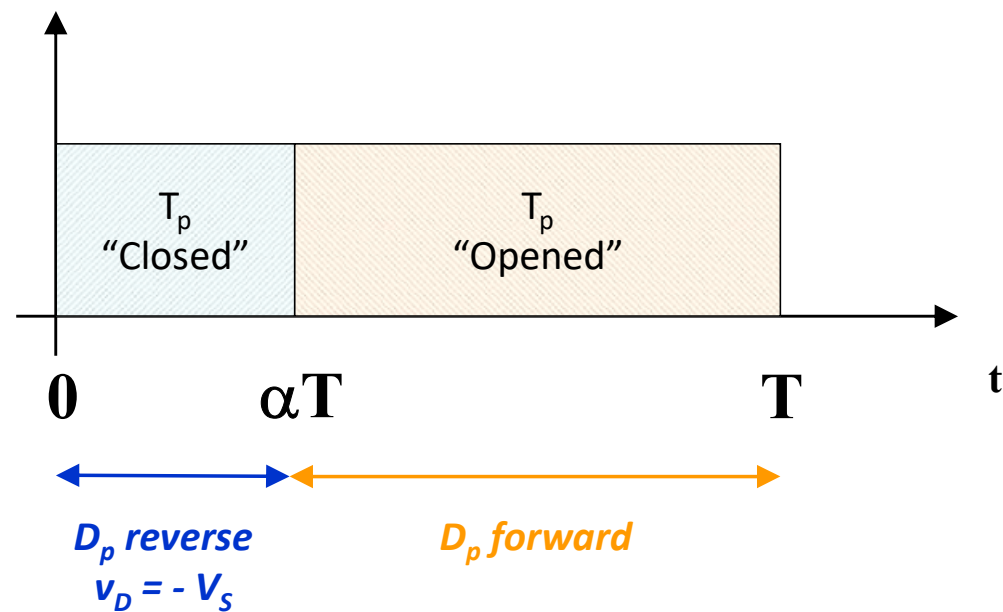
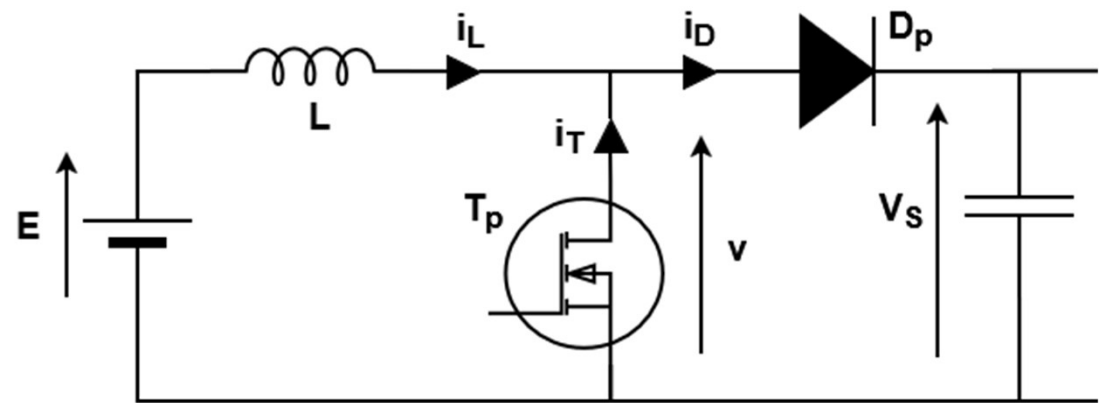
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode



Introduction

Direct conversion

Buck chopper

Boost chopper

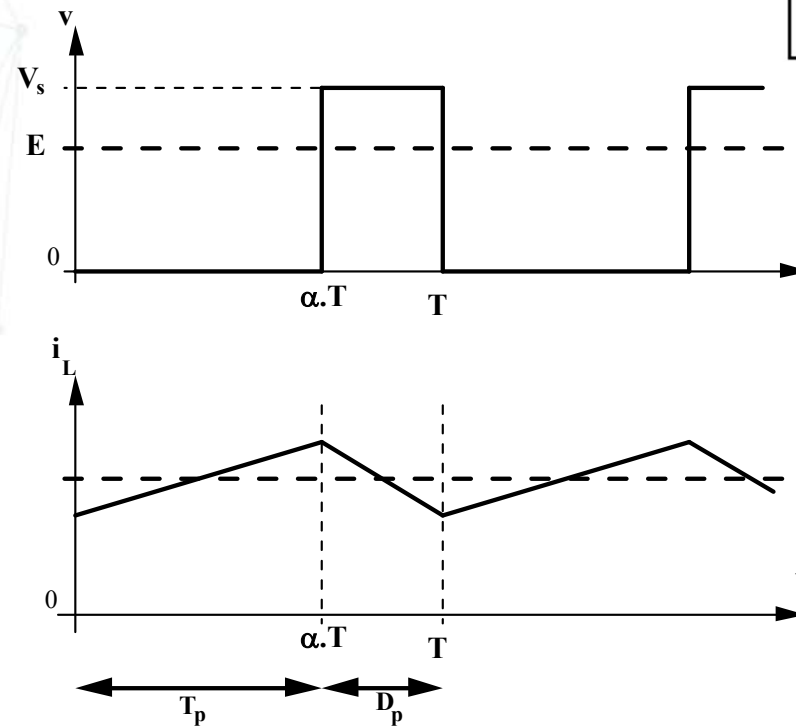
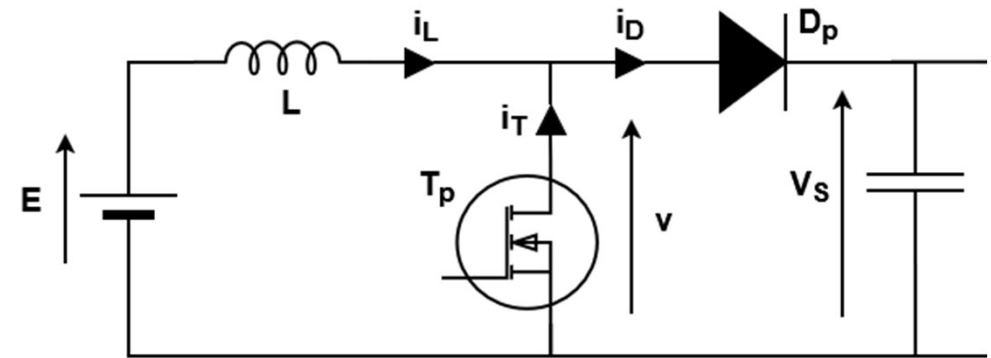
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Chronograms



$$E < V_s$$

- $0 < t < \alpha T$

$$\frac{di}{dt} = \frac{E}{L}$$

- $\alpha T < t < T$

$$\frac{di}{dt} = \frac{E - V_s}{L}$$

BOOST

Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

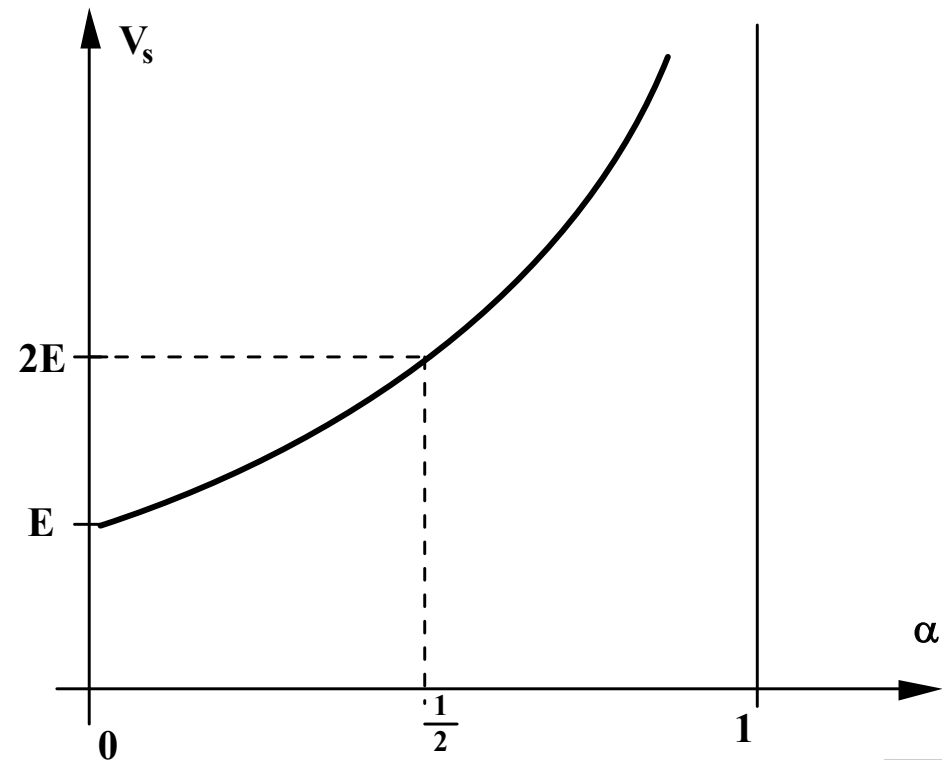
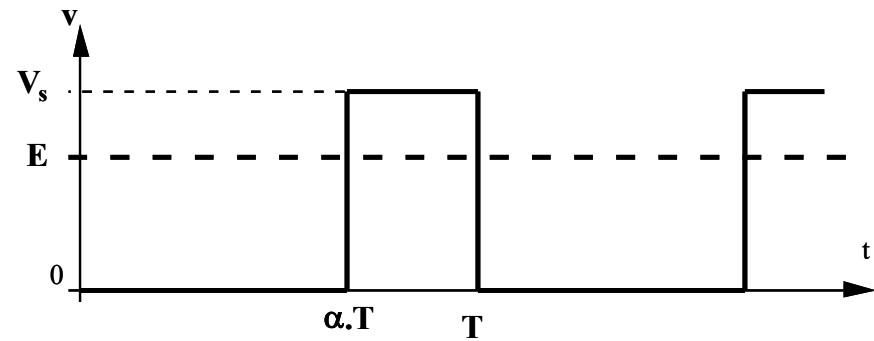
Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode:
Output voltage calculation

$$V_s = \frac{E}{1 - \alpha}$$



Introduction

Direct conversion

Buck chopper

Boost chopper

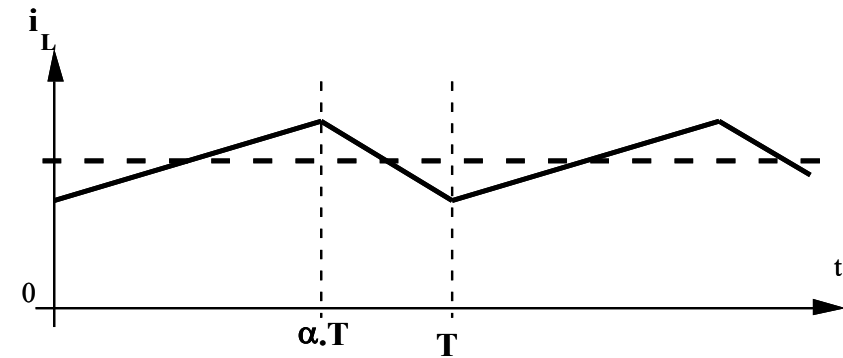
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode:
Current ripple calculation



Current ripple i_L

$$\Delta I_L = \alpha \cdot \frac{E \cdot T}{L}$$

Maximum ripple ($\alpha = 1$)

$$\Delta I_{L \max} = \frac{E \cdot T}{L}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

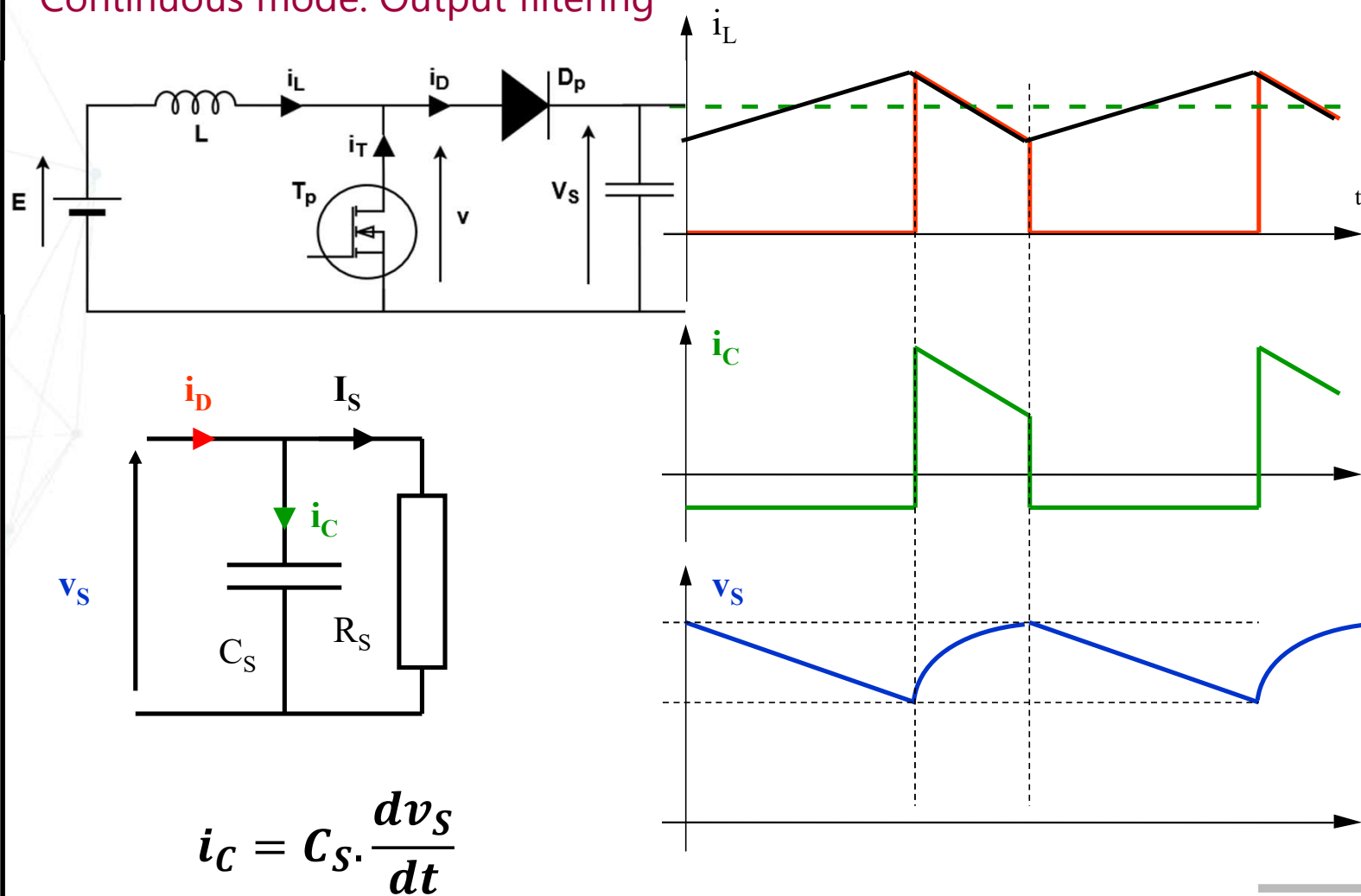
Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper

Continuous mode: Output filtering



Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

Inverters

Closed loop
chopper control

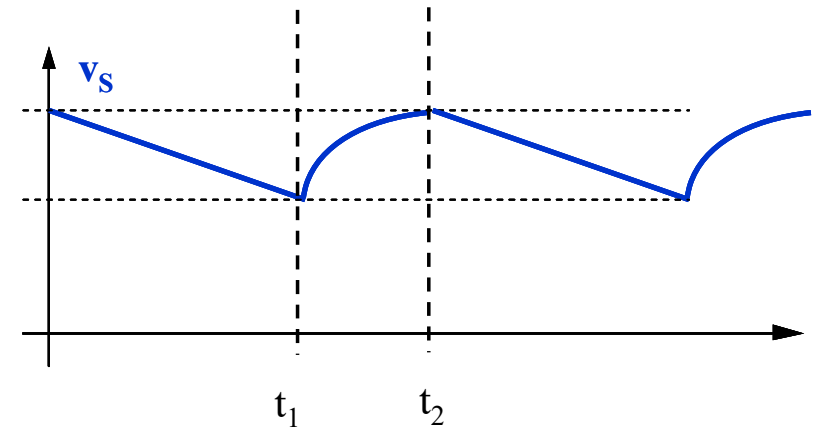
Energy storage
based chopper

Continuous mode: Output filtering

$$\Delta V_S = \frac{1}{C_S} \cdot \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{I_S \cdot \alpha T}{C_S}$$

$$\text{or } I_S = \frac{V_S}{R_S} = \frac{E}{(1-\alpha)R_S}$$

$$\Delta V_S = \frac{ET}{R_S C_S} \cdot \frac{\alpha}{1-\alpha}$$



$$C_S = \frac{\alpha \cdot E \cdot T}{(1 - \alpha) \cdot R_S \cdot \Delta V_S}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

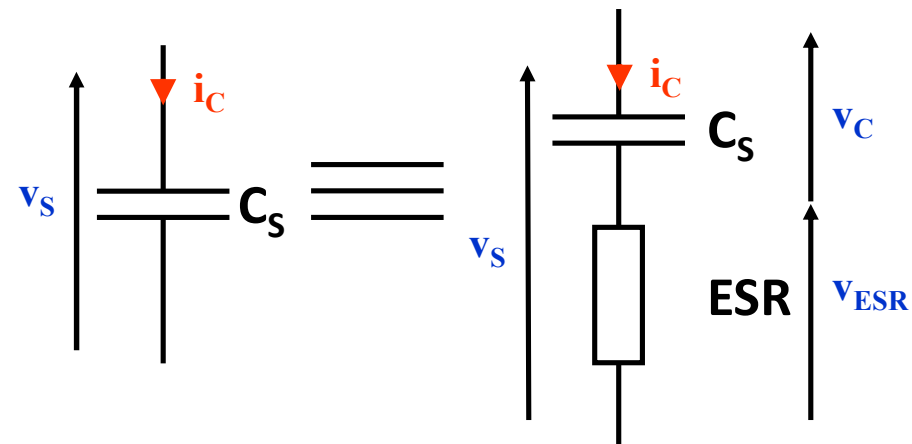
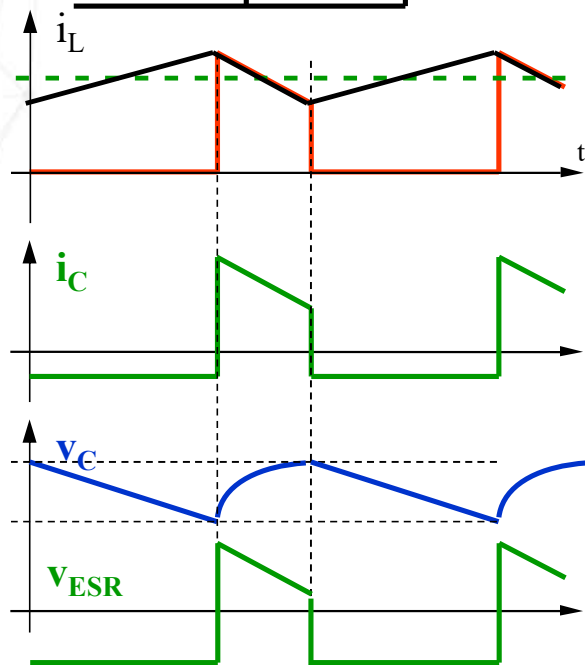
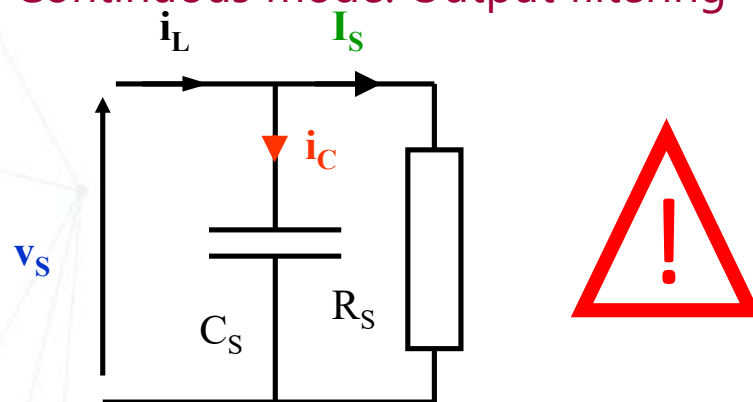
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Continuous mode: Output filtering



$$v_S = v_C + v_{ESR}$$

$$\Delta v_S < \Delta v_C + \Delta v_{ESR}$$

For high value of C_S ,

$$\Delta v_S \approx \Delta v_{ESR} = ESR \cdot \Delta i_C = ESR \cdot \Delta i_L$$

$$ESR < \frac{\Delta v_S}{\Delta i_L}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

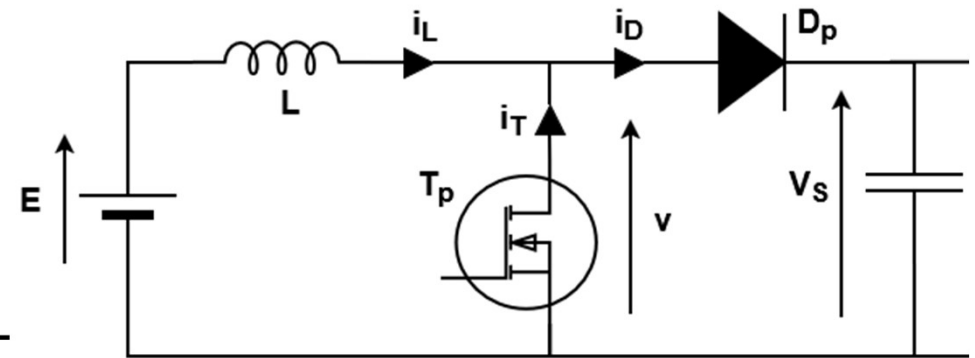
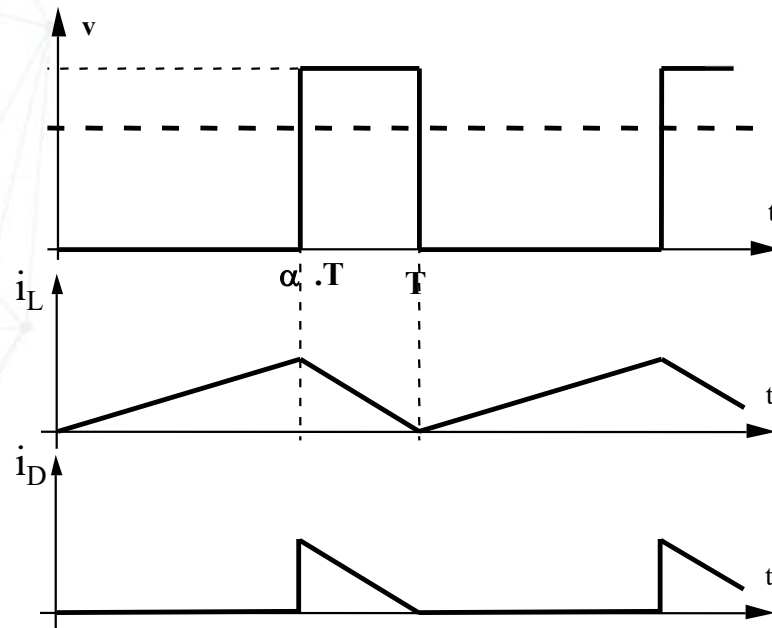
Reversible choppers

Inverters

Closed loop chopper control

Energy storage based chopper

Boundary mode: $i_L(T)=0$



$$I_S = \langle i_D \rangle = \frac{1}{2} \cdot I_{L_{\max}} \cdot (1 - \alpha) = \frac{ET}{2L} \cdot \alpha(1 - \alpha)$$

$$V_S = \frac{E}{1 - \alpha}$$

$$I_S = \frac{ET}{2L} \cdot \frac{E}{V_S} \cdot \left(1 - \frac{E}{V_S}\right)$$

Introduction

Direct conversion

Buck chopper

Boost chopper

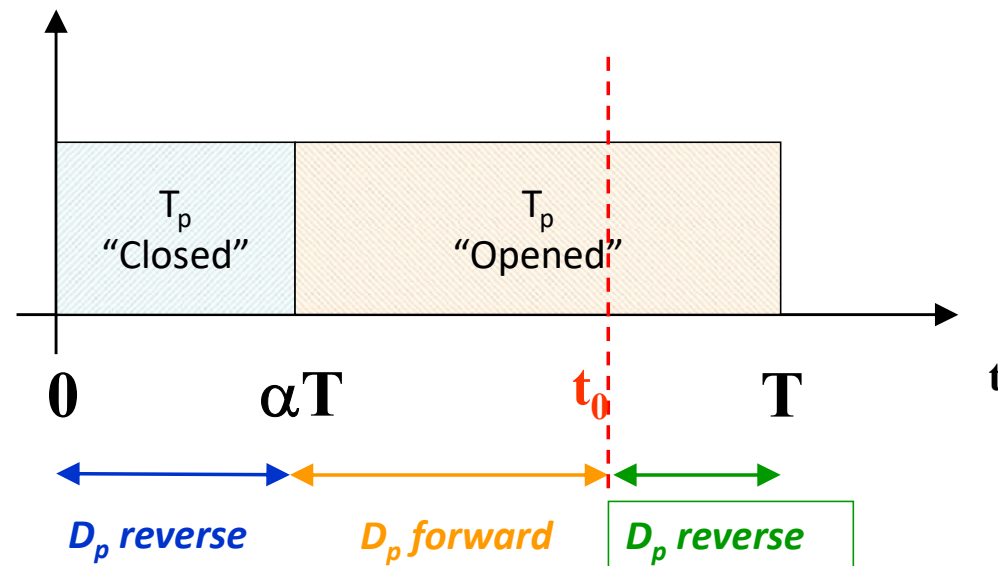
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode



At $t_0 = \alpha'.T$: diode switches to reverse mode

Introduction

Direct conversion

Buck chopper

Boost chopper

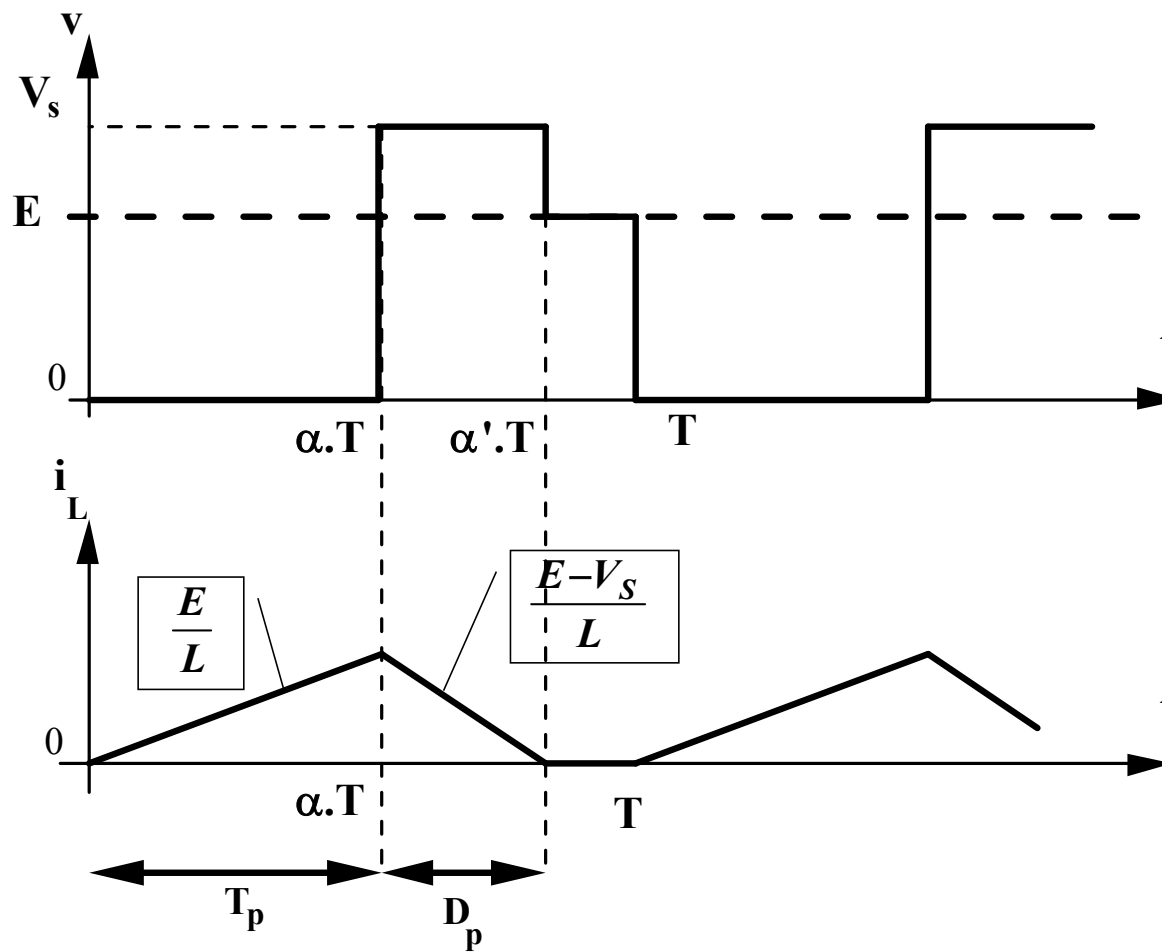
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode: Chronograms



Introduction

Direct conversion

Buck chopper

Boost chopper

Reversible
choppers

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Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode: calculation

$$\frac{E}{L} \alpha T = \frac{V_S - E}{L} (\alpha' - \alpha) T \quad \Rightarrow \quad V_S = \frac{\alpha'}{\alpha' - \alpha} E$$

$$I_S = \langle i_D \rangle = \frac{1}{2} \cdot I_{L \max} \cdot (\alpha' - \alpha) = \frac{ET}{2L} \cdot \alpha (\alpha' - \alpha)$$

$$\Rightarrow \quad V_S = E + \frac{\alpha^2 E^2 T}{2L I_S}$$

Introduction

Direct conversion

Buck chopper

Boost chopper

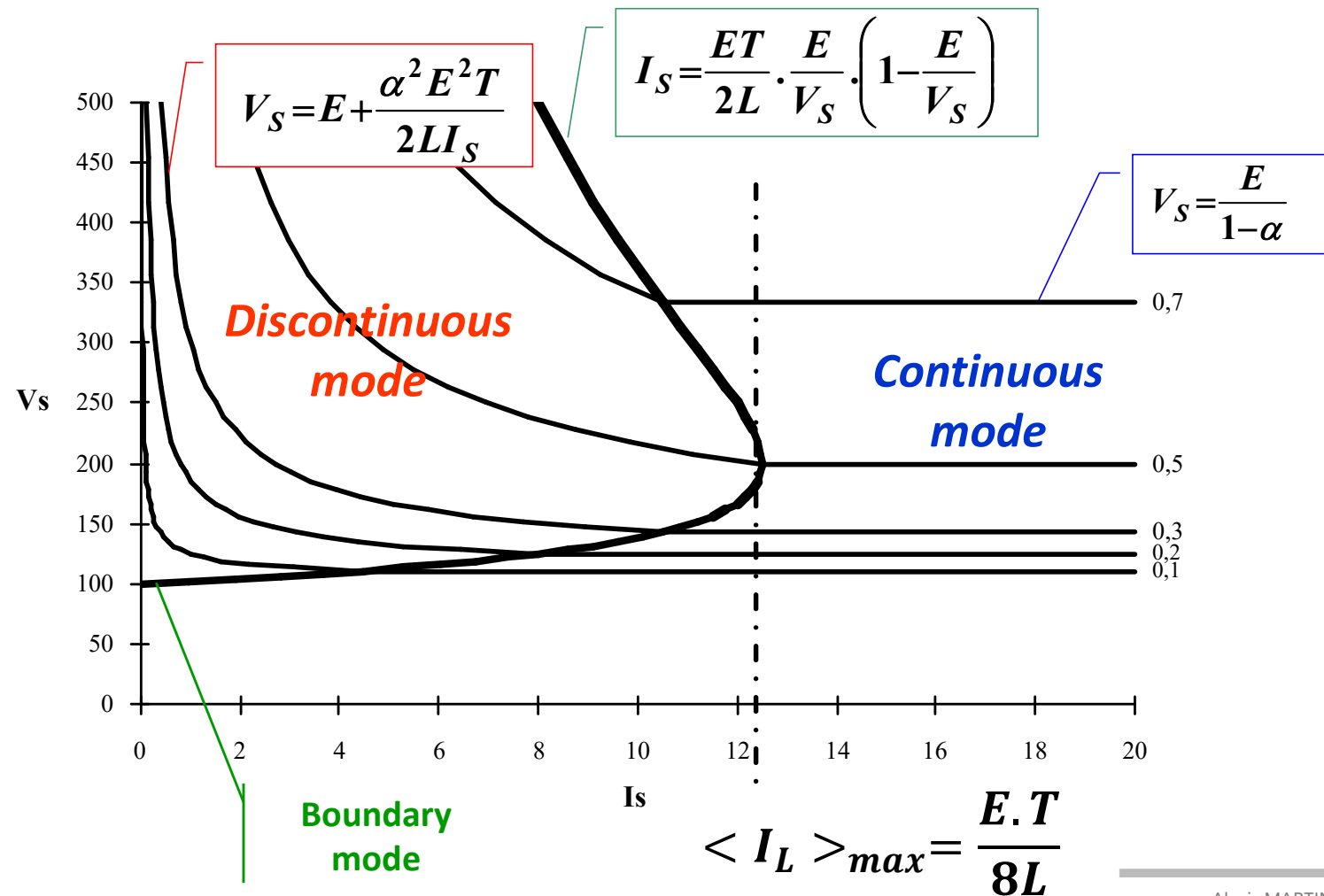
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Discontinuous mode: output characteristics



Introduction

Direct conversion

Buck chopper

Boost chopper

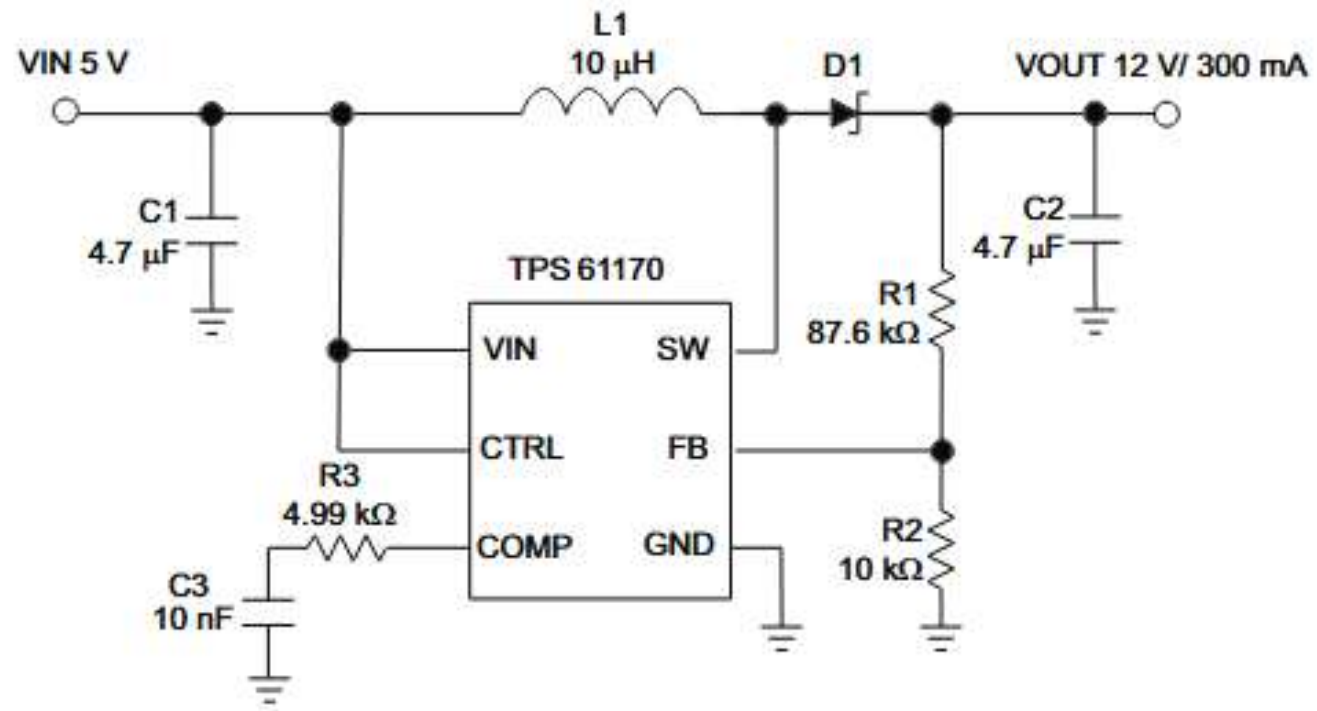
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Example : TPS61170



Introduction

Direct conversion

Buck chopper

Boost chopper

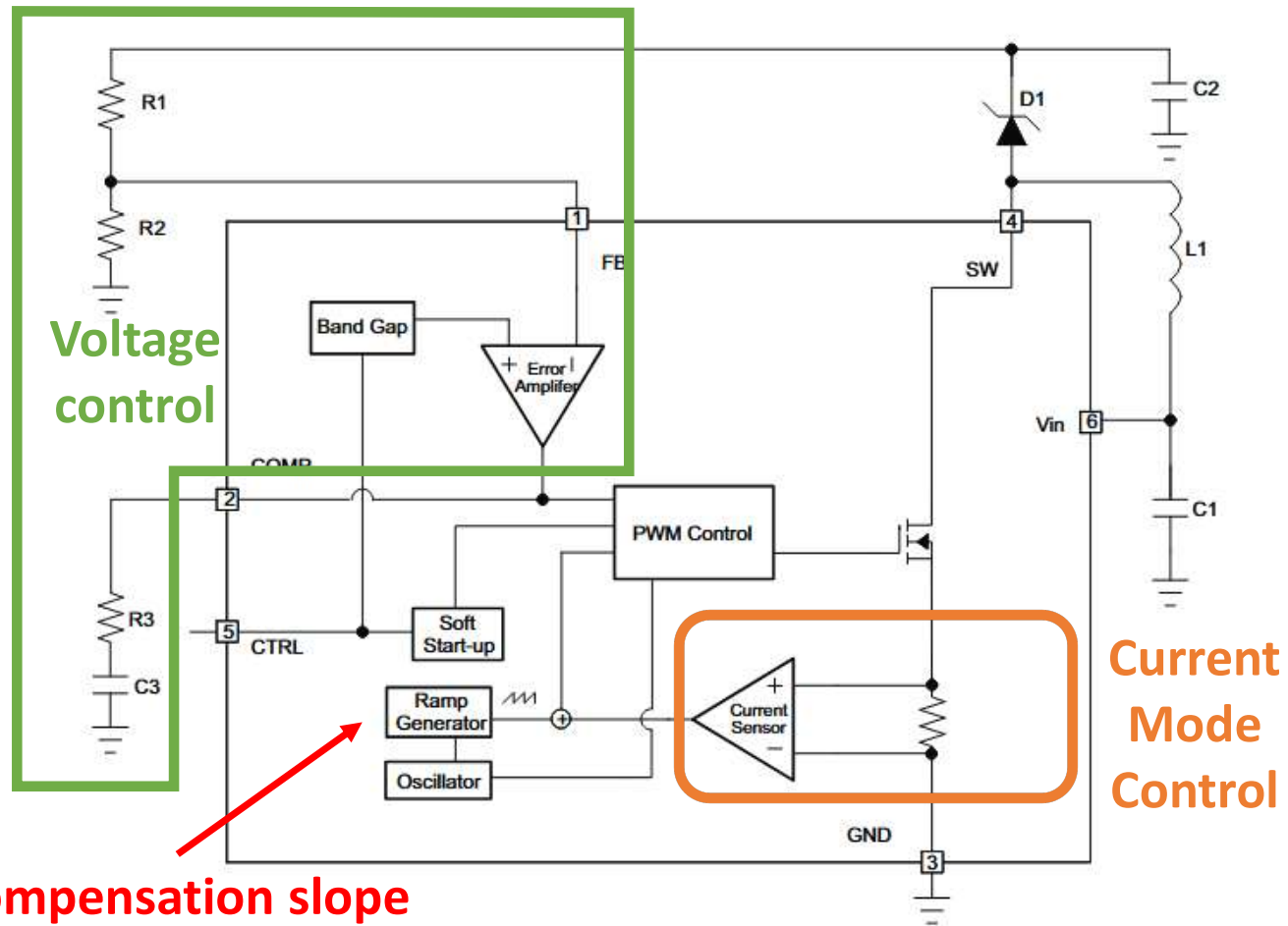
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Example : TPS61170



Compensation slope
(for stabilization)

Current
Mode
Control

Introduction

Direct conversion

Buck chopper

Boost chopper

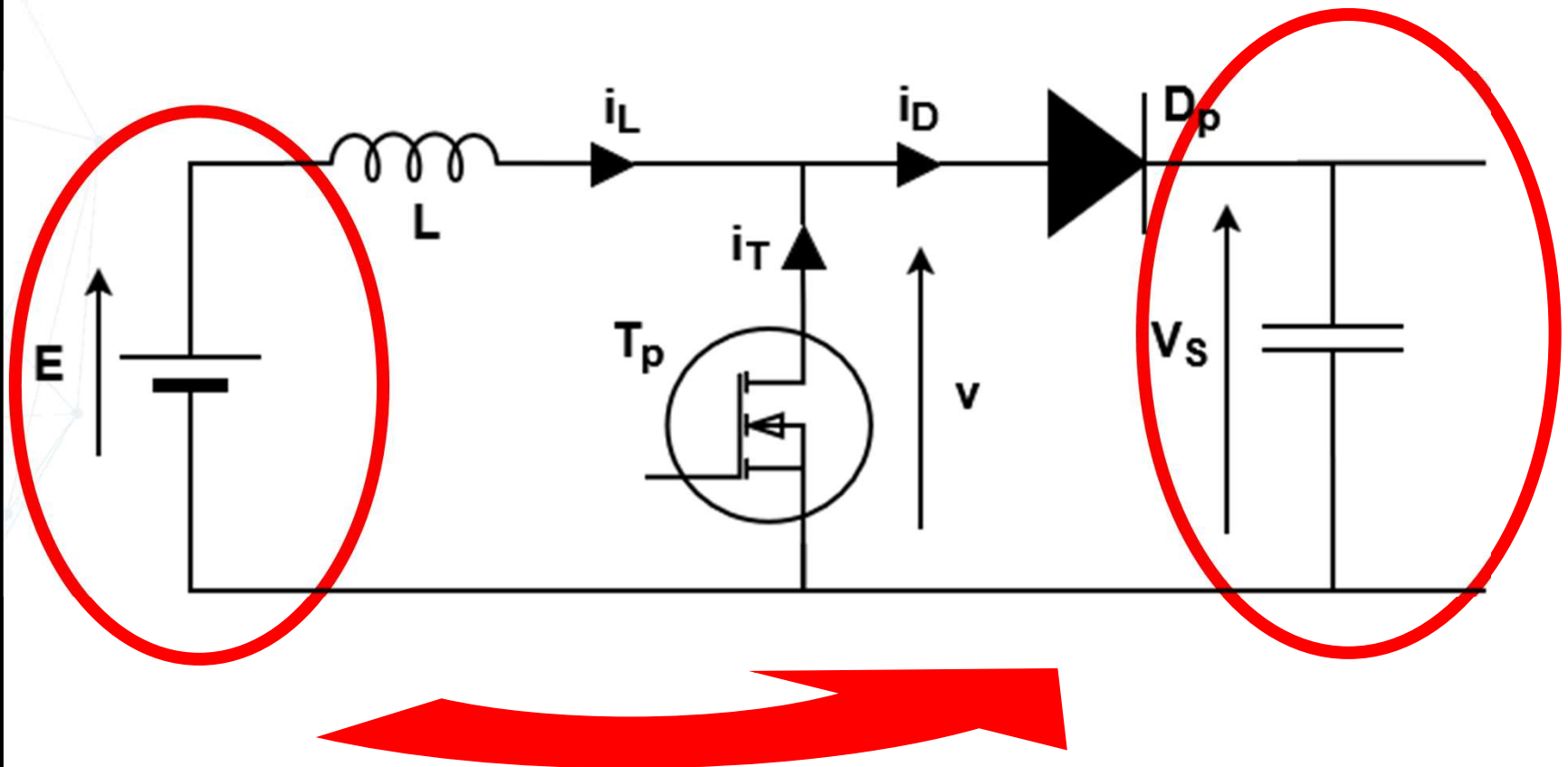
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reminder : buck chopper



Introduction

Direct conversion

Buck chopper

Boost chopper

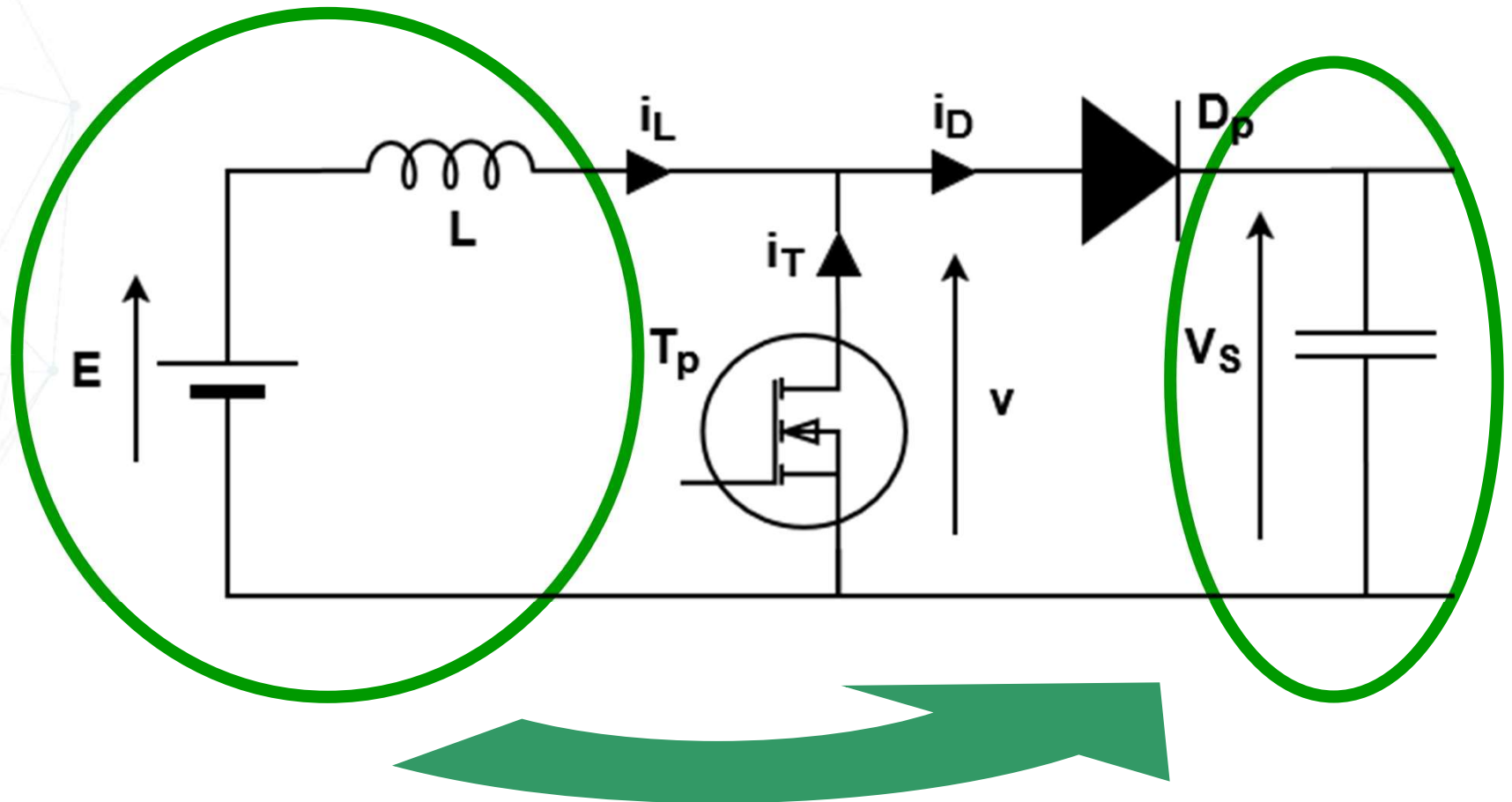
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reminder : boost chopper



Introduction

Direct conversion

Buck chopper

Boost chopper

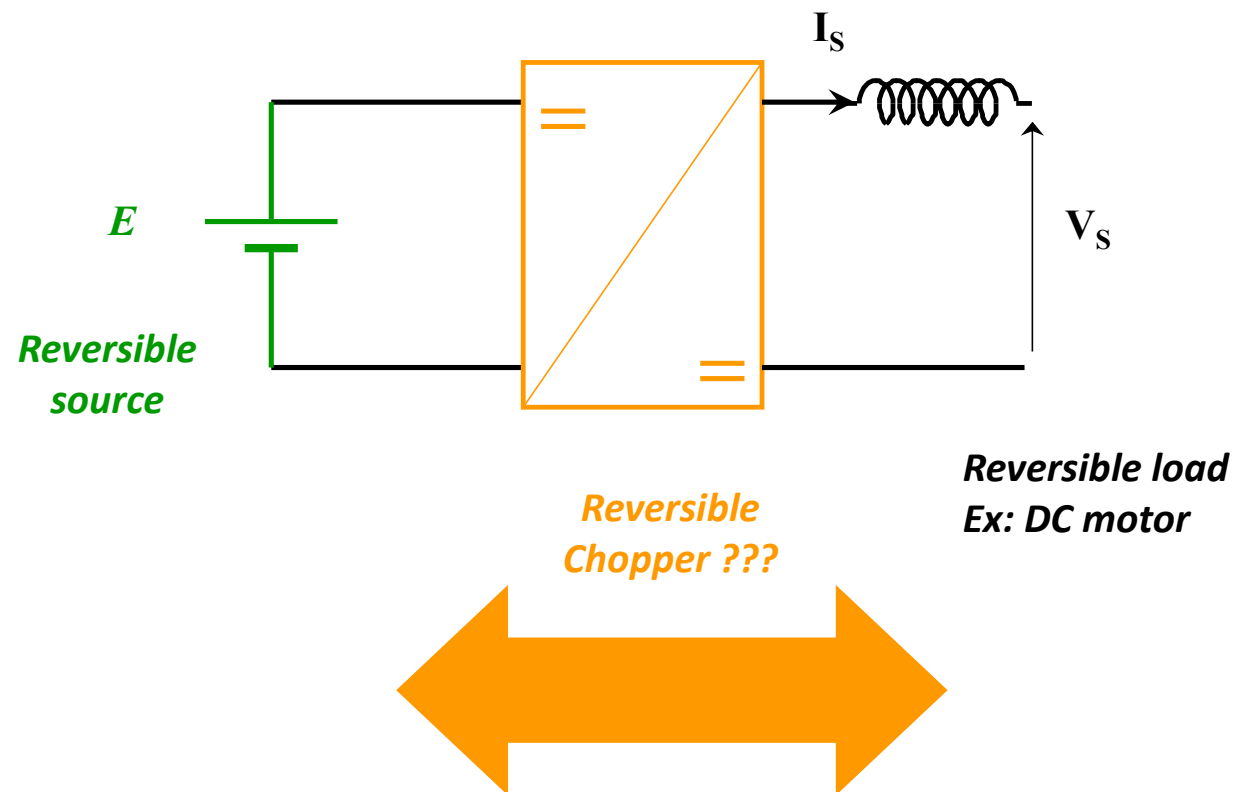
Reversible
choppers

Inverters

Closed loop
chopper control

Energy storage
based chopper

Linked sources



Introduction

Direct conversion

**Reversible
choppers**

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reversible Chopper

Introduction

Direct conversion

**Reversible
choppers**

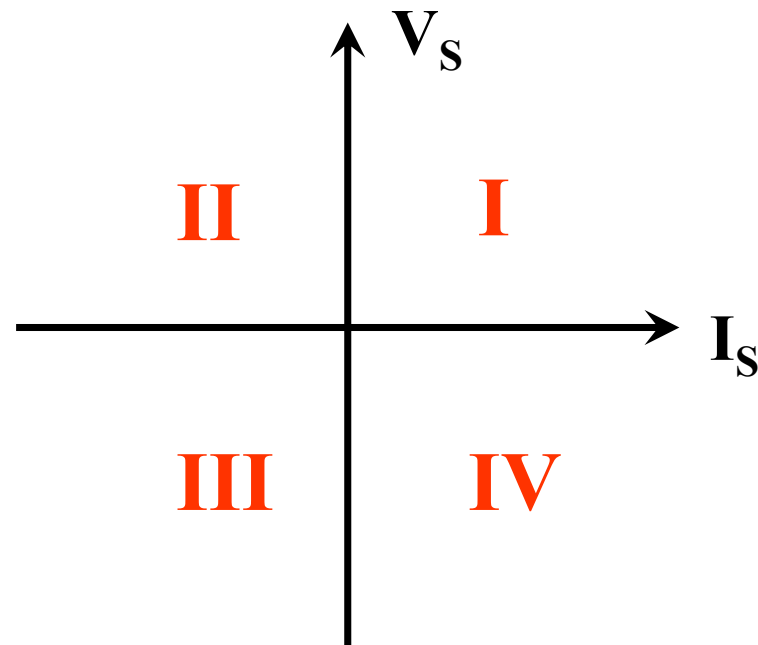
2 quadrants chopper
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Reversible choppers:
quadrants



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

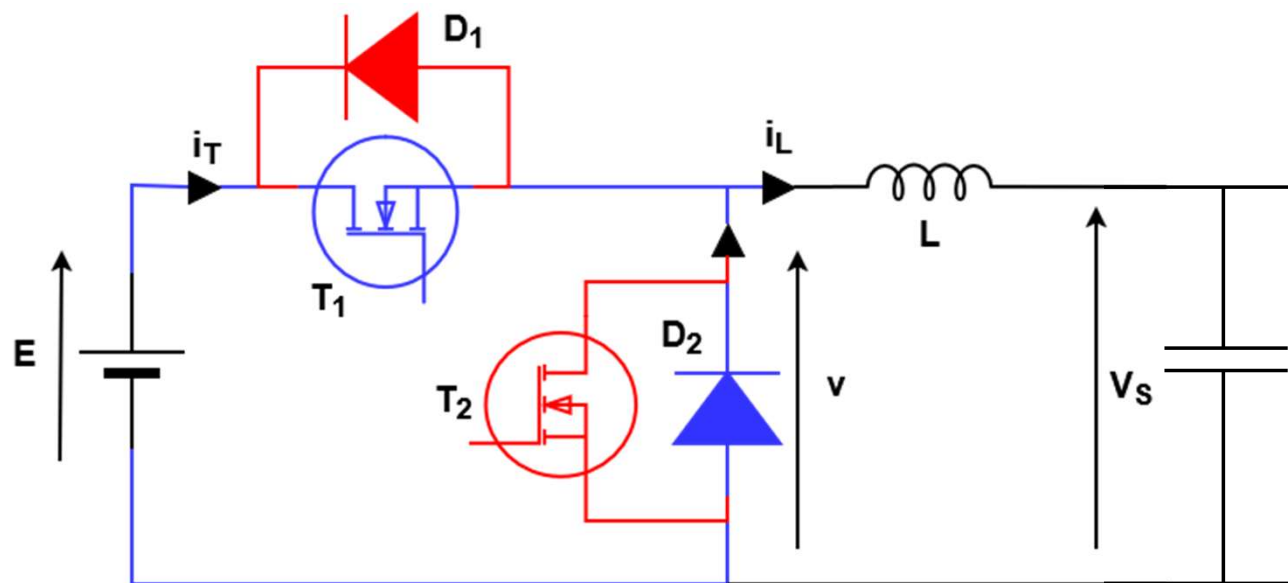
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants reversible chopper



Buck chopper

Boost chopper

Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

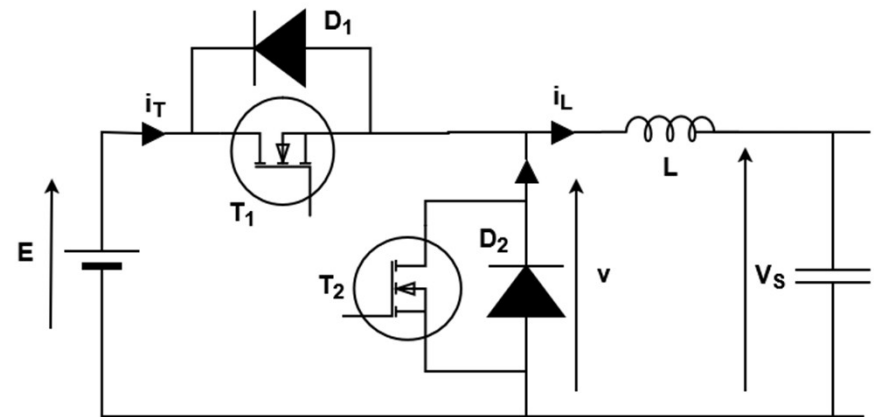
4 quadrants chopper

Inverters

Closed loop
chopper control

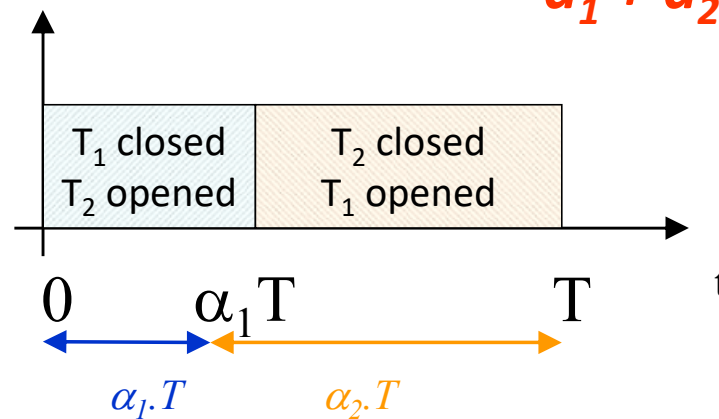
Energy storage
based chopper

Two quadrants chopper:
control



2 switches => 2 duty cycles
Complementary control

$$a_1 + a_2 = 1$$



Introduction

Direct conversion

Reversible
choppers

2 quadrants chopper

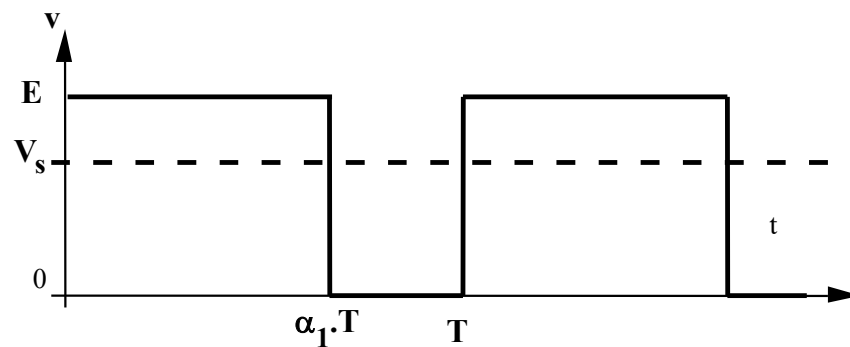
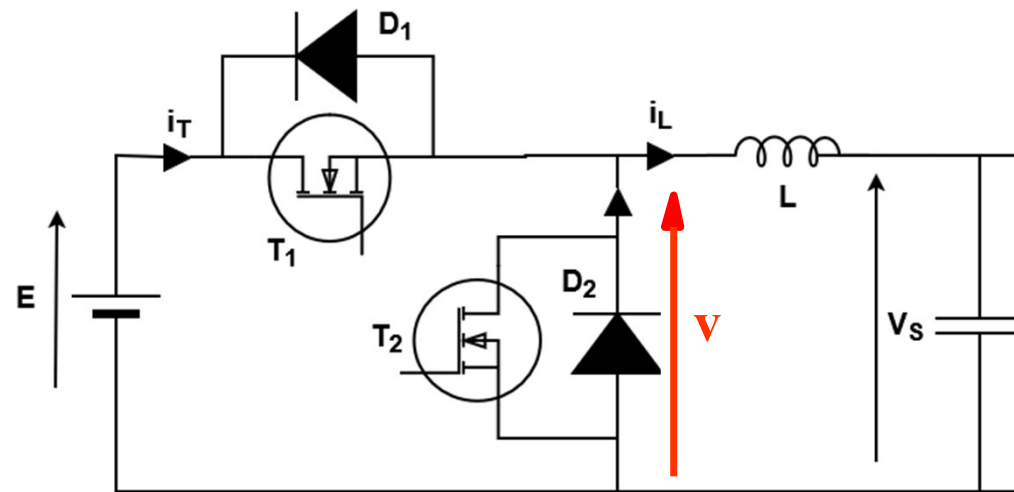
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: output voltage



$$V_s = \alpha_1 \cdot E$$

Introduction

Direct conversion

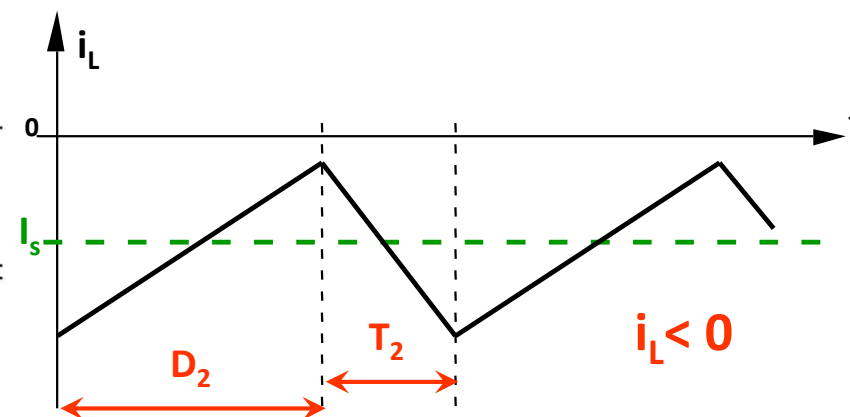
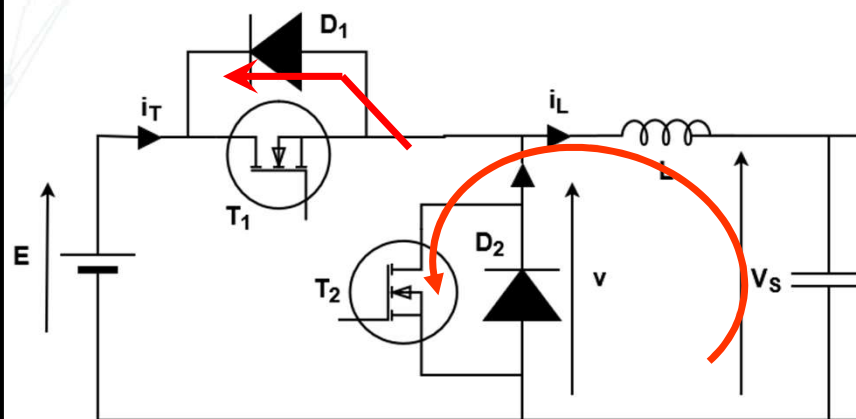
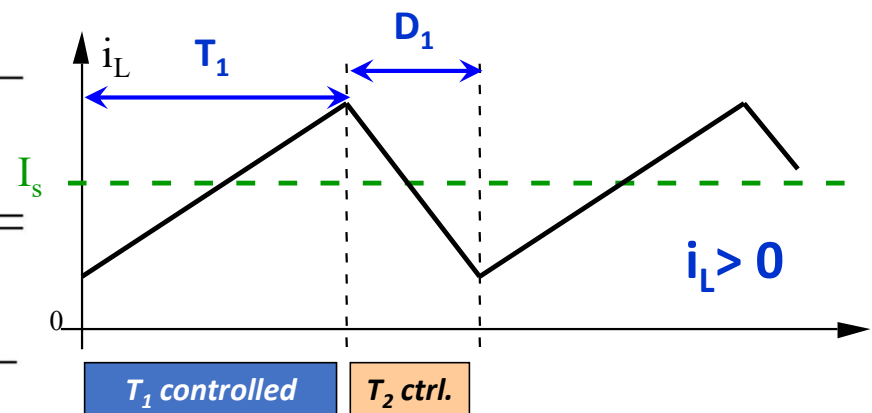
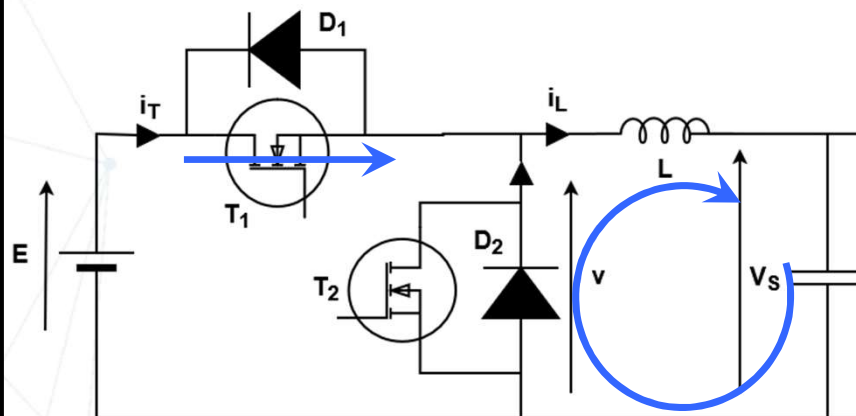
**Reversible
choppers**
2 quadrants chopper
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: output current



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

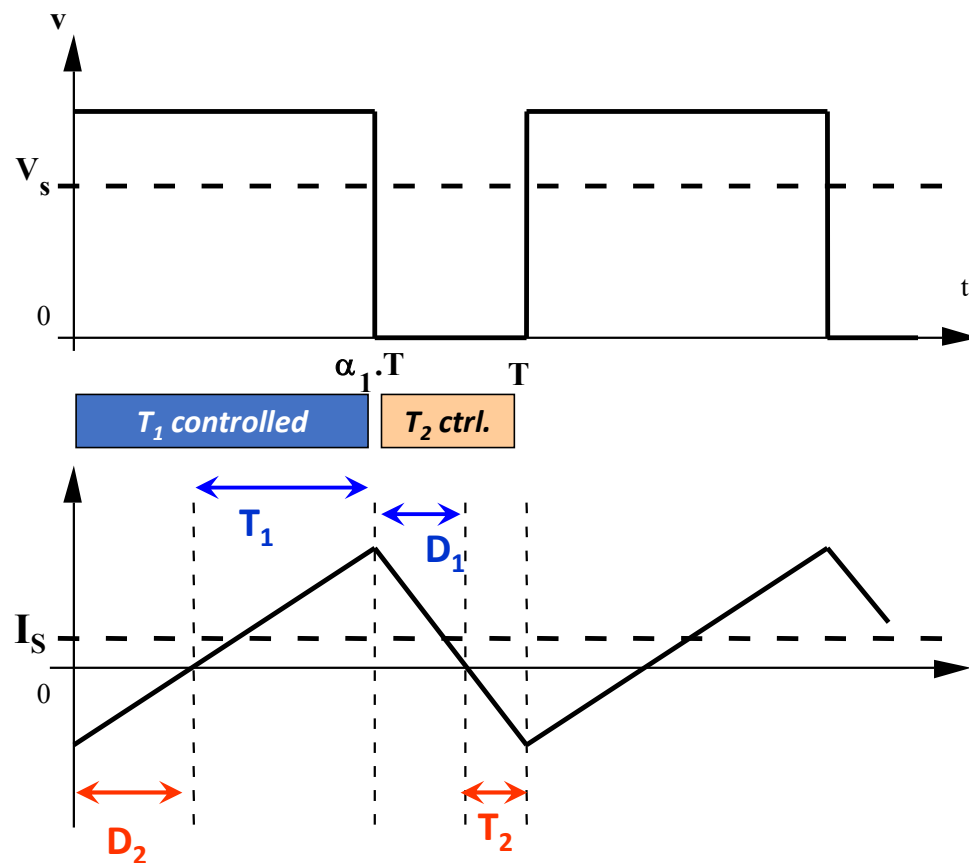
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: Arbitrary output current



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

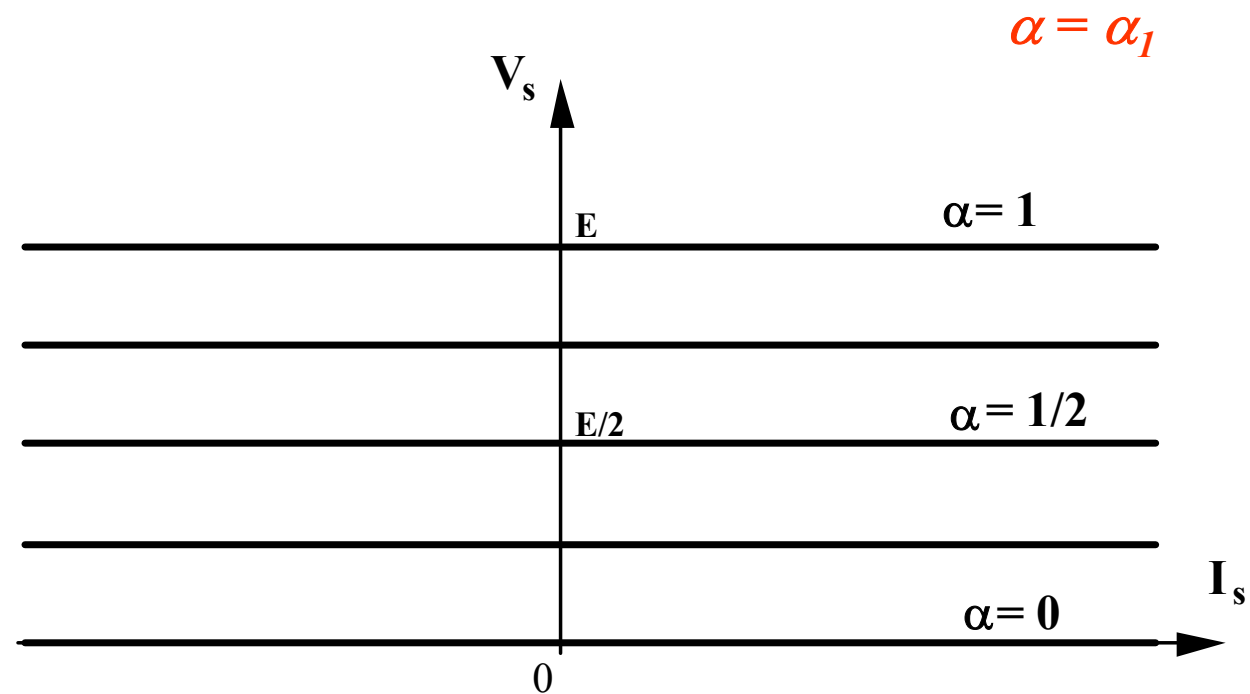
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

Two quadrants chopper: Output characteristics grid



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

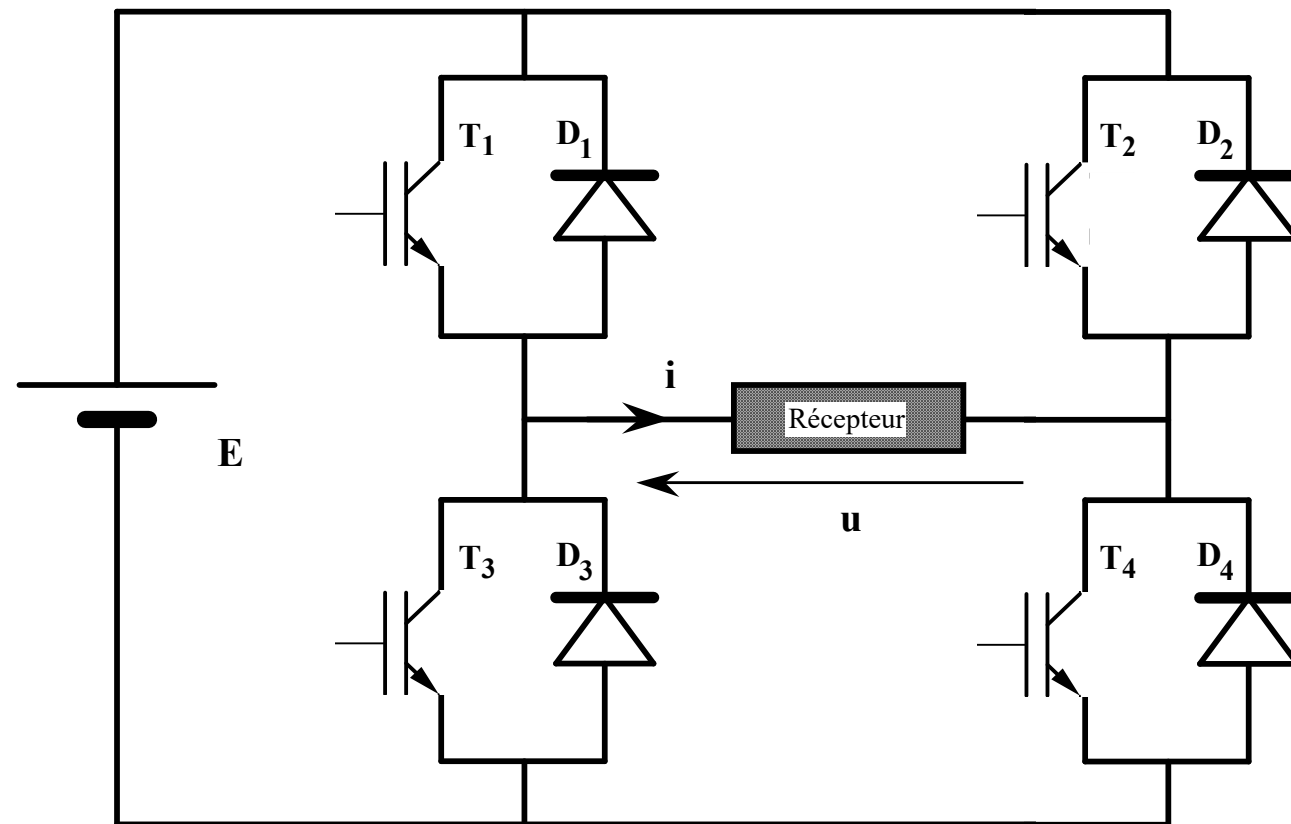
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

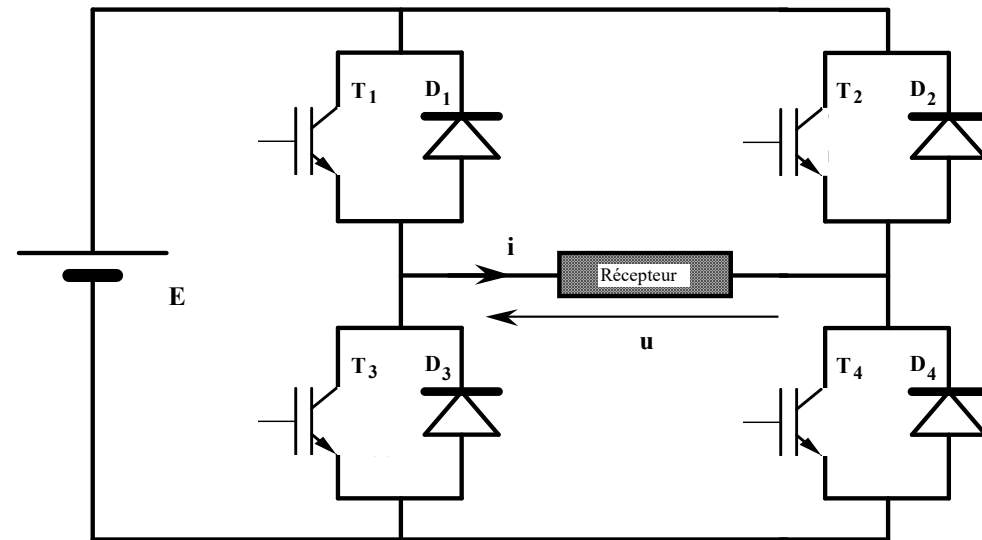
4 quadrants chopper

Inverters

Closed loop
chopper control

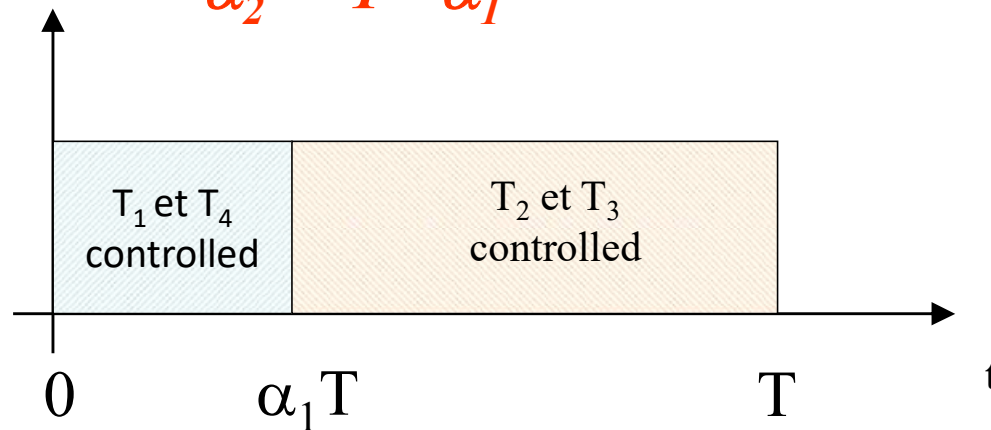
Energy storage
based chopper

4 quadrants chopper:
Synchronous complementary control



$$\alpha_1 = \alpha_4 \text{ et } \alpha_2 = \alpha_3$$

$$\alpha_2 = 1 - \alpha_1$$



Introduction

Direct conversion

**Reversible
choppers**

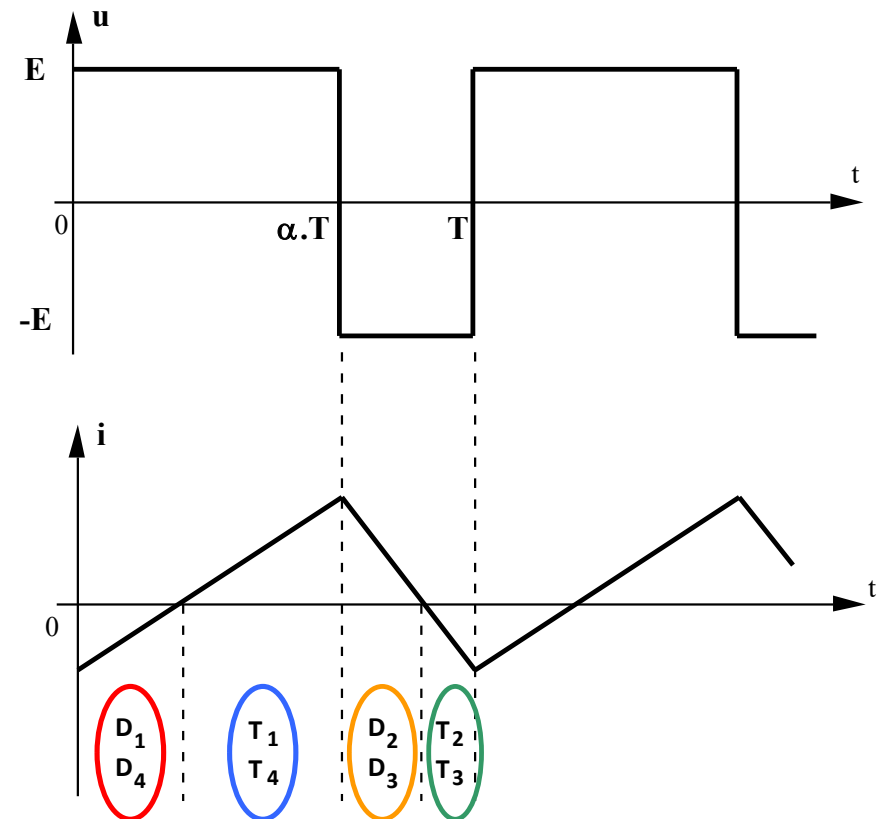
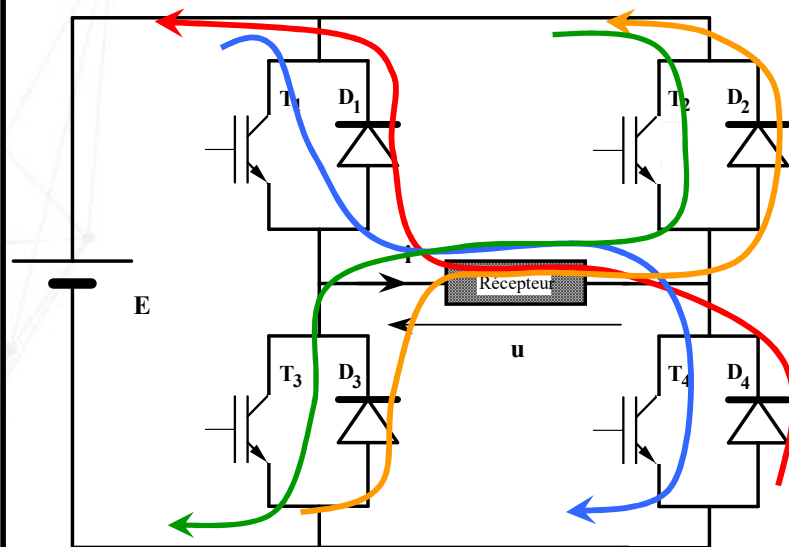
2 quadrants chopper
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper:
Synchronous complementary control (chronograms)



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

4 quadrants chopper

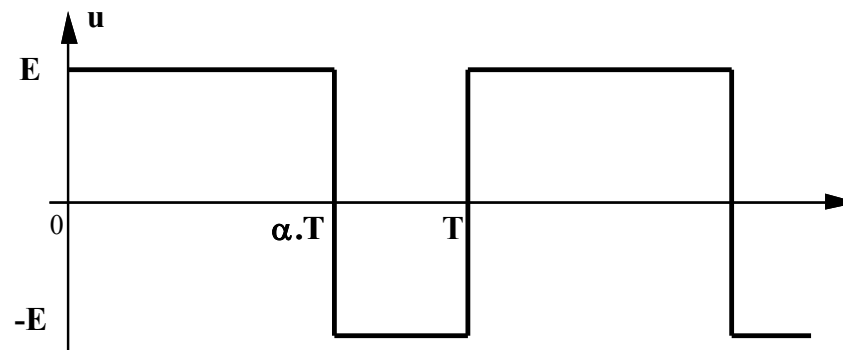
Inverters

Closed loop
chopper control

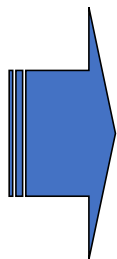
Energy storage
based chopper

4 quadrants chopper:

Synchronous complementary control (Average output voltage)



$$\langle u \rangle = (2\alpha - 1).E$$



***High voltage ripple: $2.E$
Low switching frequency***

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

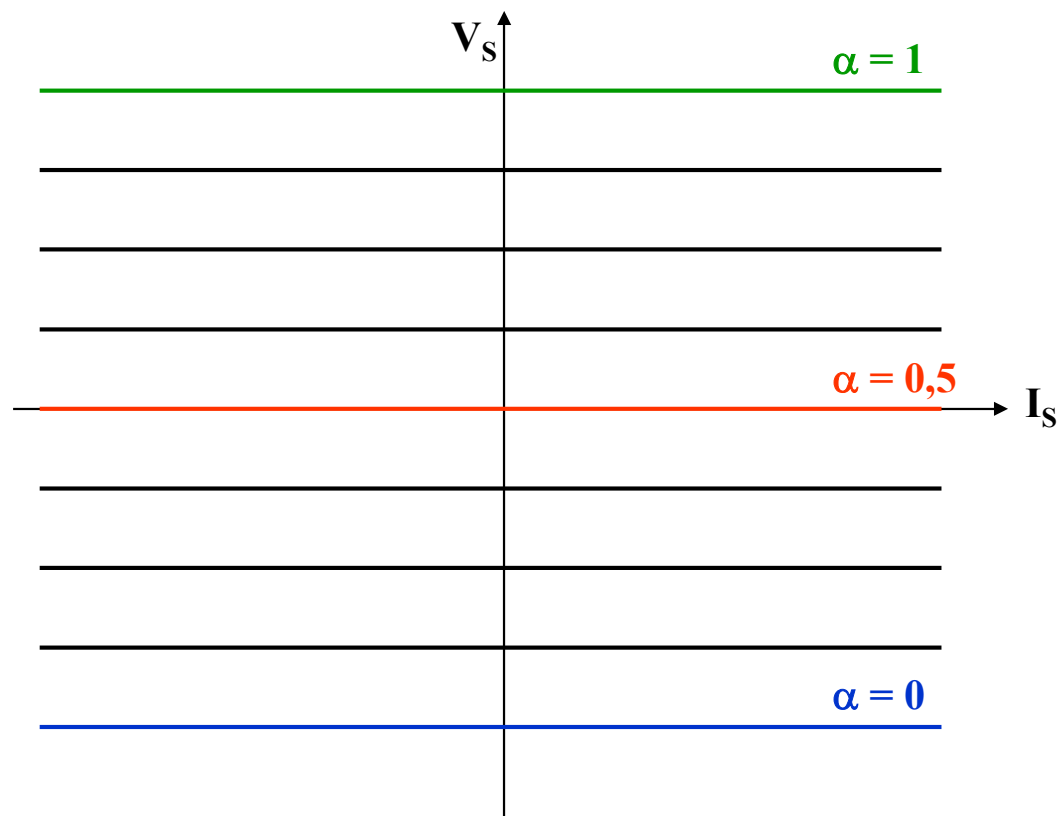
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Output characteristics



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

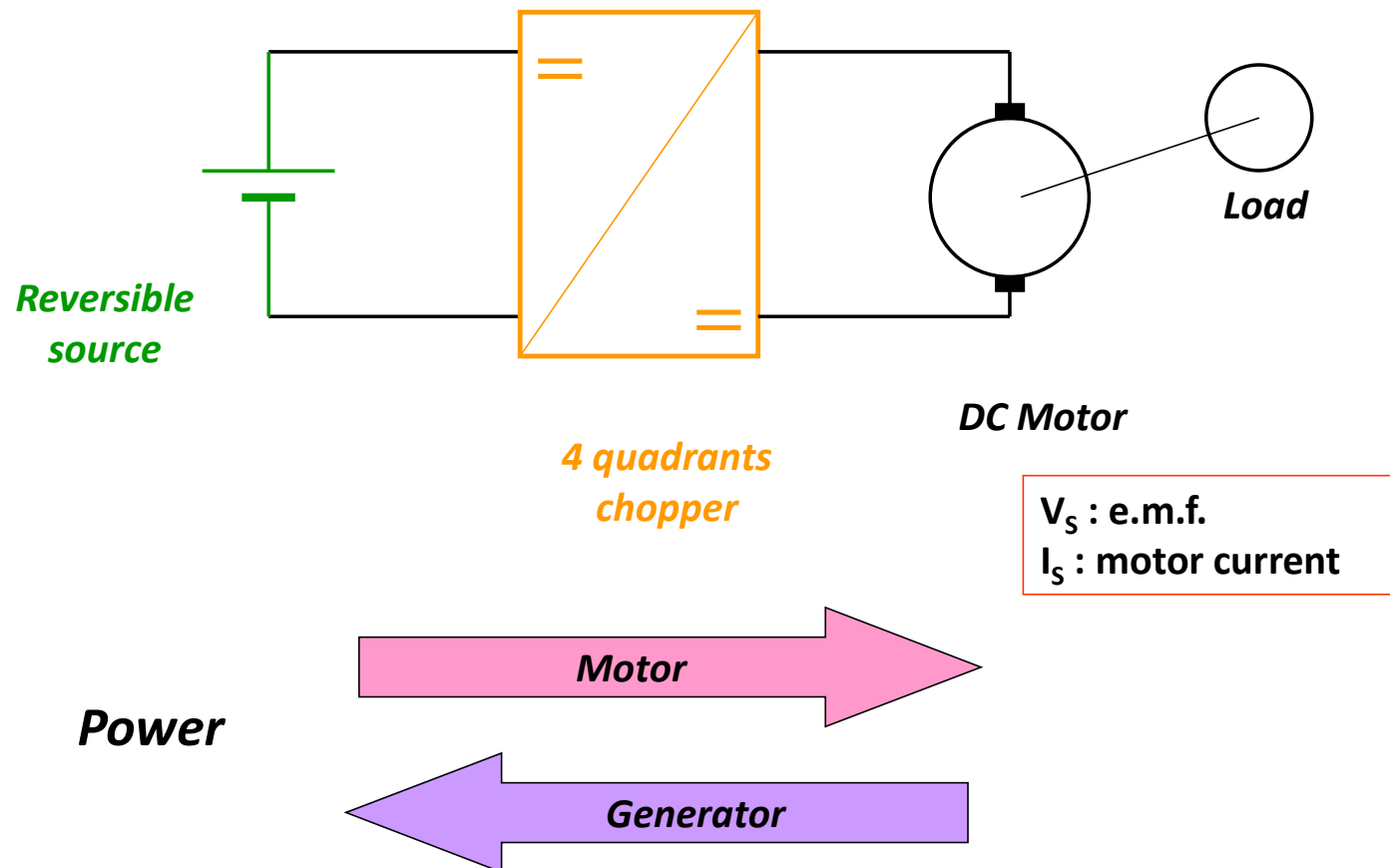
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: control of a DC motor



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

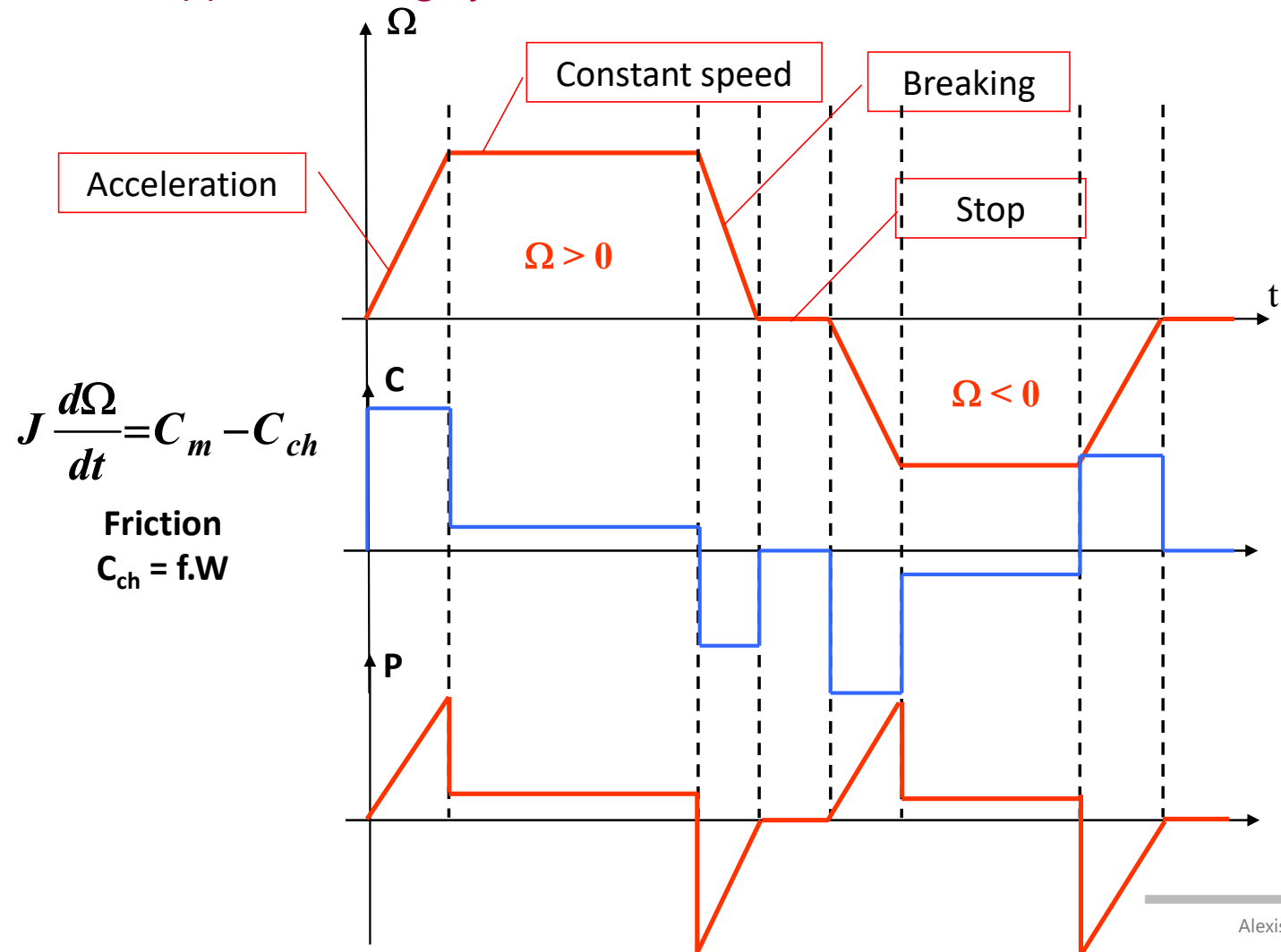
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: working cycle



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

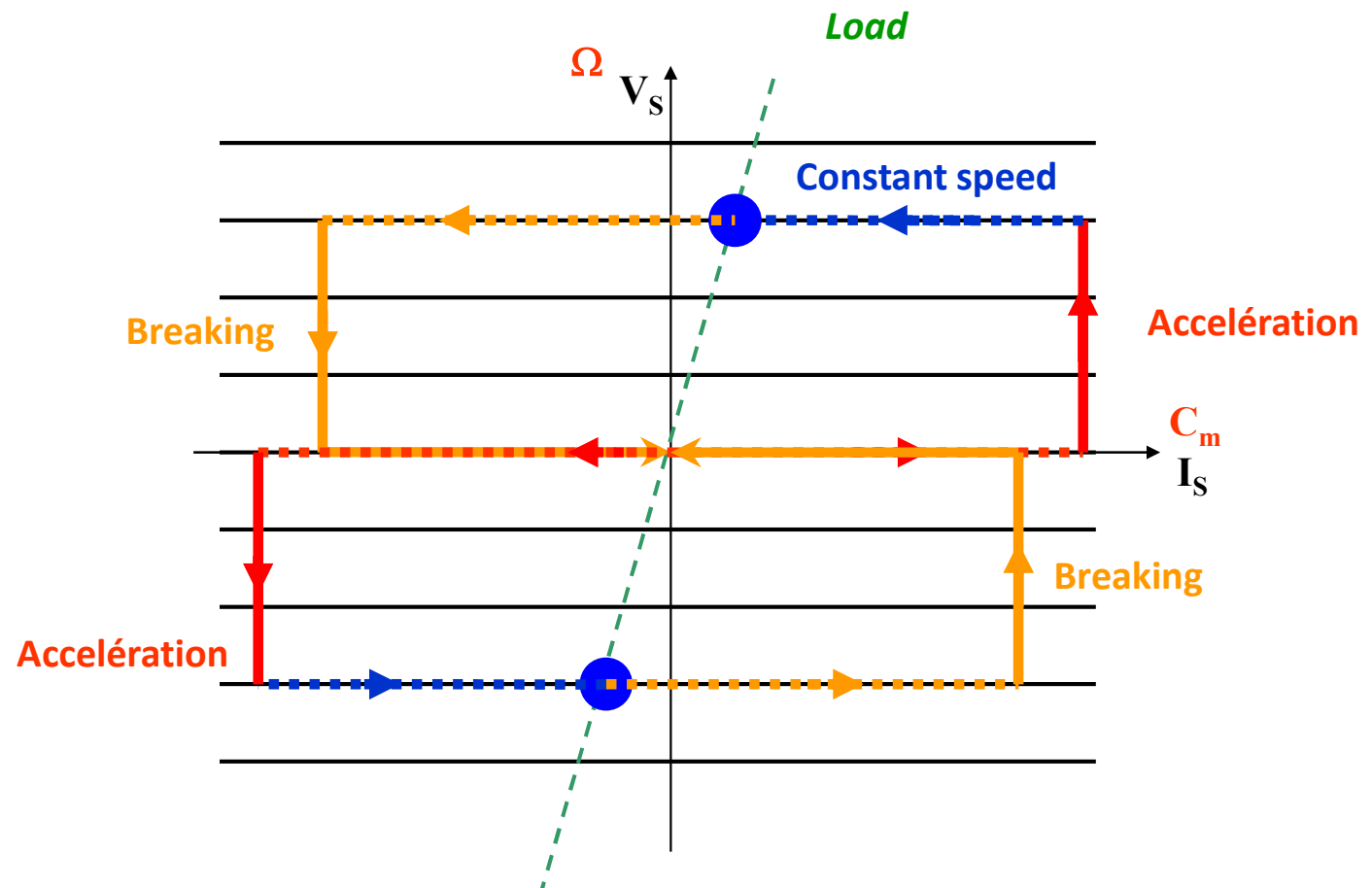
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Operating point



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

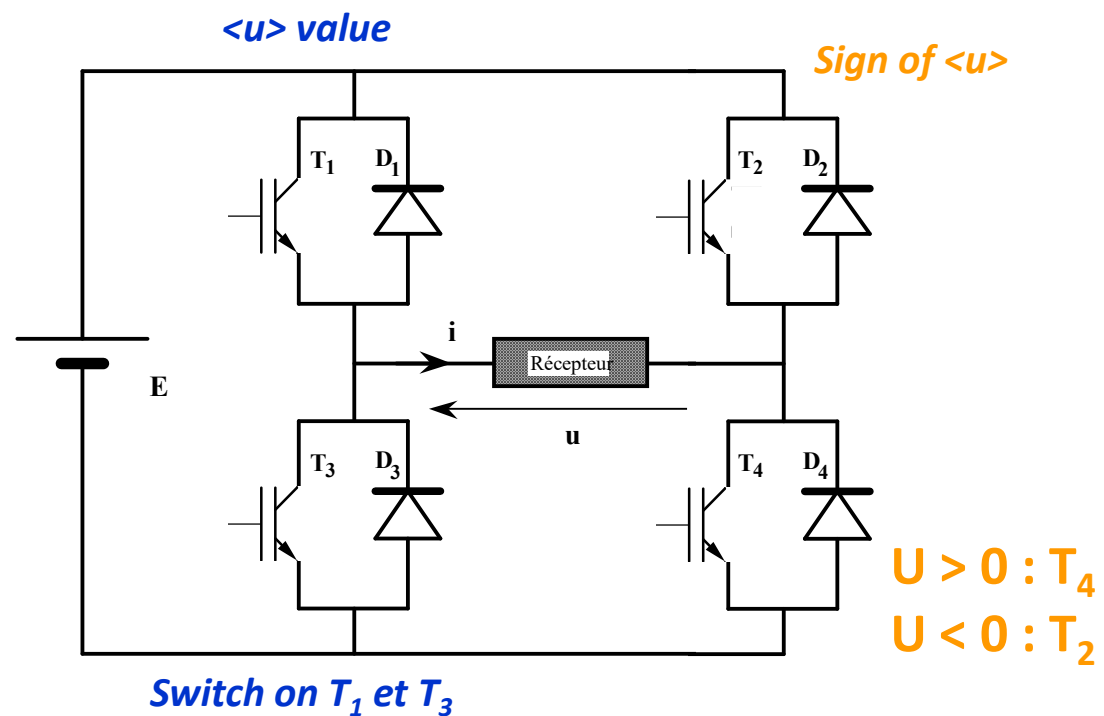
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: additional control



Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

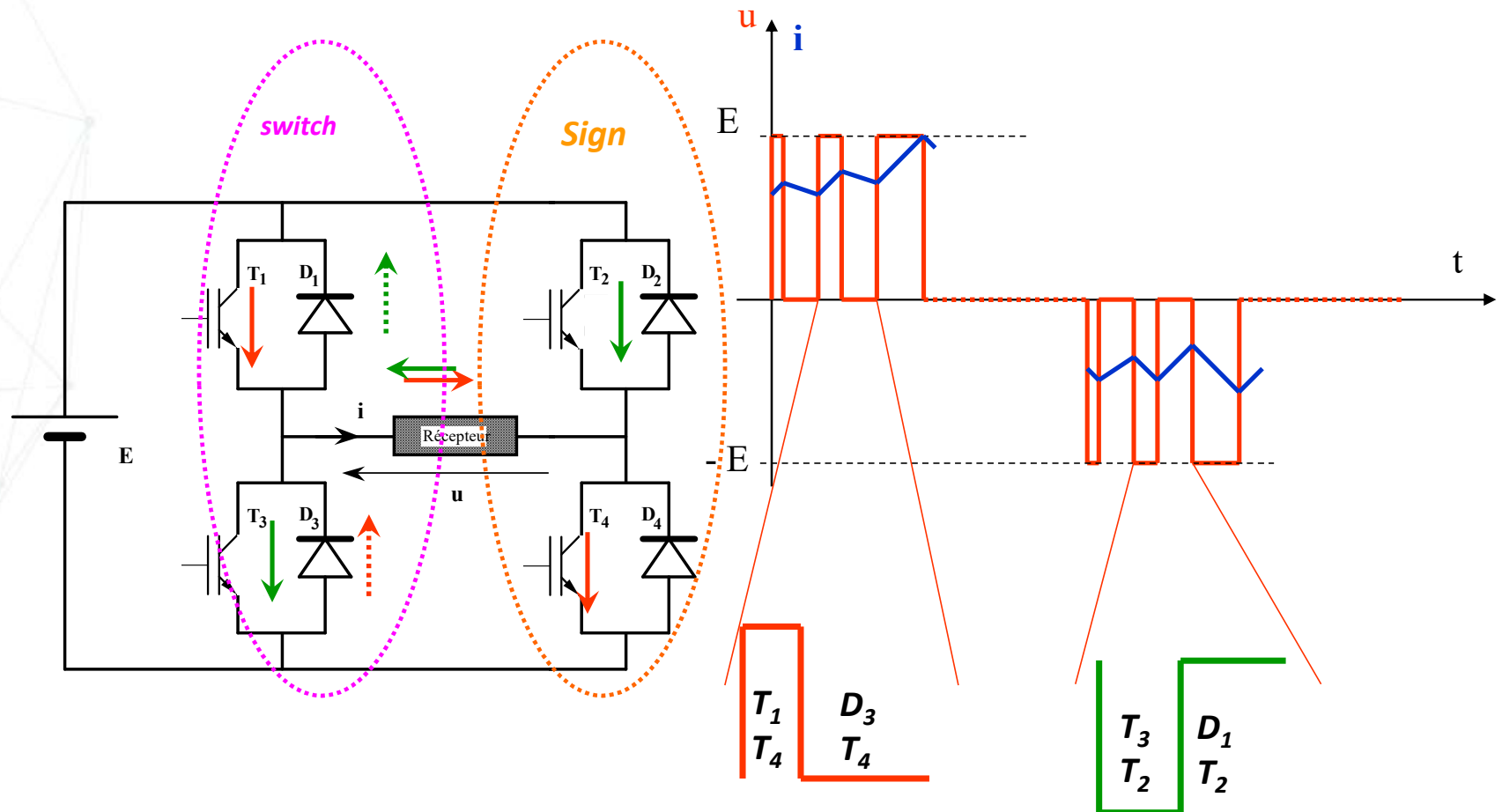
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: additional control



Introduction

Direct conversion

**Reversible
choppers**

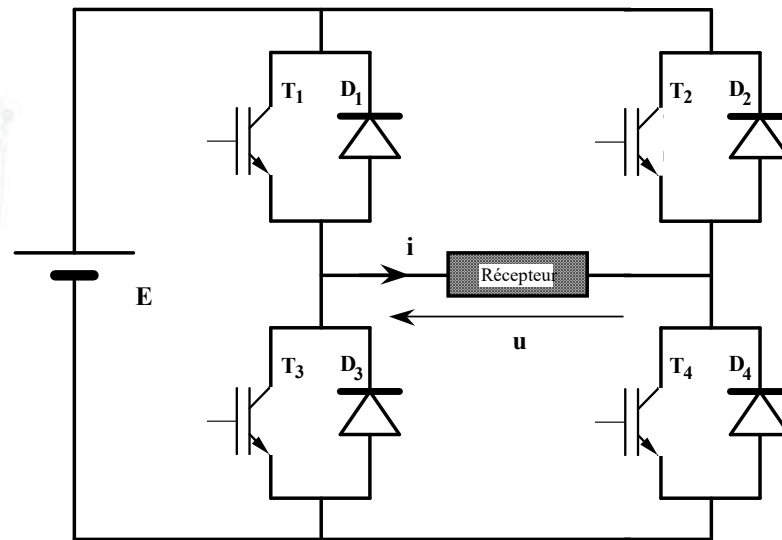
2 quadrants chopper
4 quadrants chopper

Inverters

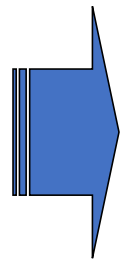
Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: additional control



$U = \pm a.E$
Average value during
a switching time T_d



Voltage ripple reduced: E
Switching only on T_1/T_3 : losses reduced
Instabilities around 0V !!

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

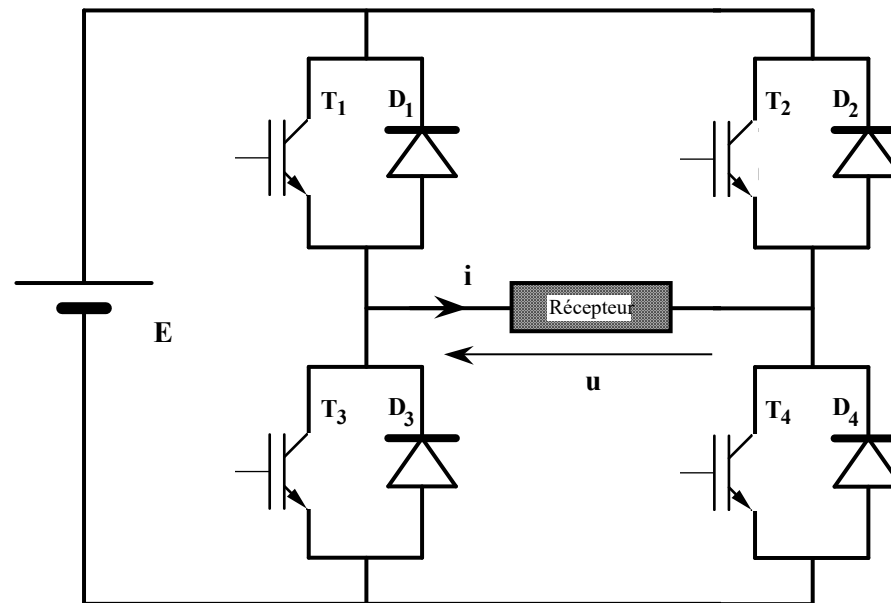
4 quadrants chopper

Inverters

Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



$$\alpha_1 = \alpha_4$$

$$\alpha_2 = \alpha_3$$

$$\alpha_3 = 1 - \alpha_1$$

$$\alpha_4 = 1 - \alpha_2$$

Switching on T_1 and T_3 at $t = 0$

Switching on T_2 and T_4 at $t = T/2$

Introduction

Direct conversion

**Reversible
choppers**

2 quadrants chopper

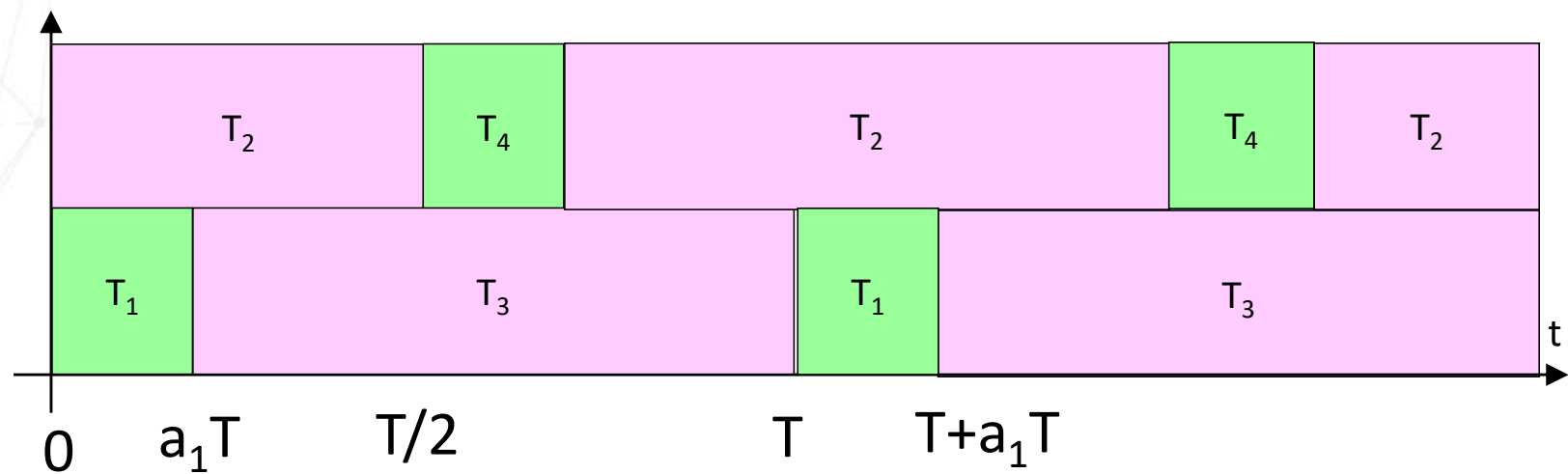
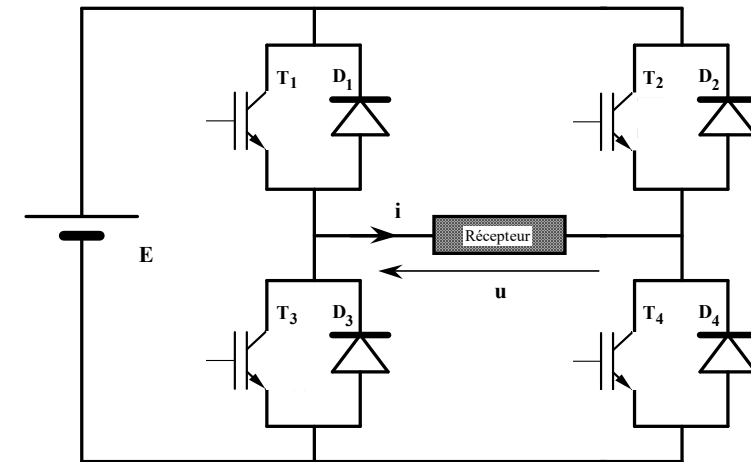
4 quadrants chopper

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4 quadrants chopper: Shifted complementary control



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Direct conversion

**Reversible
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2 quadrants chopper

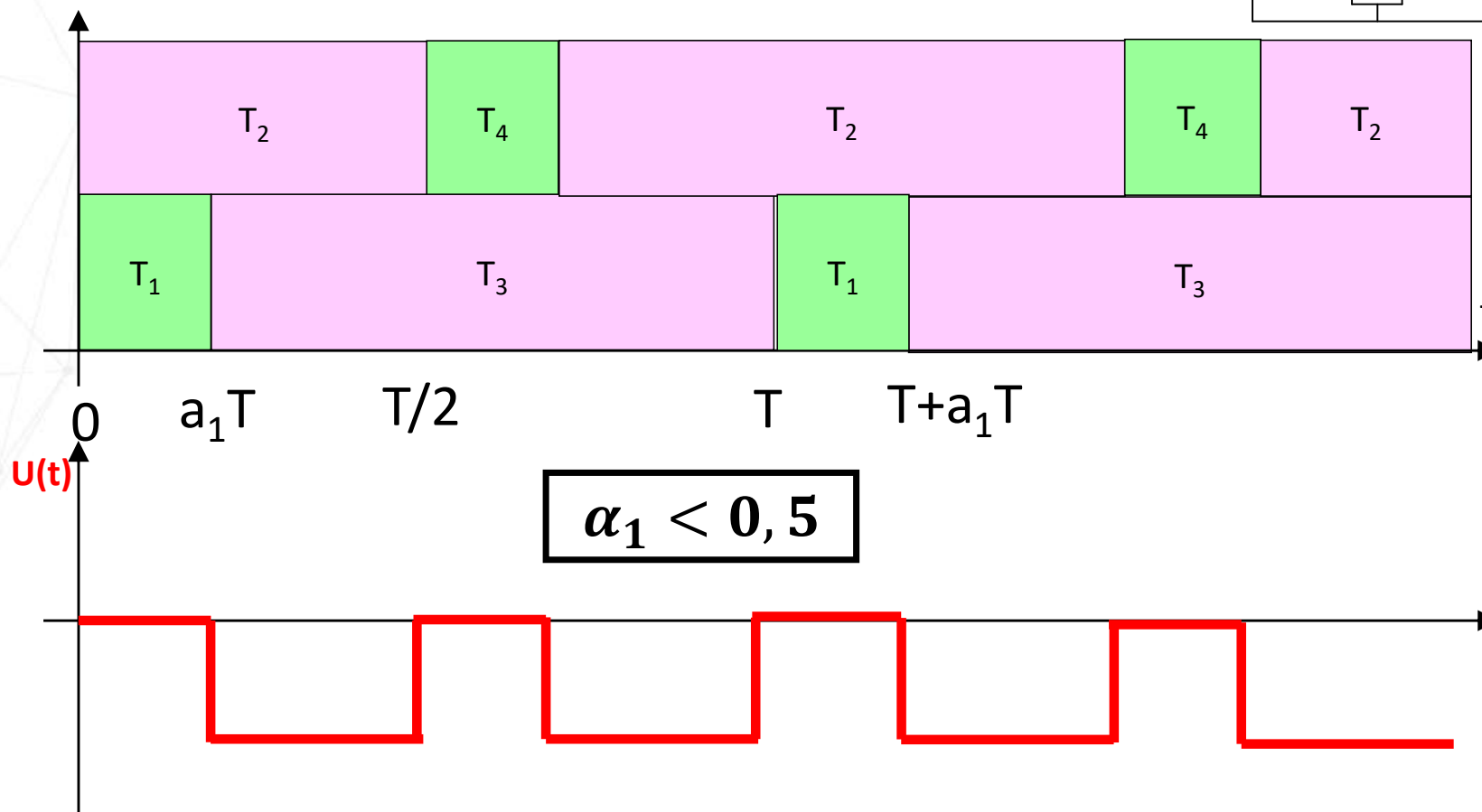
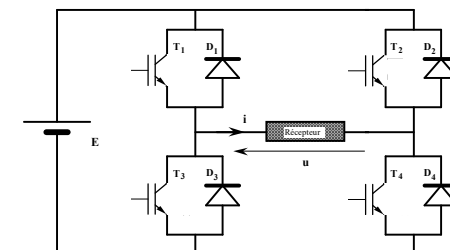
4 quadrants chopper

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4 quadrants chopper: Shifted complementary control



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Direct conversion

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2 quadrants chopper

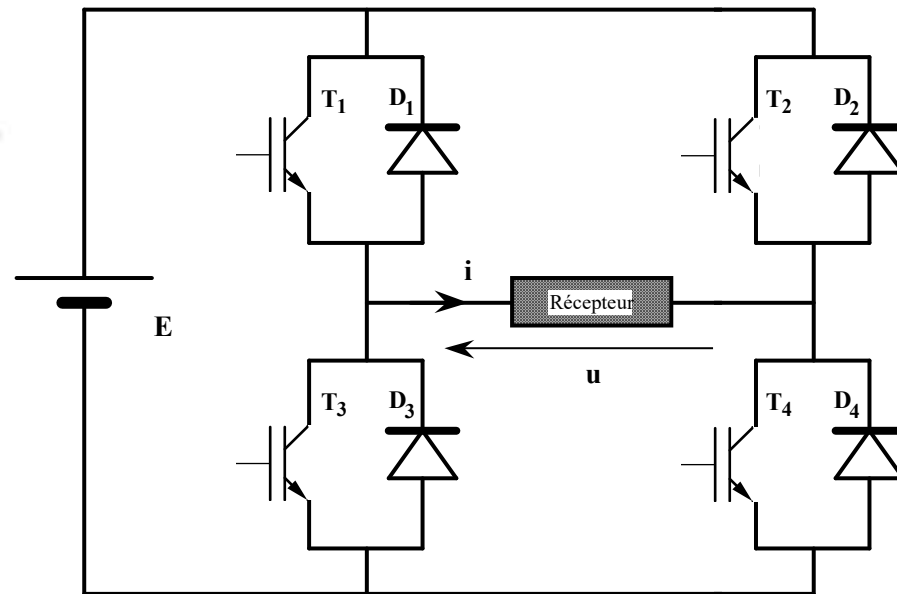
4 quadrants chopper

Inverters

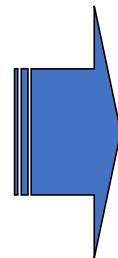
Closed loop
chopper control

Energy storage
based chopper

4 quadrants chopper: Shifted complementary control



$$U = (2a-1).E$$



Voltage ripple reduced: E
High switching frequency
=> PWM

Introduction

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2 quadrants chopper

4 quadrants chopper

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Closed loop
chopper control

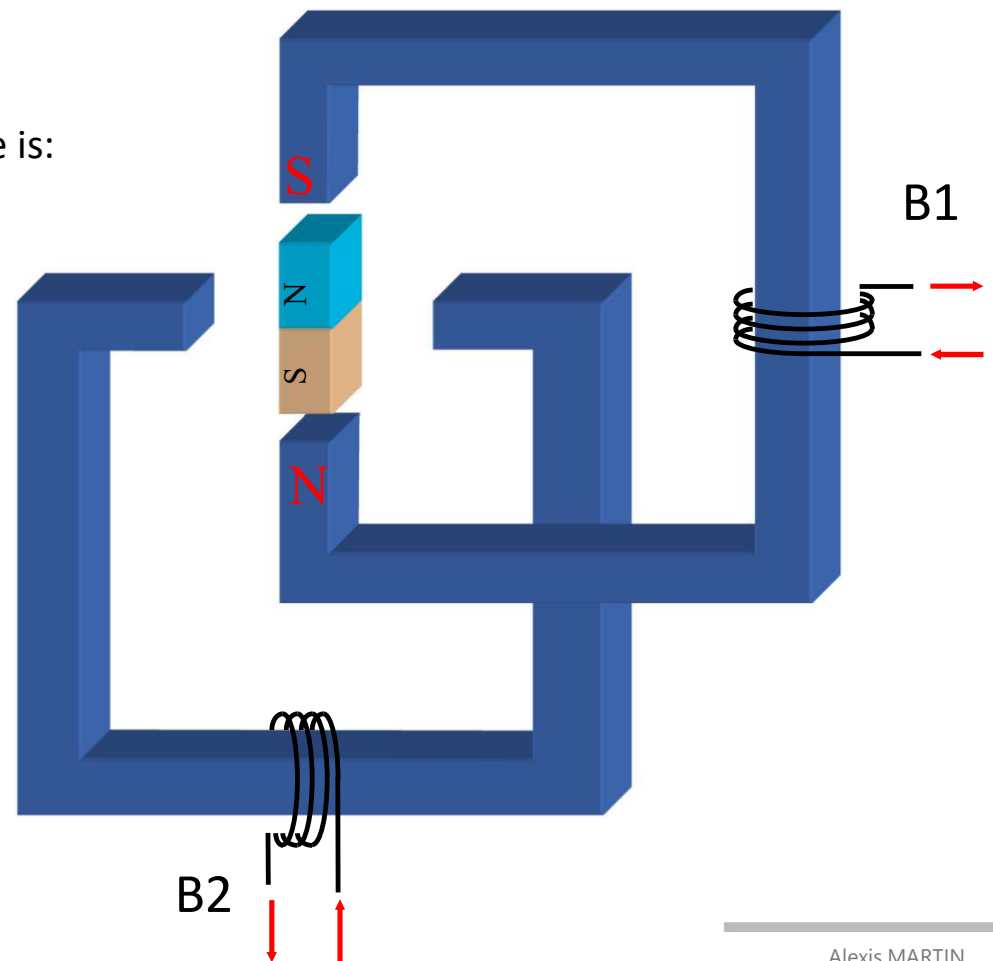
Energy storage
based chopper

4 quadrants chopper: Stepper motor control

Operation principle: full step

For a clockwise rotation, the sequence is:

B1	B2
+I	0
0	+I
-I	0
0	-I



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choppers**

2 quadrants chopper

4 quadrants chopper

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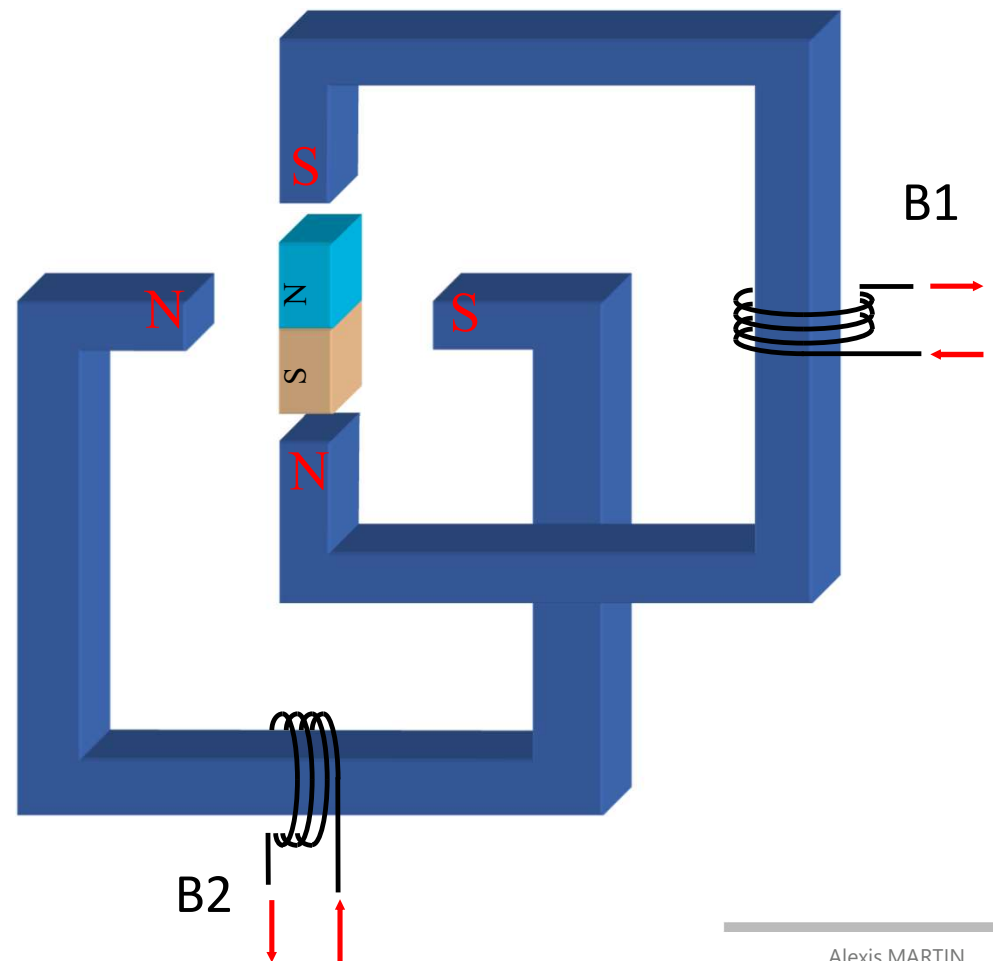
Energy storage
based chopper

4 quadrants chopper: Stepper motor control

Operation principle: half step

For a clockwise rotation:

B1	B2
I+	0
I+	I+
0	I+
I-	I+
I-	0
I-	I-
0	I-
I+	I-



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Inverter

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Single phase inverter

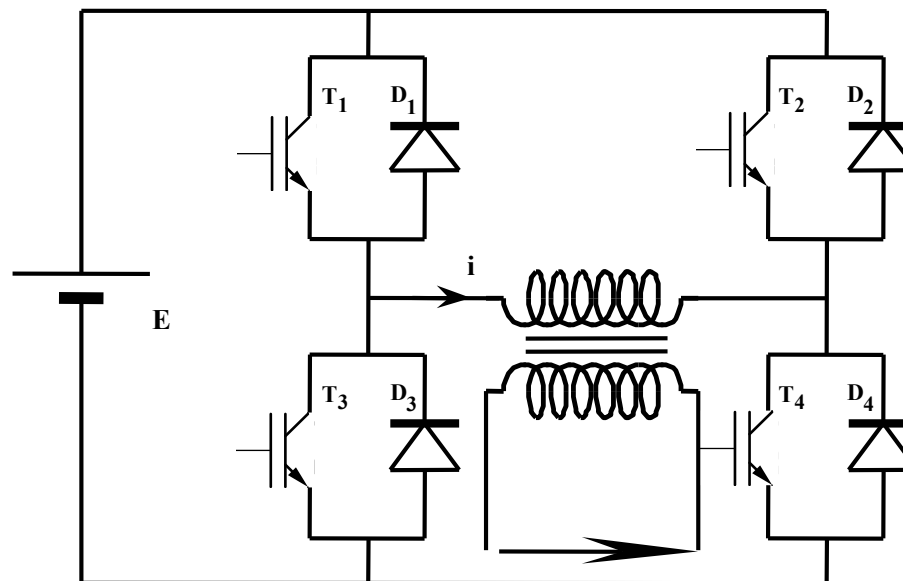
Three phase inverter

Three phase motor
control

Closed loop
chopper control

Energy storage
based chopper

Single phase inverter



$$\alpha = \frac{1}{2} + \delta_{\alpha} \sin(\omega t)$$

Introduction

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Single phase inverter

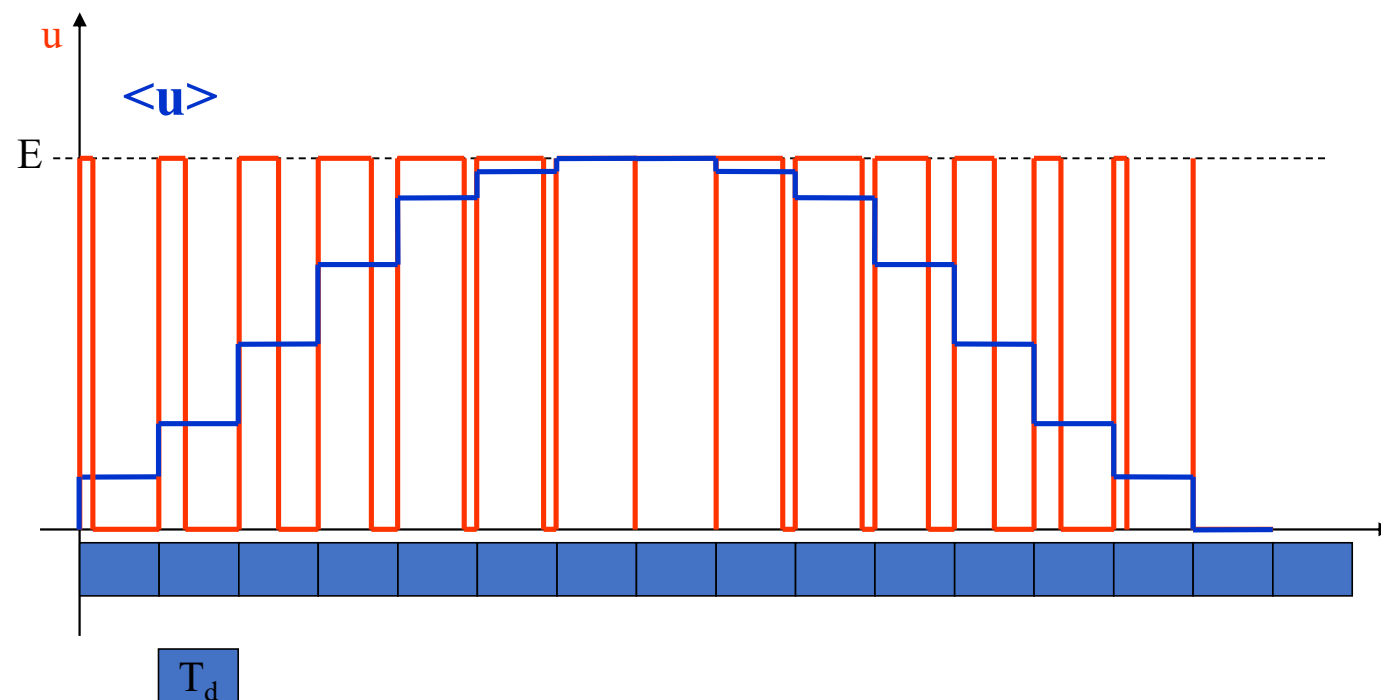
Three phase inverter

Three phase motor
control

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Energy storage
based chopper

Single phase inverter: PWM control



Introduction

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Single phase inverter

Three phase inverter

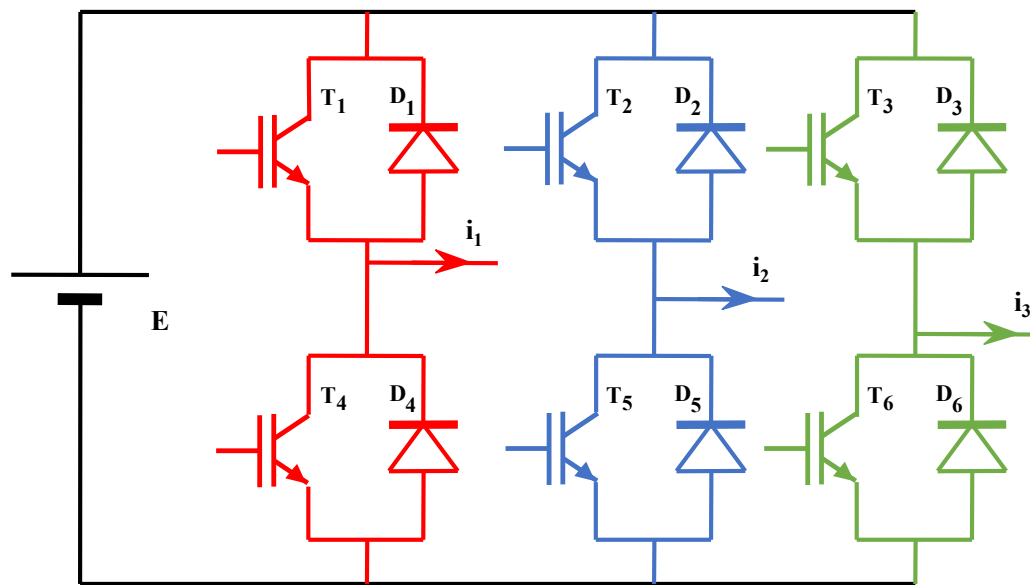
Three phase motor
control

Closed loop
chopper control

Energy storage
based chopper

Three phase inverter

$$\alpha_2 = \frac{1}{2} + \delta_\alpha \sin(\omega t - \frac{2\pi}{3})$$



$$\alpha_1 = \frac{1}{2} + \delta_\alpha \sin(\omega t)$$

$$\alpha_3 = \frac{1}{2} + \delta_\alpha \sin(\omega t + \frac{2\pi}{3})$$

Introduction

Direct conversion

Reversible
choppers

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Single phase inverter

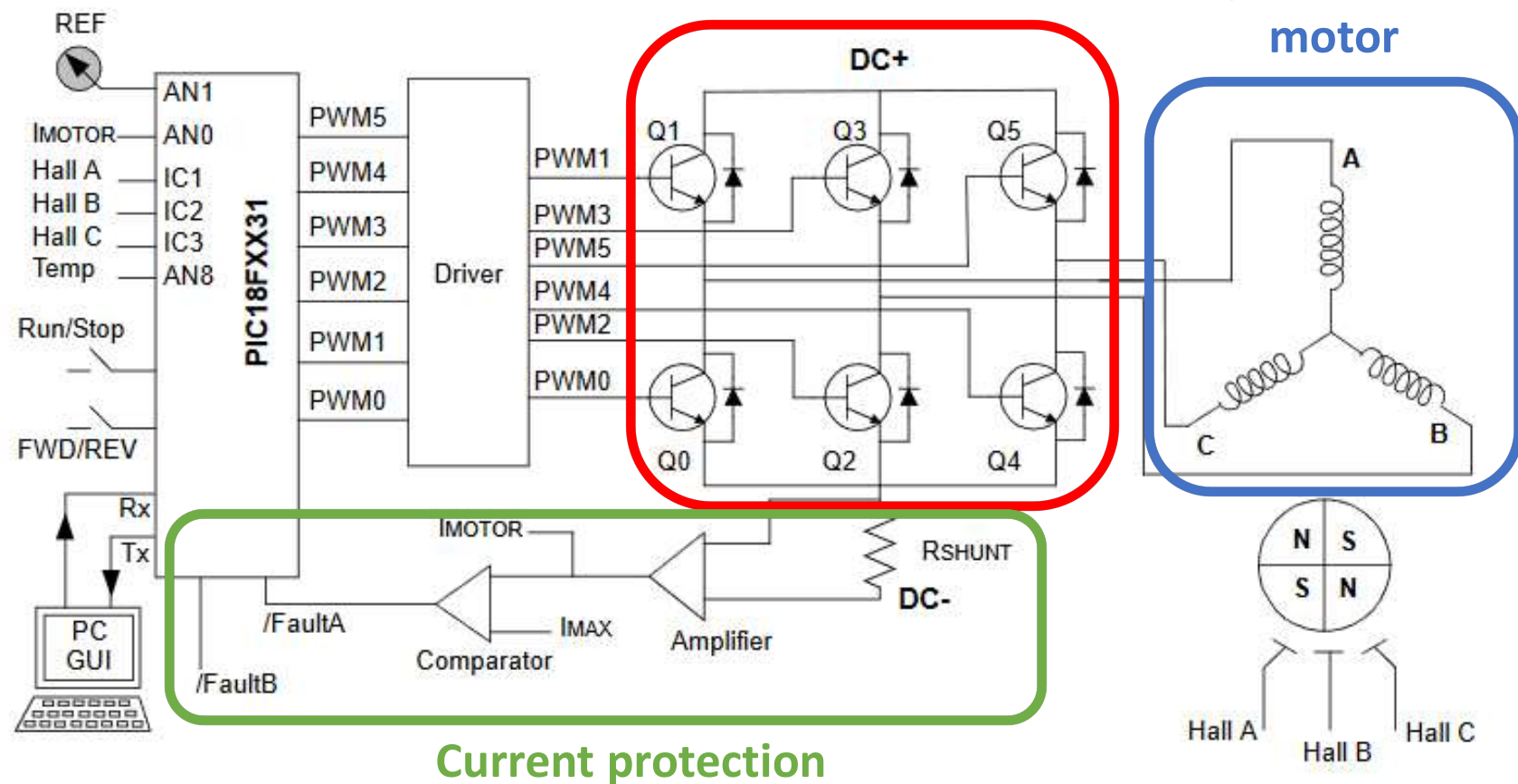
Three phase inverter

**Three phase motor
control**

Closed loop
chopper control

Energy storage
based chopper

3 Phases motor control : BLDC



AN899 (Microchip) : Brushless DC Motor Control Using PIC18FXX31 MCUs

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Direct conversion

Reversible
choppers

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Single phase inverter

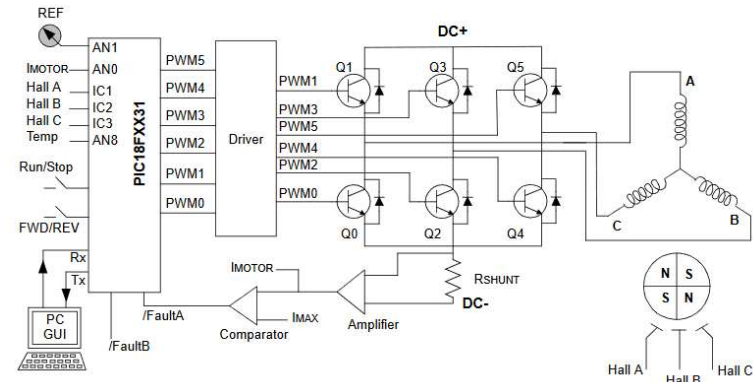
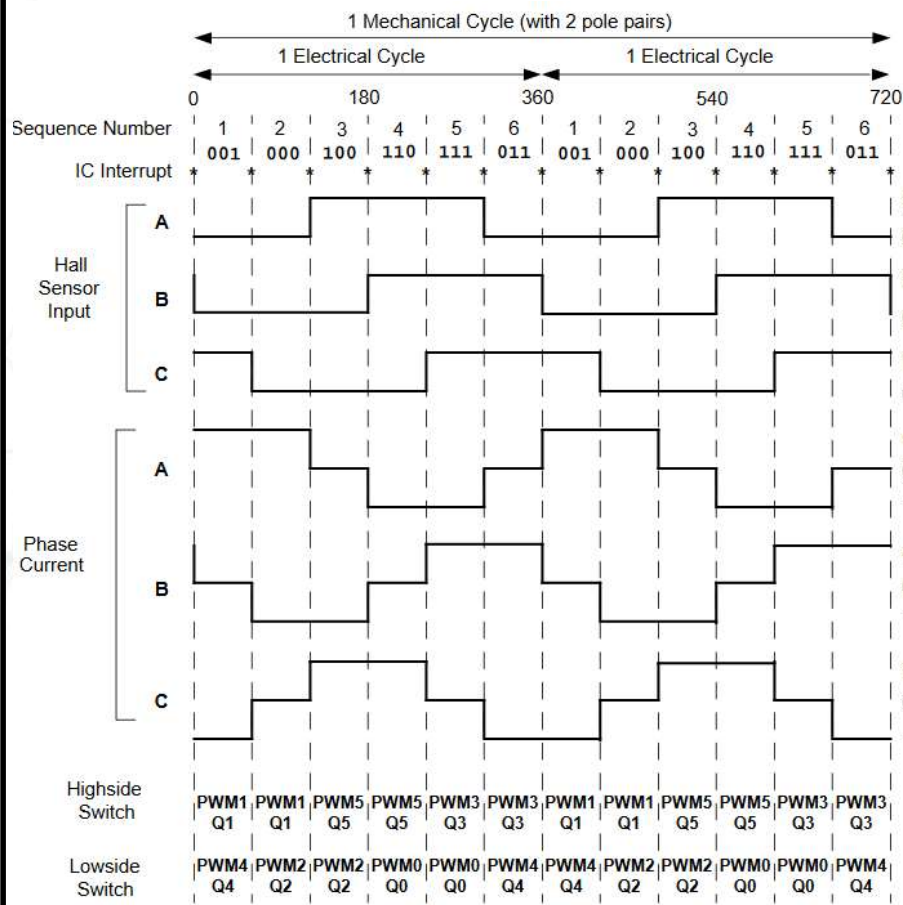
Three phase inverter

Three phase motor
control

Closed loop
chopper control

Energy storage
based chopper

3 Phases motor control : BLDC



Sequence Number	Hall Sensor Input			Active PWMs		Phase Current		
	A	B	C			A	B	C
1	0	0	1	PWM1(Q1)	PWM4(Q4)	DC+	Off	DC-
2	0	0	0	PWM1(Q1)	PWM2(Q2)	DC+	DC-	Off
3	1	0	0	PWM5(Q5)	PWM2(Q2)	Off	DC-	DC+
4	1	1	0	PWM5(Q5)	PWM0(Q0)	DC-	Off	DC+
5	1	1	1	PWM3(Q3)	PWM0(Q0)	DC-	DC+	Off
6	0	1	1	PWM3(Q3)	PWM4(Q4)	Off	DC+	DC-

AN899 (Microchip) : Brushless DC Motor Control Using PIC18FXX31 MCUs

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**Closed loop
chopper control**

Energy storage
based chopper

Closed loop chopper control

Introduction

Direct conversion

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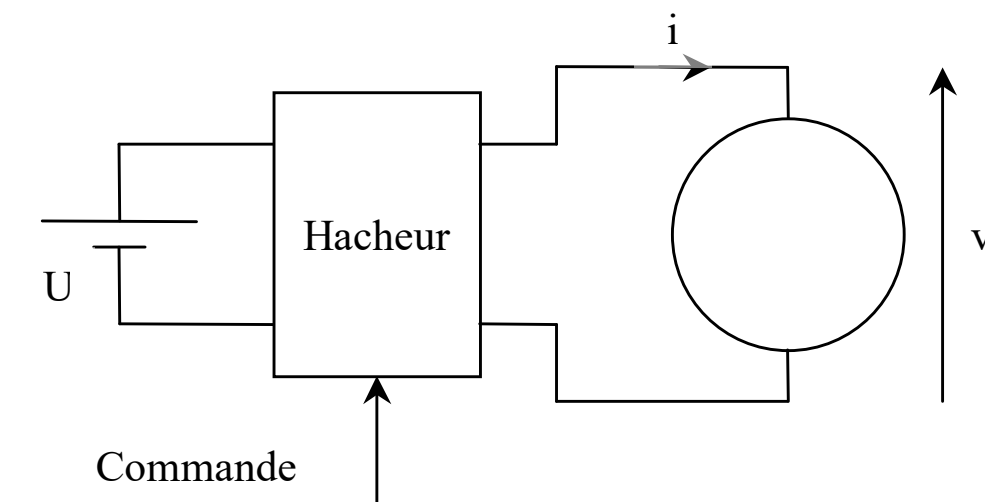
Models

Current control

Speed control

Energy storage
based chopper

Closed loop chopper control



Possible control : speed / torque

Voltage / current

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DC motor model

Electrical equation : $u = e + r \cdot i + L \cdot \frac{di}{dt}$

Mechanical equation : $J \frac{d\Omega}{dt} + f\Omega = \gamma_m - \gamma_r$

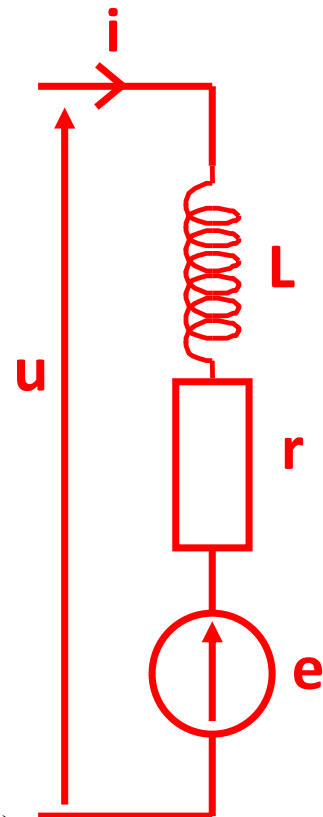
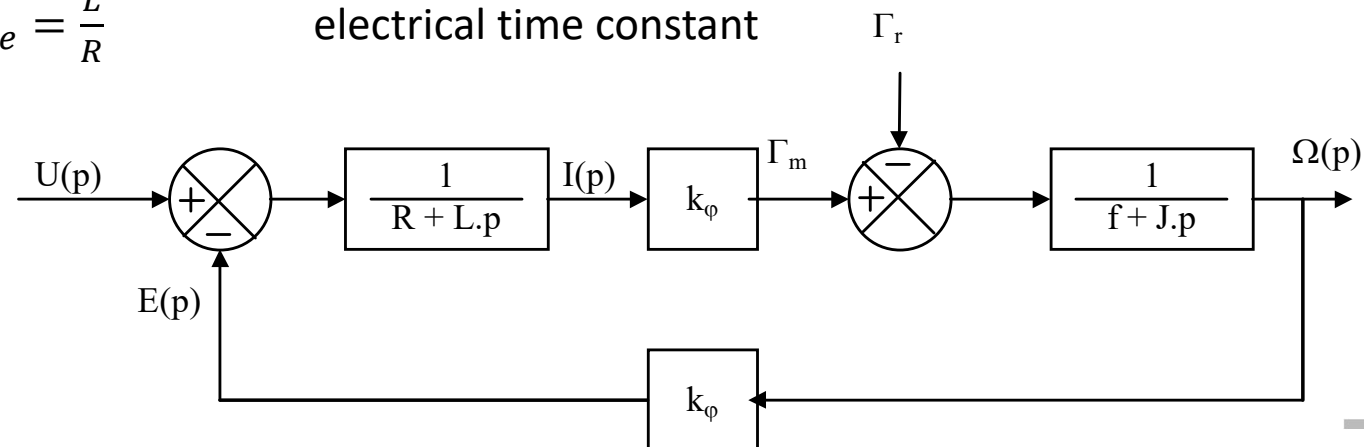
Coupling equations : $e = k_\phi \cdot \Omega \quad \gamma_m = k_\phi \cdot i$

$$\tau_{em} = \frac{J \cdot R}{k_\phi^2 + R \cdot f}$$

electromechanical time constant

$$\tau_e = \frac{L}{R}$$

electrical time constant



Introduction

Direct conversion

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**Closed loop
chopper control**

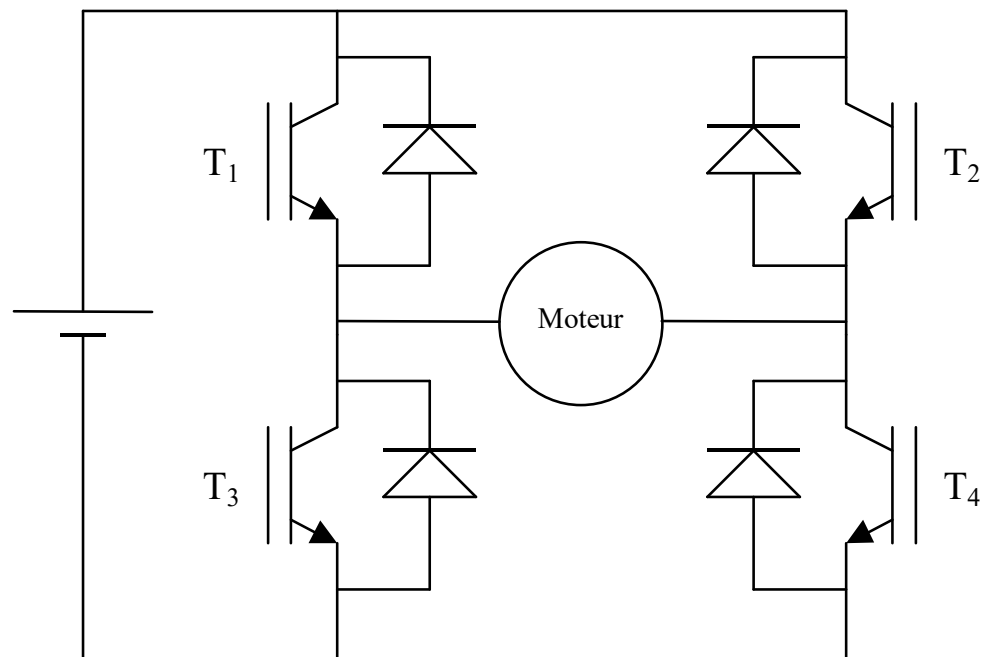
Models

Current control

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Energy storage
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Reversible structure



Introduction

Direct conversion

Reversible
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**Closed loop
chopper control**

Models

Current control

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Chopper model

$$G_h(p) = \frac{U(p)}{\alpha(p)} = 2.V_{DC} \cdot e^{-\frac{T_d}{2} \cdot p}$$

(Average delay $T_d/2$)

$$G_h(p) \approx \frac{G_0}{1 + \tau_h \cdot p} \quad \text{with} \quad \tau_h = \frac{T_d}{2} \quad \text{and} \quad G_0 = 2.V_{DC}$$

(linearization)

Introduction

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**Closed loop
chopper control**

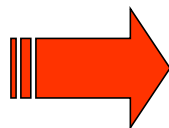
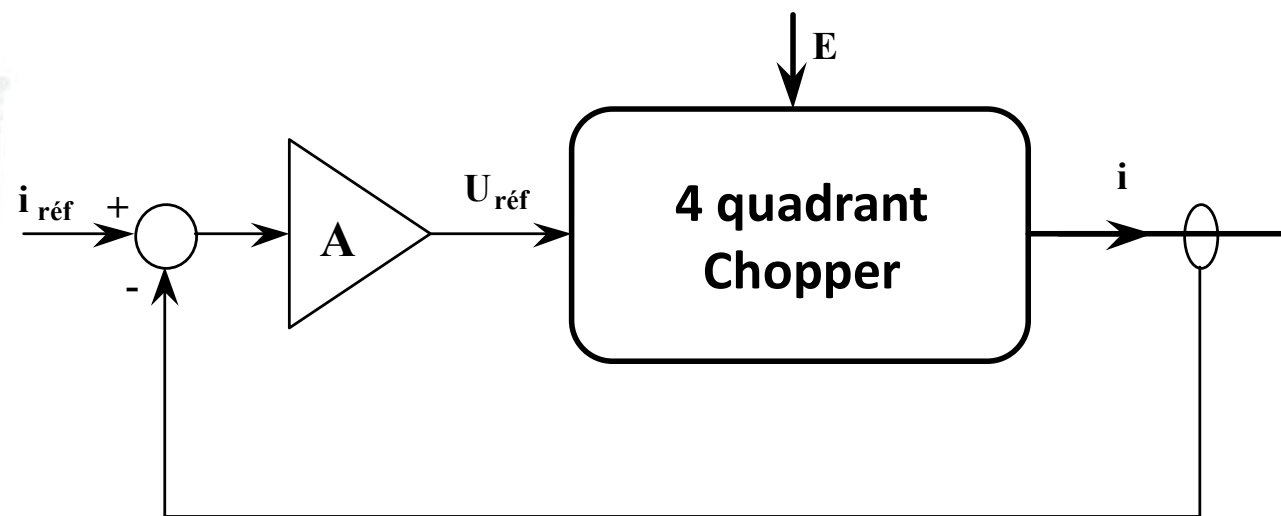
Models

Current control

Speed control

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Current control



Torque control of a motor

Introduction

Direct conversion

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choppers

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**Closed loop
chopper control**

Models

Current control

Speed control

Energy storage
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Current loop

Armature current regulation

- *Current regulation*
- *Torque control ($G = k_{\phi} \cdot I$)*
- *Better transient response*
- *Need a high performance sensor*
- *Simple Regulation (P.I.)*

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Direct conversion

Reversible
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**Closed loop
chopper control**

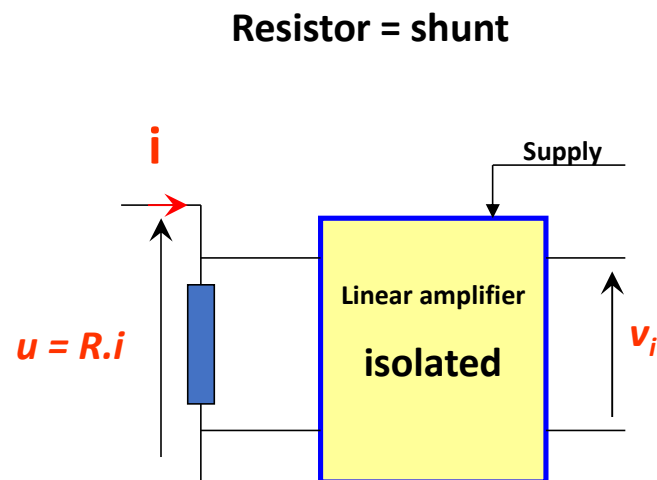
Models

Current control

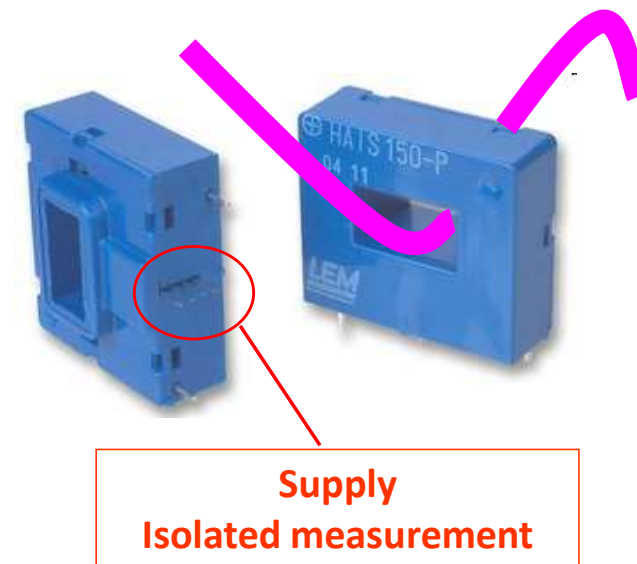
Speed control

Energy storage
based chopper

Current loop: Current sensor



Hall effect sensor



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Direct conversion

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**Closed loop
chopper control**

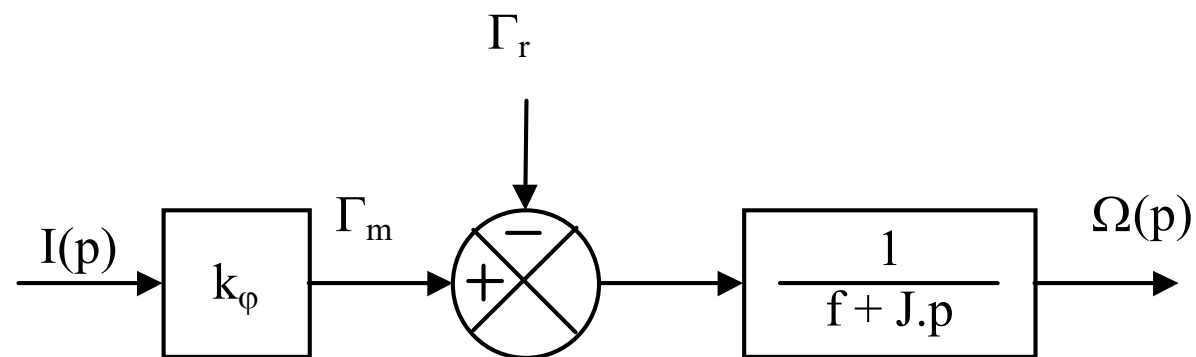
Models

Current control

Speed control

Energy storage
based chopper

DC motor: Current control



$$G_i(p) = \frac{\Omega(p)}{I(p)} = \frac{k_\varphi}{f + J.p}$$

Introduction

Direct conversion

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choppers

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**Closed loop
chopper control**

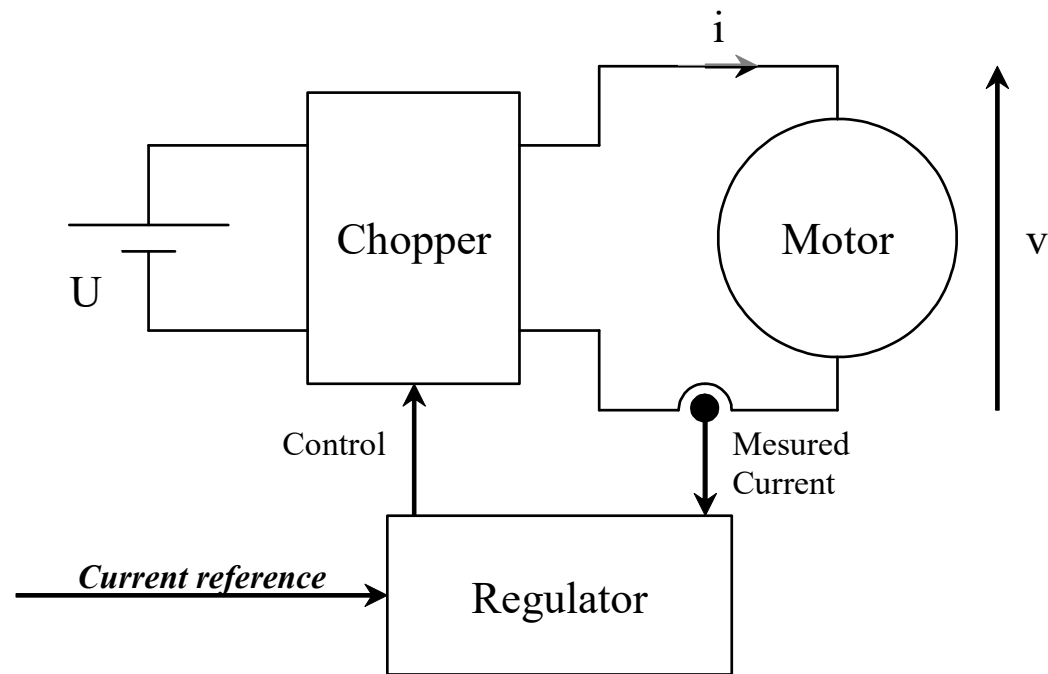
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Current loop



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Control criteria

Accuracy : no static error

Speed : time response enhancement

Unconditional stability (high margin)

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Direct conversion

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Closed loop
chopper control

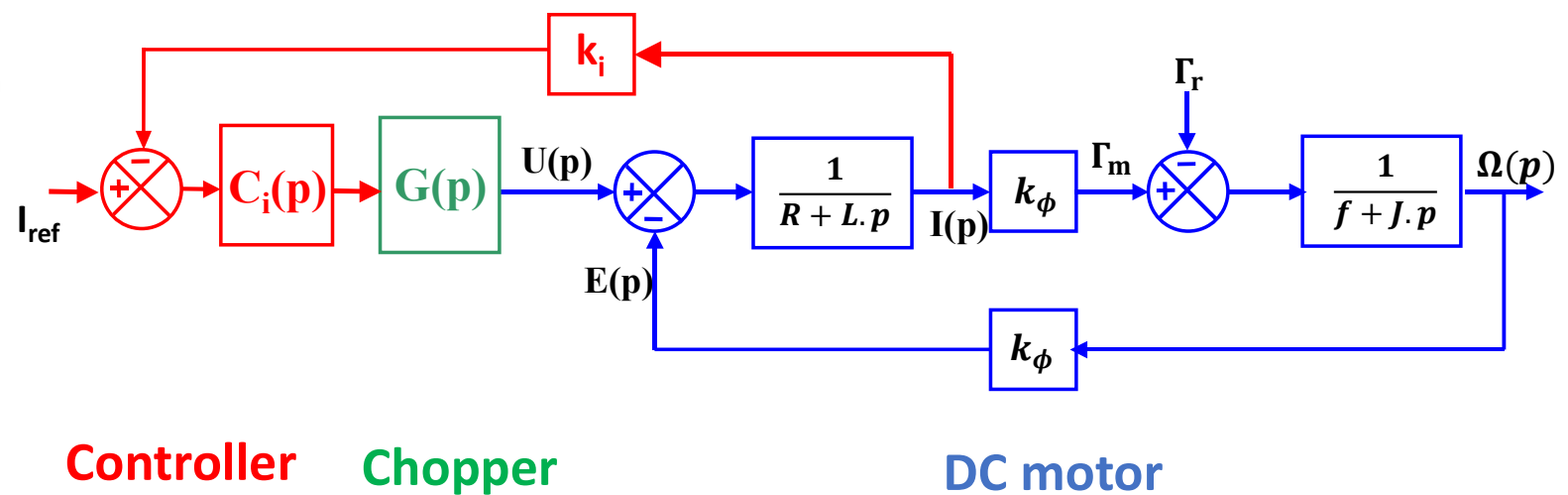
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Current Control



= > *Controller choice: $C_i(p)$*

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Direct conversion

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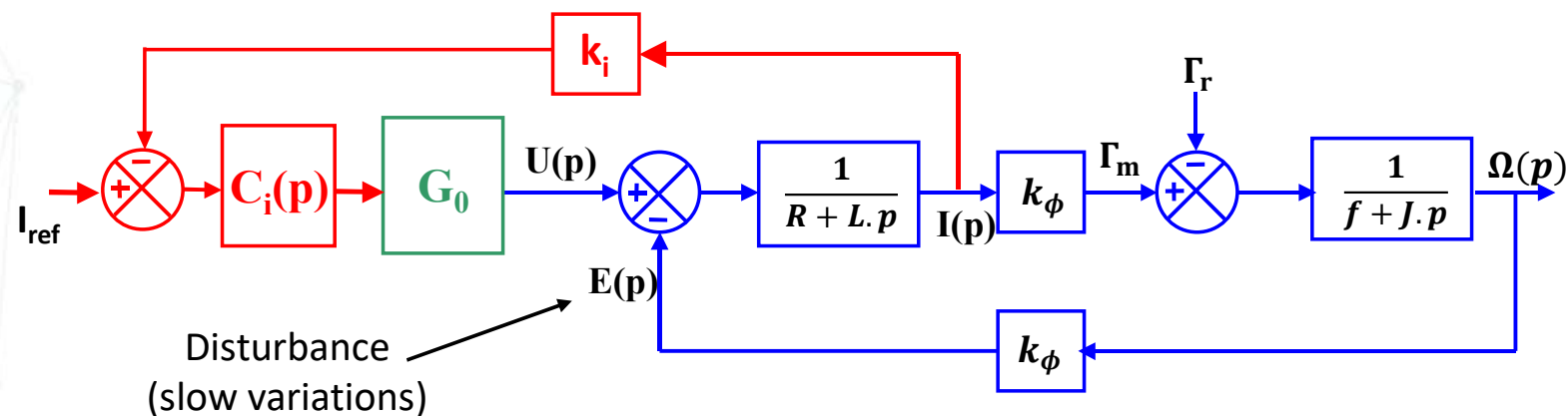
Models

Current control

Speed control

Energy storage
based chopper

P.I. control: simplified chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{NCOL}(p) = \frac{G_0 / R}{1 + \tau_e p}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \tau_e$$

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P.I. control: simplified chopper

$$H_{CL}(p) = \frac{k_i \cdot G_0 / R}{k_i \cdot G_0 / R + \tau_e p} = \frac{1}{1 + \tau_{BF} p}$$

$$\tau_{BF} = \frac{\tau_e}{\frac{k_i G_0}{R}}$$

1st order without error, faster

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**Closed loop
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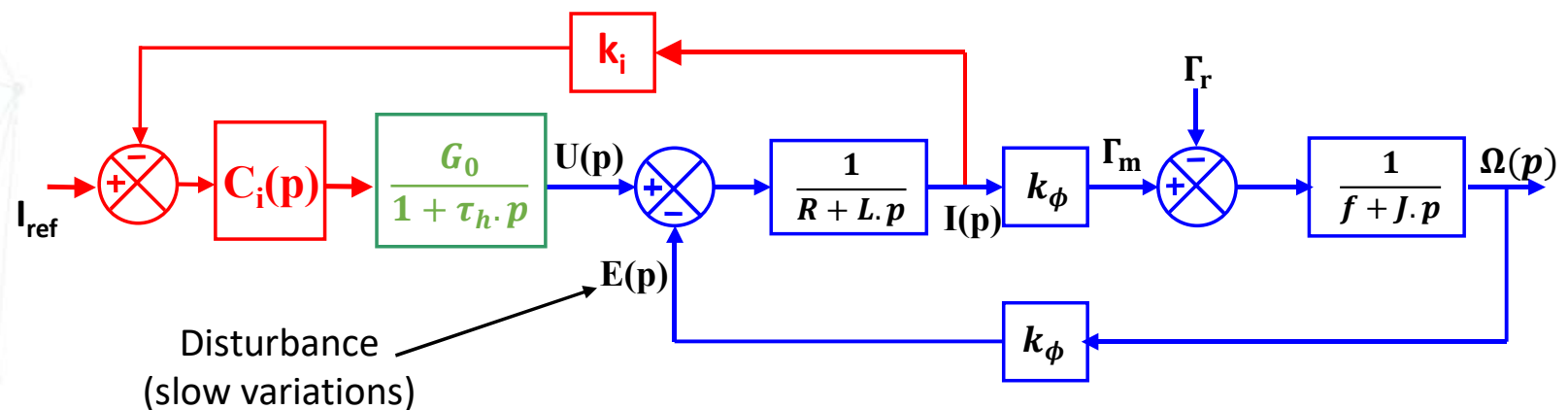
Models

Current control

Speed control

Energy storage
based chopper

P.I. control: first order chopper



$$C_i(p) = k_i \cdot \frac{1 + \tau_i p}{\tau_i p}$$

$$H_{OL}(p) = \frac{G_0 / R}{(1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$H_{OL}(p) = \frac{k_i \cdot (1 + \tau_i p) G_0 / R}{\tau_i p (1 + \tau_h p) \cdot (1 + \tau_e p)}$$

$$\Rightarrow \tau_i = \max(\tau_h, \tau_e)$$

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**Closed loop
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based chopper

Closed loop

$$H_{OL}(p) = \frac{k_i G_0 / R}{\tau_{he} p (1 + \tau_{he})}$$

$$H_{CL}(p) = \frac{1}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

Second ordre without error

Choose of m : damping $[0,7 .. 1]$

Choose of ω_0 : bandwidth

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Models

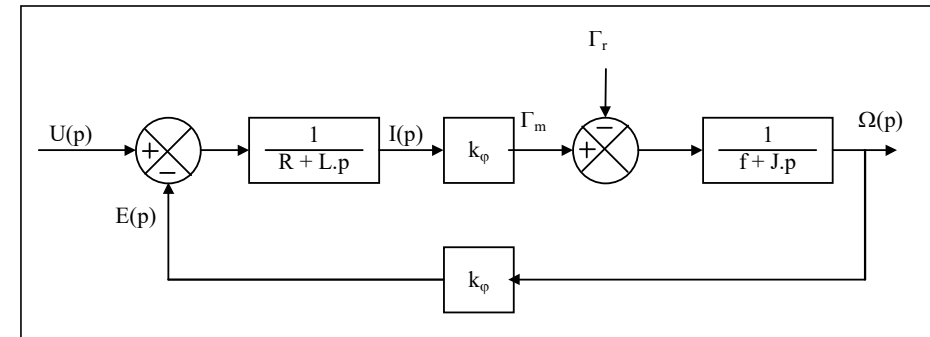
Current control

Speed control

Energy storage
based chopper

DC motor: Speed / voltage transfer function

$$G'_u(p) = \frac{\Omega(p)}{U(p)} = \frac{k_\phi}{k_\phi^2 + (R + L.p).(f + J.p)}$$



$$G_u(p) = \frac{\Omega(p)}{U(p)} = \frac{k_\phi}{k_\phi^2 + Rf} \cdot \frac{1}{(1 + \tau_e p).(1 + \tau_{em} p)}$$

$$\tau_{em} = \frac{J.R}{k_\phi^2 + R.f} \quad \text{electromechanical time constant}$$

$$\tau_e = \frac{L}{R} \quad \text{electrical time constant}$$

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Direct conversion

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**Closed loop
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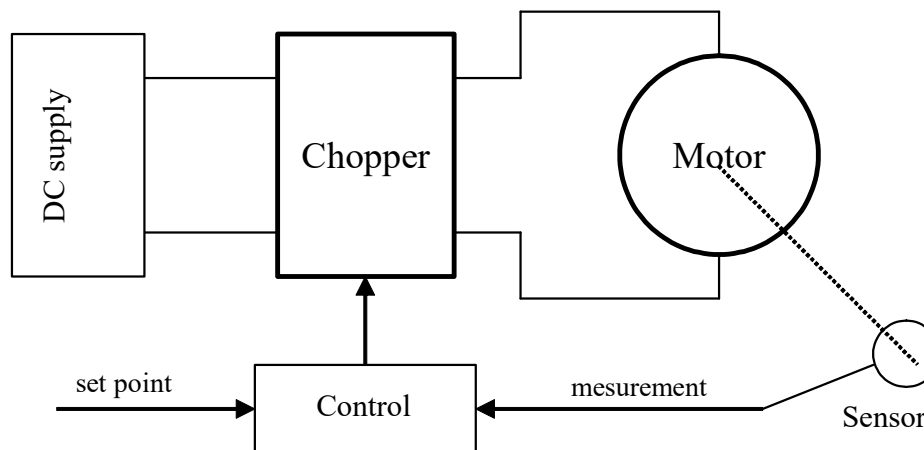
Models

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Speed control

Energy storage
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Speed loop



$$G_{hv}(p) = \frac{V_{\Omega mes}(p)}{V_{\Omega ref}(p)} = \frac{G_{0\Omega}}{1 + 2m \frac{p}{\omega_0} + \frac{p^2}{\omega_0^2}}$$

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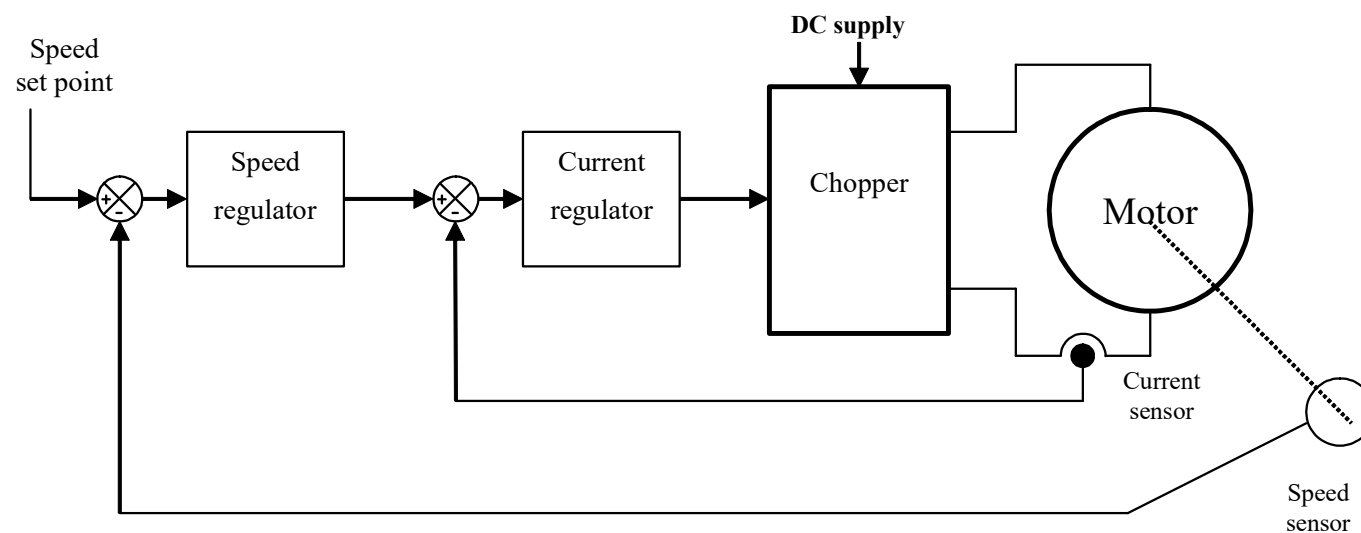
Models

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Nested loop



Internal loop faster than external loop

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Indirect Choppers

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Direct conversion

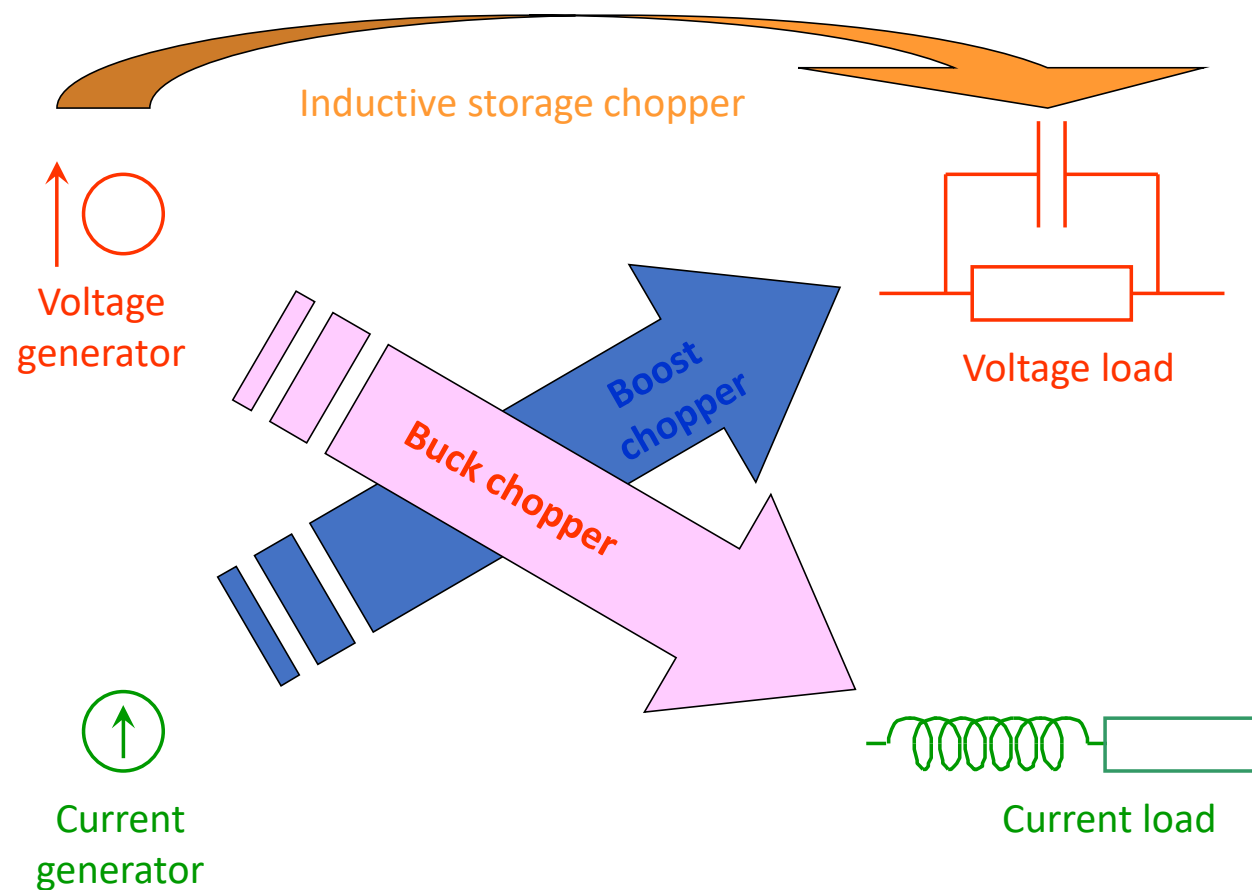
Reversible
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Indirect chopper: principle



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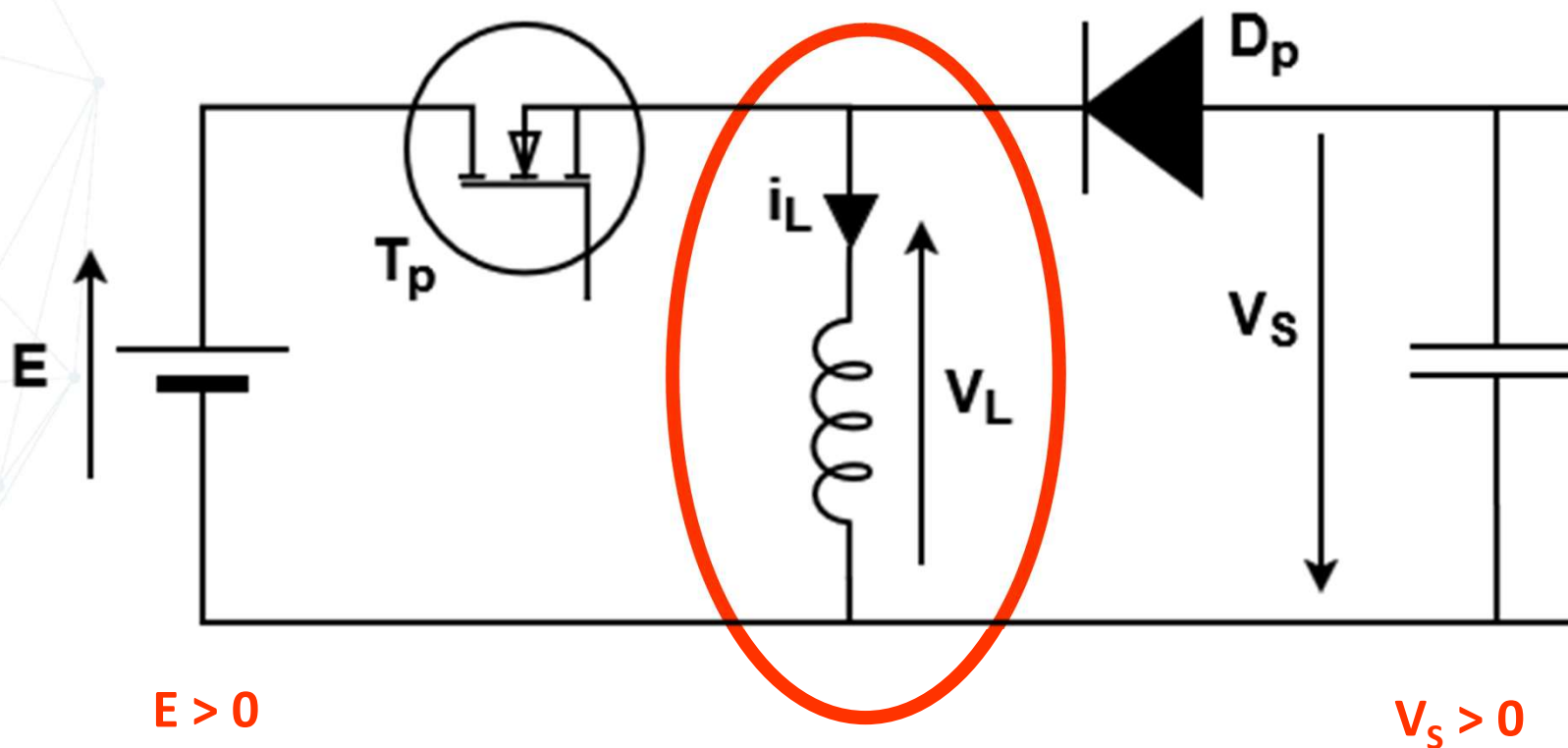
Energy storage
based chopper

Buck-Boost

Cuk

SEPIC

Buck-Boost structure



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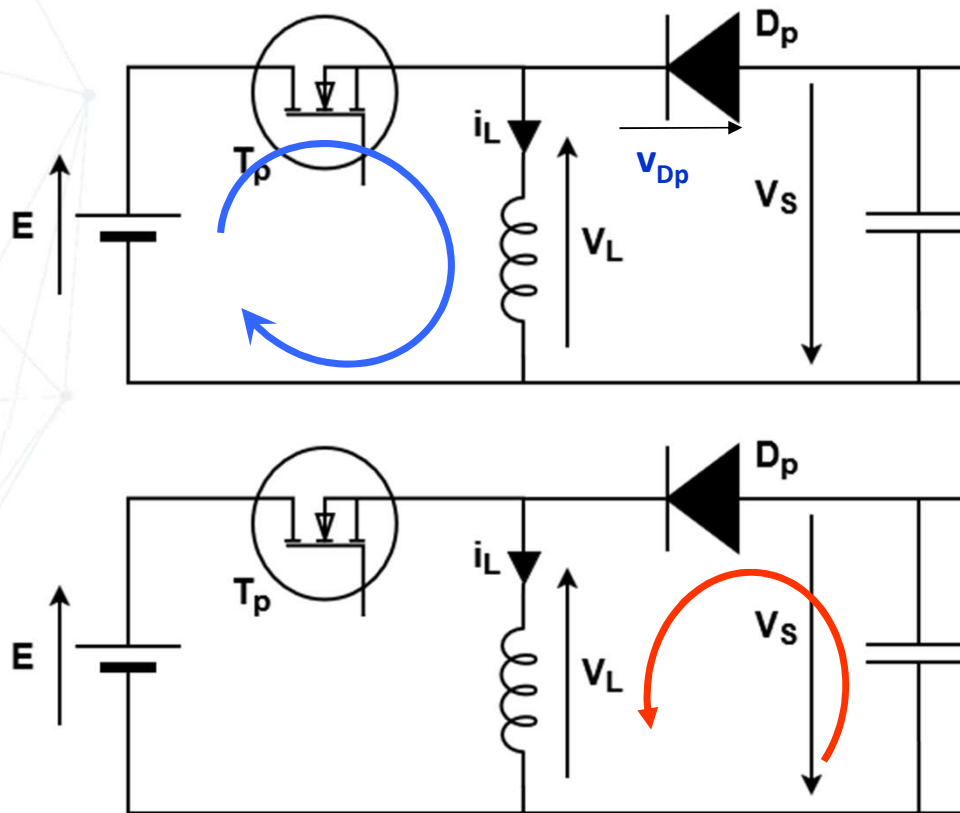
Energy storage
based chopper

Buck-Boost

Cuk

SEPIC

Buck-boost: Continuous mode ($i_L > 0$)



$0 < t < \alpha T : T_p$ closed

$v_{Dp} = -E - V_s \Rightarrow D_p$ reverse

$v_L = E = L \cdot di_L/dt \Rightarrow i_L$ increasing > 0

$\alpha T < t < T : T_p$ opened

$i_L > 0 \Rightarrow D_p$ forward

$v_L = -V_s = L \cdot di_L/dt \Rightarrow i_L$ decreasing

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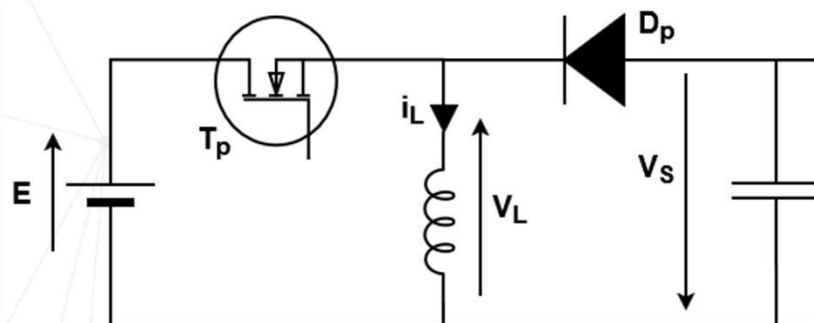
Energy storage
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Buck-Boost

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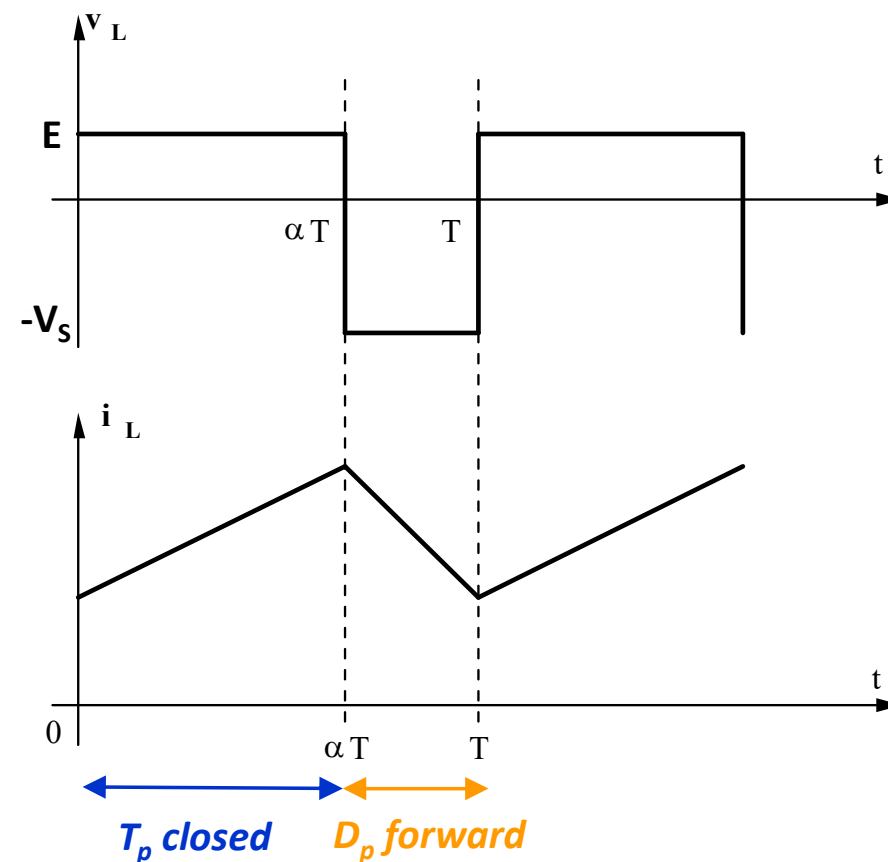
SEPIC

Buck-boost: Continuous mode (chronograms)



$$\langle v_L \rangle = 0$$

Inductive storage chopper



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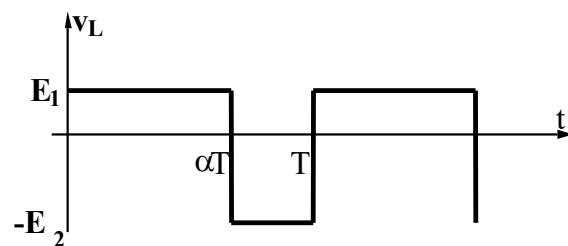
Energy storage
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Buck-Boost

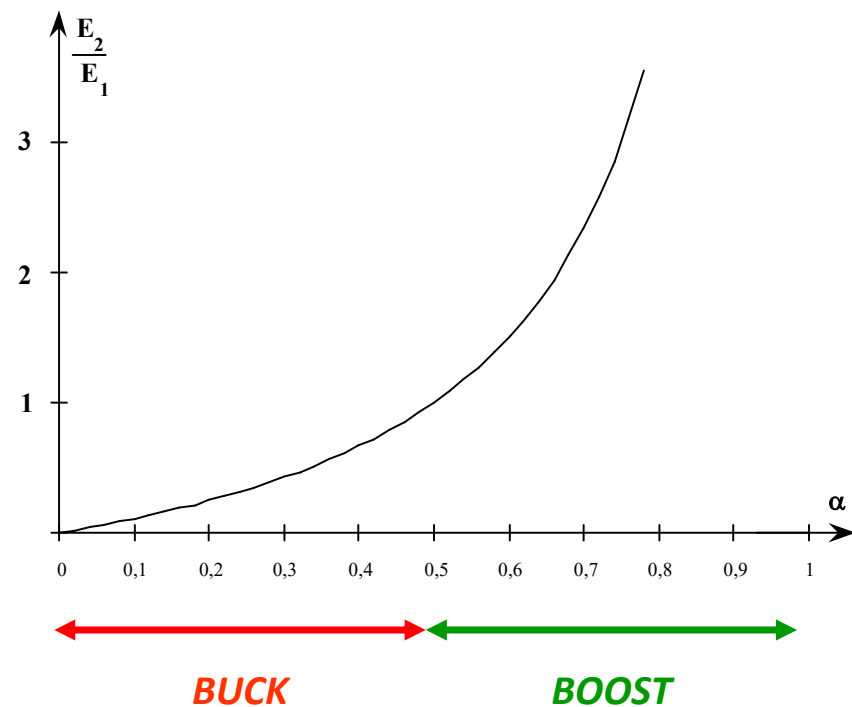
Cuk

SEPIC

Buck-boost: Continuous mode (Transfer characteristics)



$$\langle v_L \rangle = 0$$



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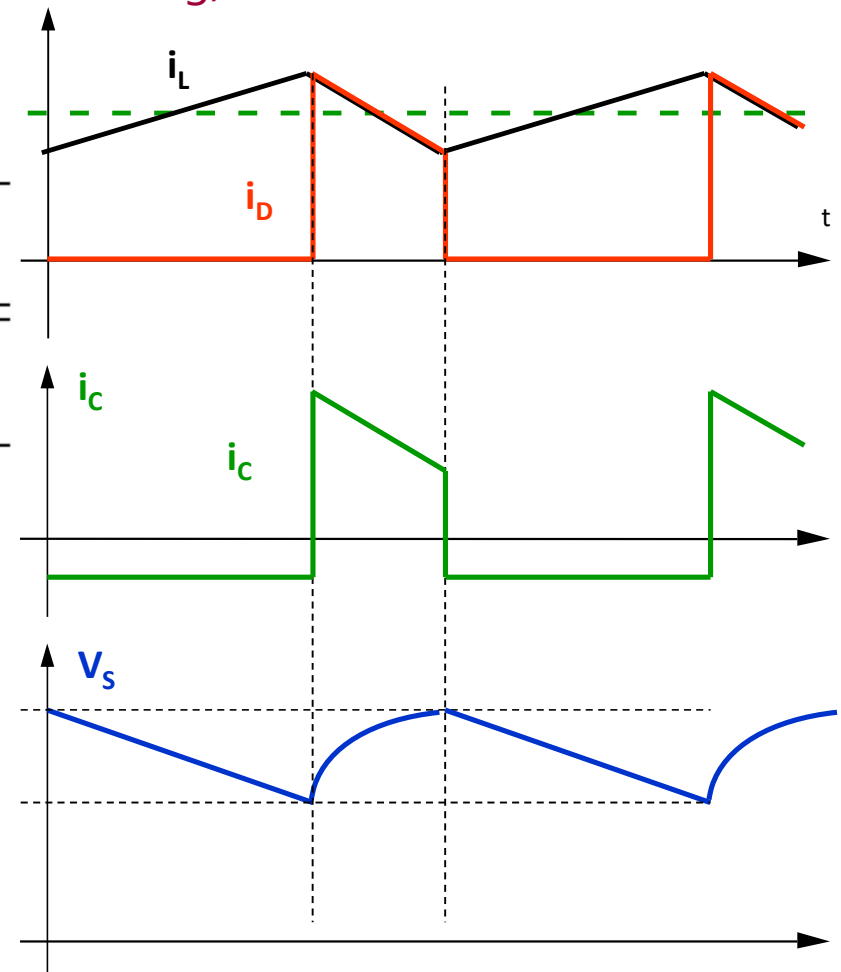
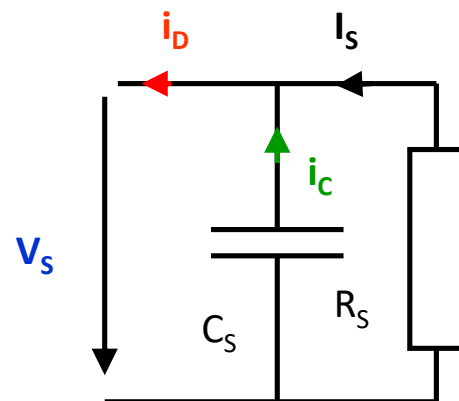
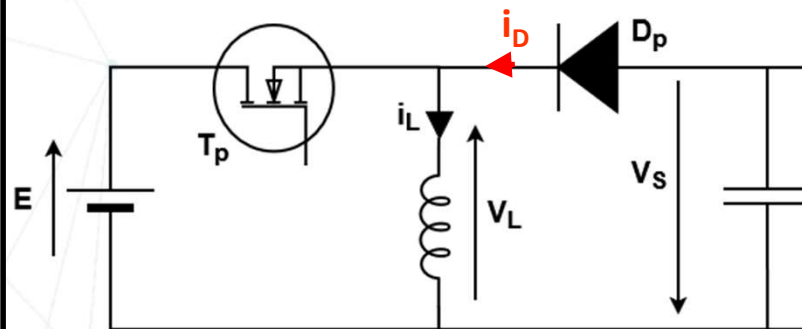
Energy storage
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Buck-boost: Continuous mode (output filtering)



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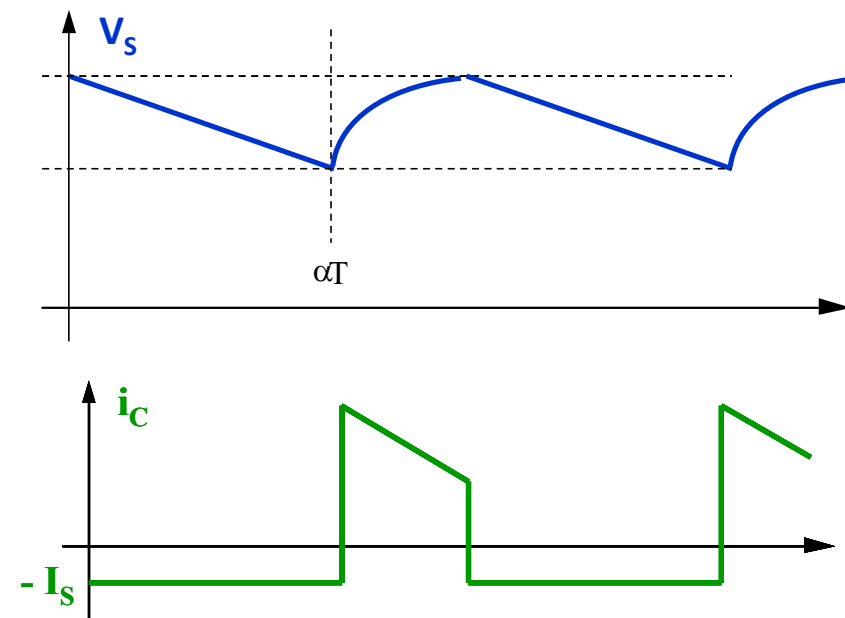
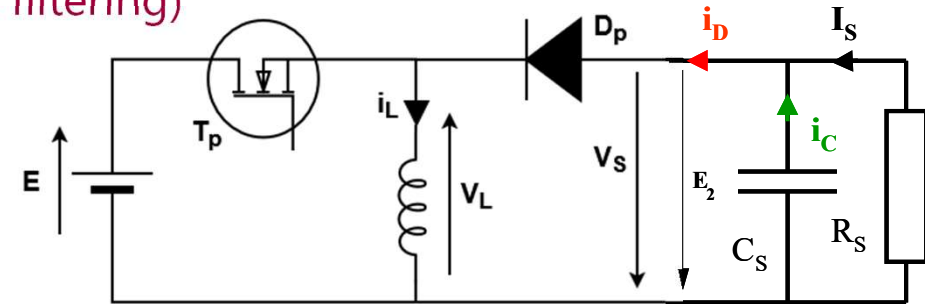
Buck-boost: Continuous mode (Output filtering)

$$\Delta V_S = \frac{1}{C_S} \cdot \int_0^{\alpha T} i_C(t) \cdot dt = \frac{I_S \cdot \alpha T}{C_S}$$

$$I_S = \frac{V_S}{R_S} = \frac{\alpha \cdot E}{(1 - \alpha) R_S}$$

$$\Delta V_S = \frac{ET}{R_S C_S} \cdot \frac{\alpha^2}{1 - \alpha}$$

$$C_S = \frac{\alpha^2 \cdot E \cdot T}{(1 - \alpha) \cdot R_S \cdot \Delta V_S}$$



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Energy storage
based chopper

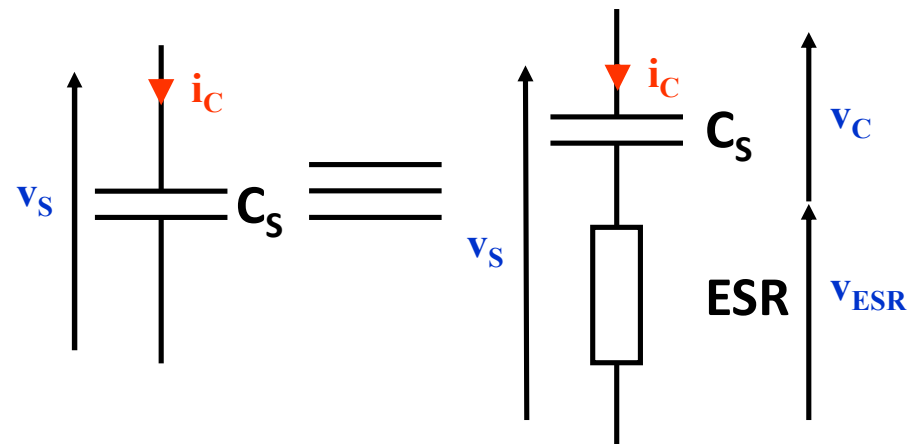
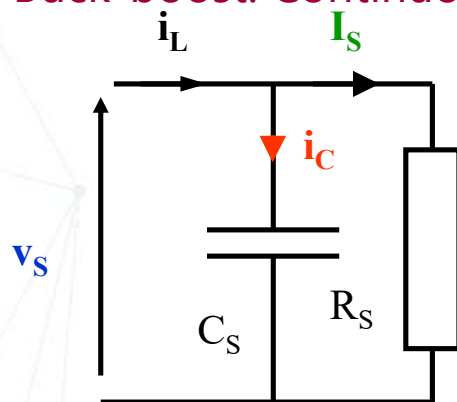
Buck-Boost

Cuk

SEPIC



Buck-boost: Continuous mode (Output filtering)



$$v_s = v_C + v_{ESR}$$

$$\Delta v_s < \Delta v_C + \Delta v_{ESR}$$

For high value of C_s ,
 $\Delta v_s \approx \Delta v_{ESR} = ESR \cdot \Delta i_c = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta V_s}{\Delta i_L}$$

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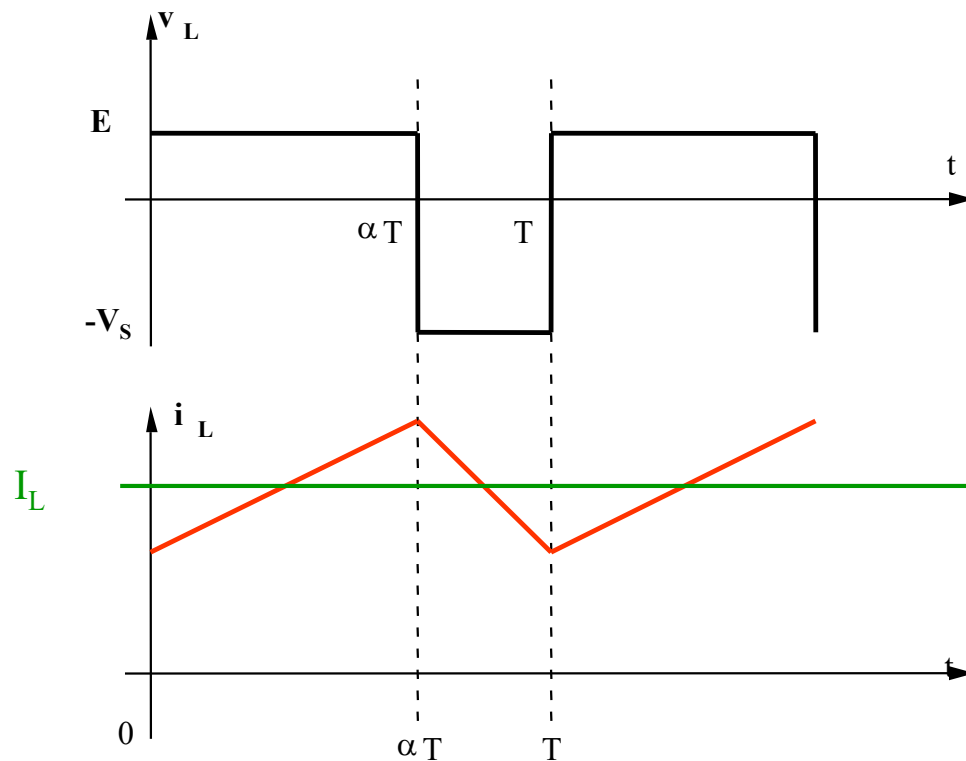
Energy storage
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Cuk

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Buck-boost: "decreasing" load



=> *Boundary mode*

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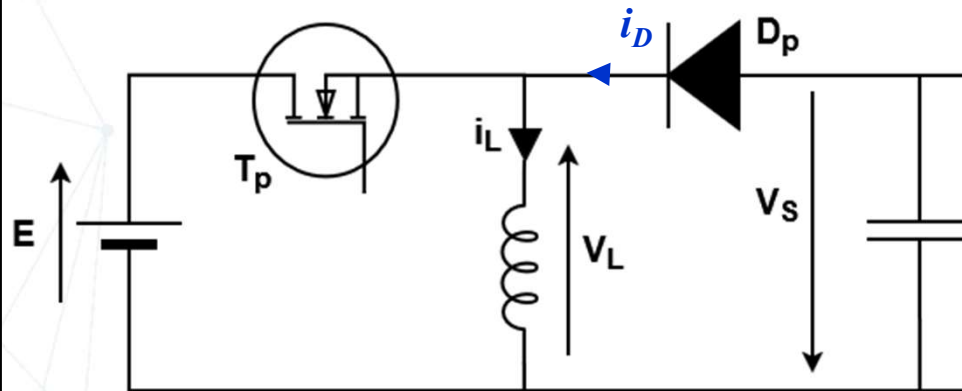
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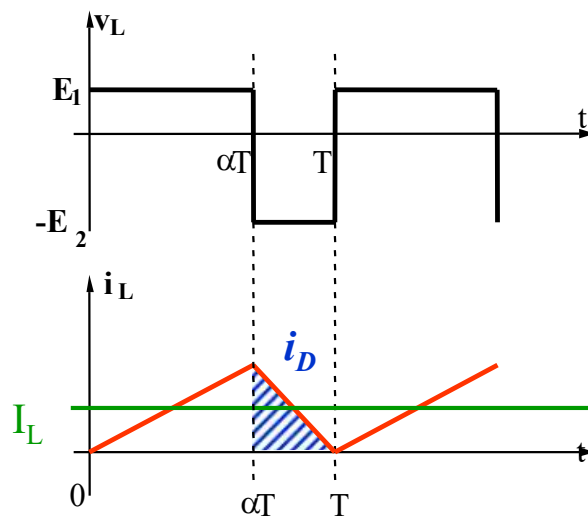
SEPIC

Buck-boost: Boundary mode



$$V_S = E \cdot \frac{\alpha}{1 - \alpha} \Rightarrow \alpha = \frac{V_S}{E + V_S}$$

$$I_{Lc} = \frac{E}{L} \frac{\alpha T}{2} = \frac{E \cdot V_S}{E + V_S} \cdot \frac{T}{2L} = \frac{I_{Lmax}}{2}$$



Average boundary mode $I_S = \langle i_D \rangle$

$$I_S = \frac{I_{Lmax}}{2} (1 - \alpha) = \frac{E^2 \cdot V_S}{(E + V_S)^2} \frac{T}{2L}$$

➡ $V_S = f(I_S)$

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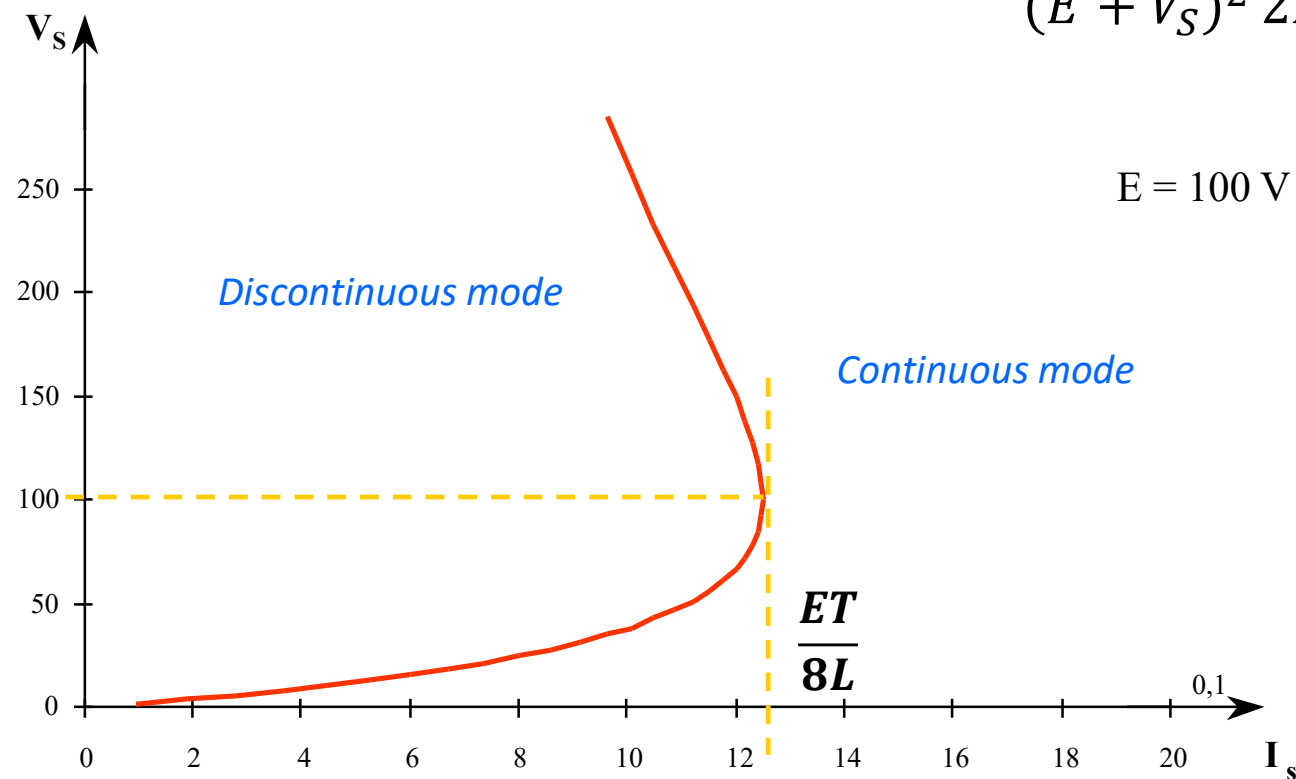
Cuk

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Buck-boost: Boundary mode (characteristics)

$$I_S = \frac{E^2 \cdot V_S}{(E + V_S)^2} \frac{T}{2L}$$

$E = 100 \text{ V}$



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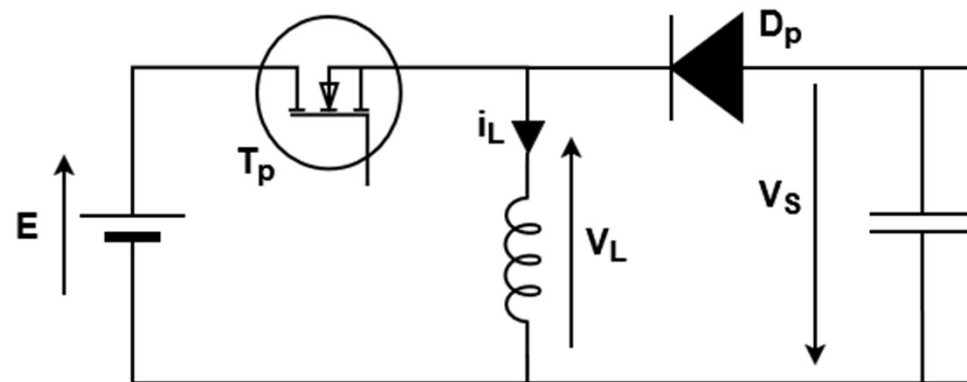
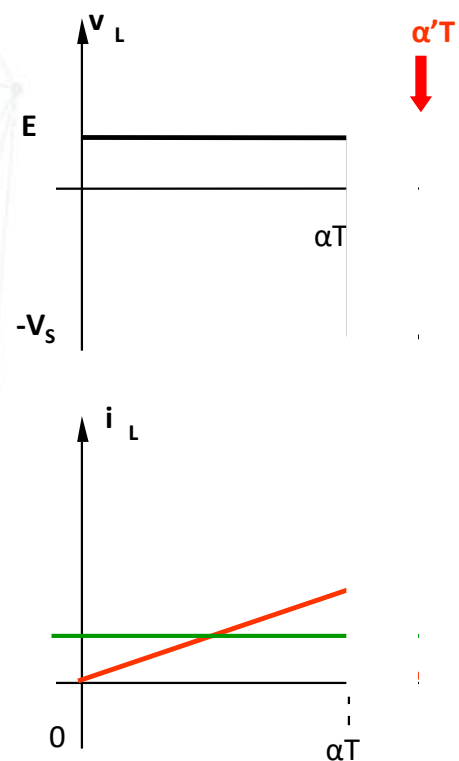
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Buck-boost: Discontinuous mode



Full study: Switched-mode power supply

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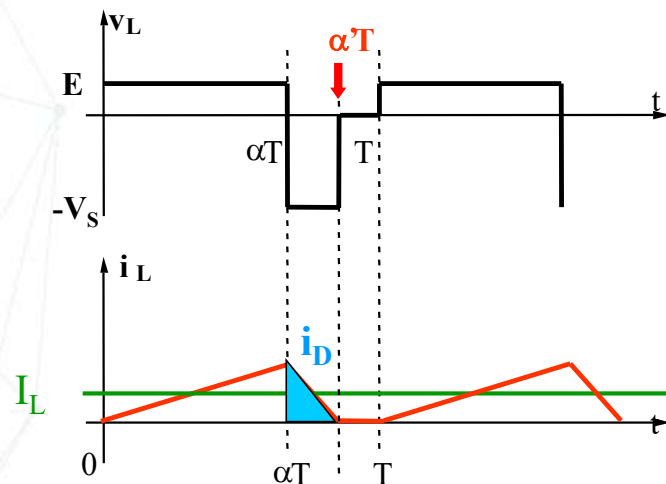
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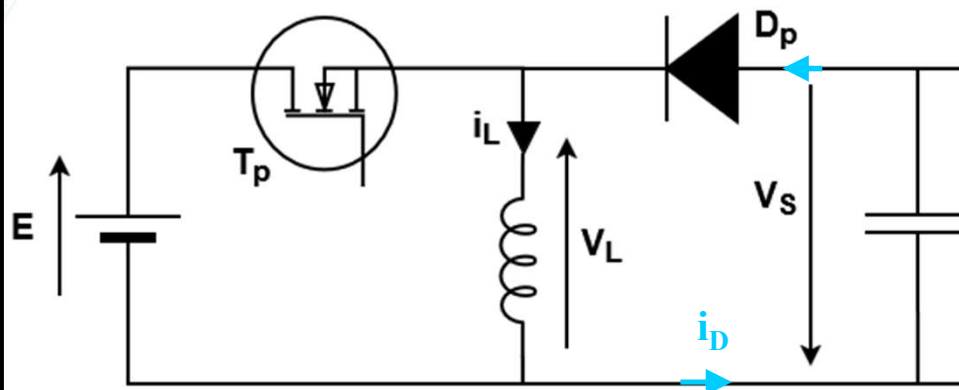
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Buck-boost: Discontinuous mode



$$\langle v_L \rangle = 0 = \alpha \cdot E - (\alpha' - \alpha) \cdot V_S \Rightarrow \alpha' = \alpha \frac{E + V_S}{V_S}$$

$$I_S = \langle i_D \rangle = (\alpha' - \alpha) \cdot \frac{I_{L\max}}{2} = (\alpha' - \alpha) \left(\alpha \frac{ET}{2L} \right)$$



$$I_S = \frac{\alpha^2 \cdot E^2 \cdot T}{2L \cdot V_S}$$

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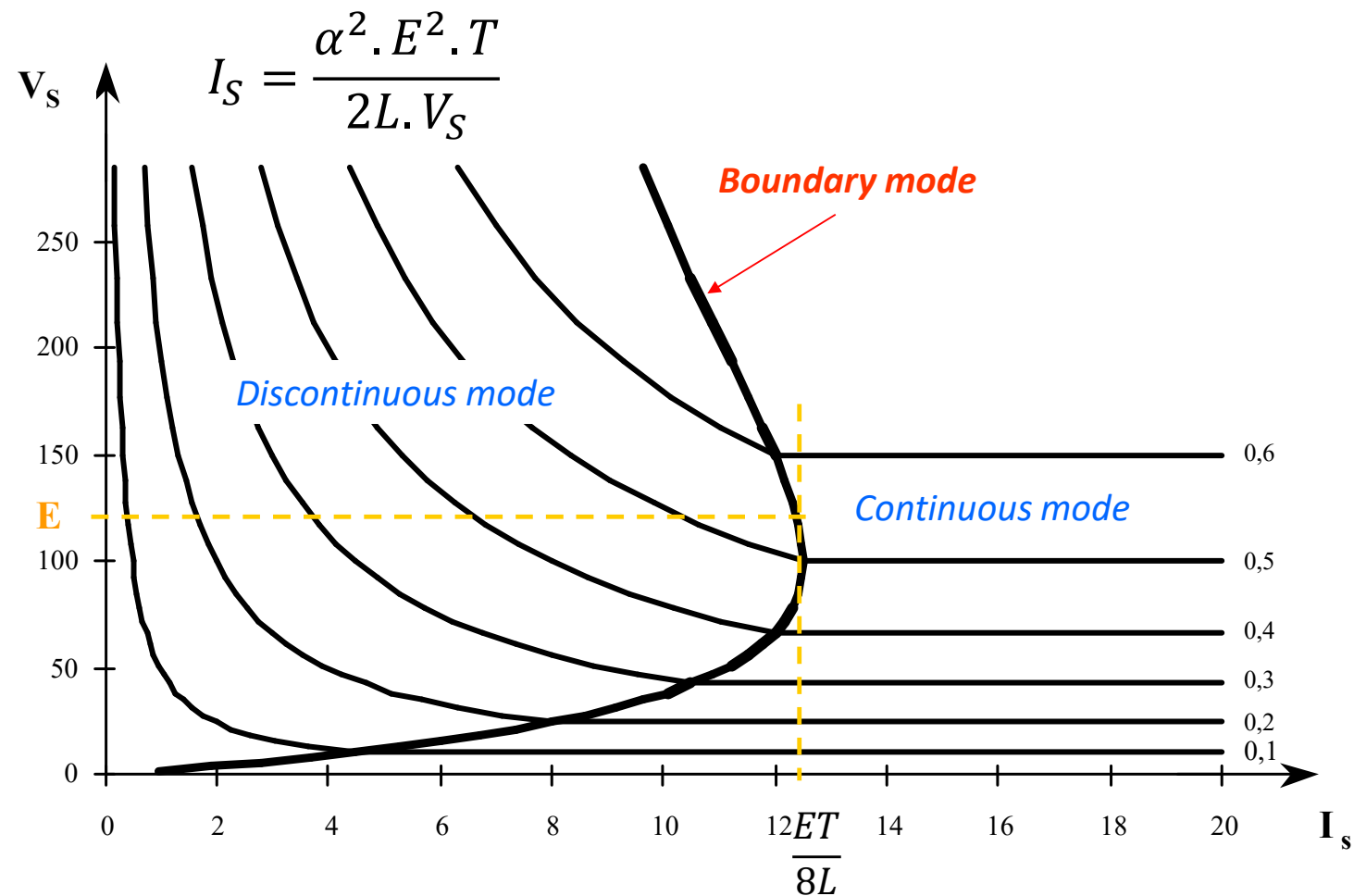
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Buck-boost: Output characteristics



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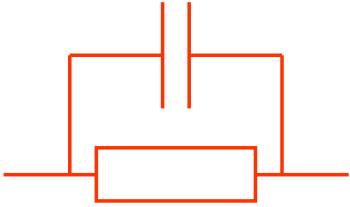
Cuk

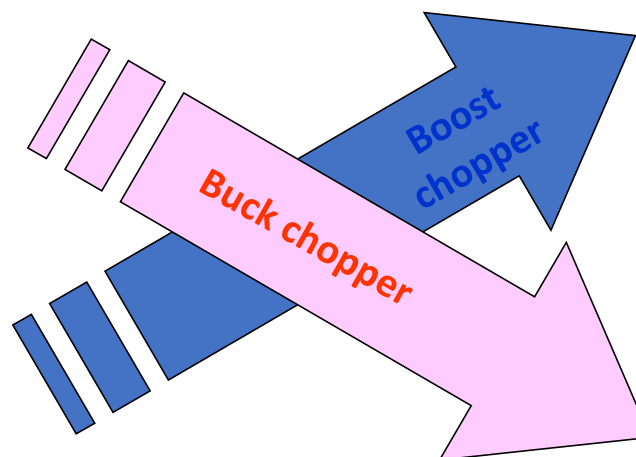
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Capacitive storage chopper


Voltage
generator


Voltage load




Current
generator


Current load

Capacitive storage chopper

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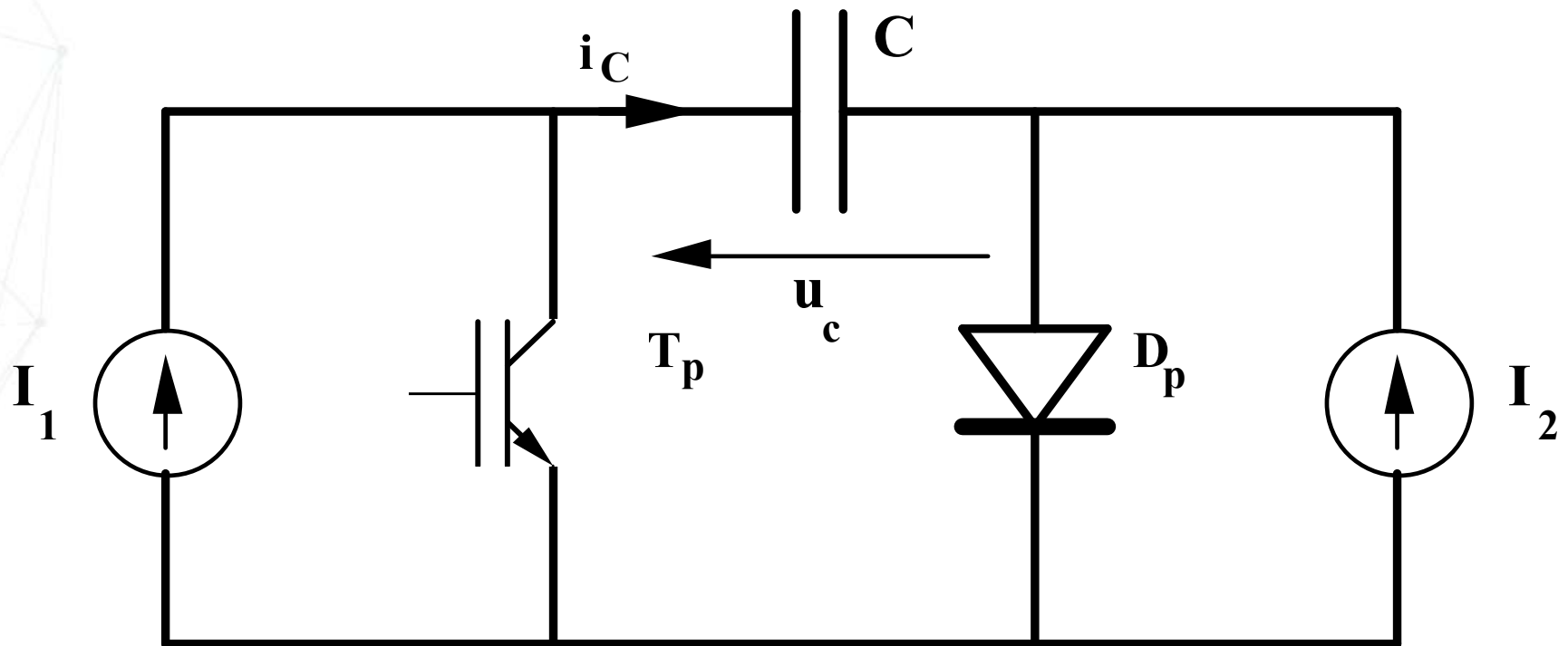
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CUK structure



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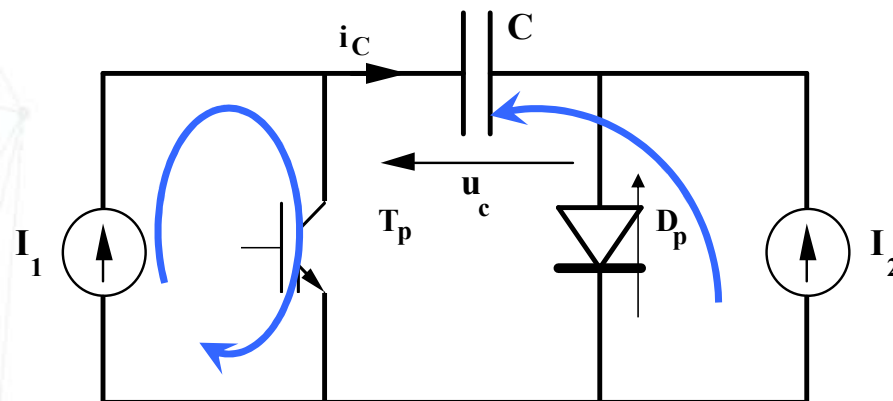
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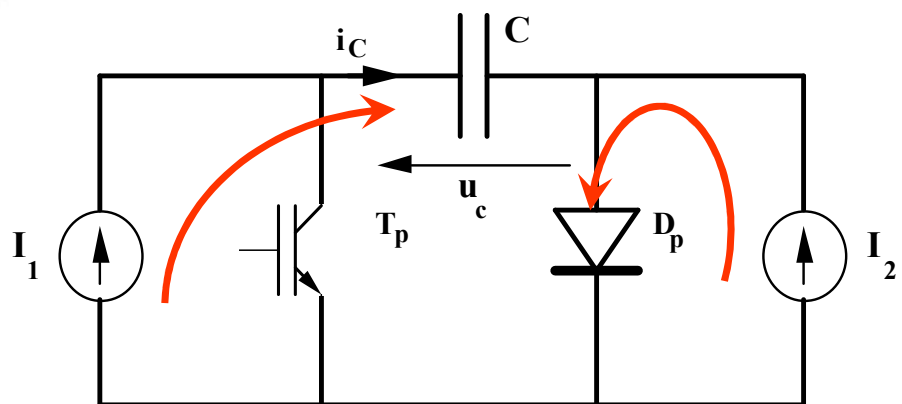
CUK structure: continuous mode ($u_c > 0$)



$0 < t < \alpha T : T_p \text{ closed}$

$v_{Dp} = -u_c \Rightarrow D_p \text{ reverse}$

$i_c = -I_2 = C \cdot du_c/dt \Rightarrow u_c \text{ decreases}$



$\alpha T < t < T : T_p \text{ opened}$

$i_c = I_1 = C \cdot du_c/dt \Rightarrow u_c \text{ increases}$

$i_D = I_1 + I_2 \Rightarrow D_p \text{ forward}$

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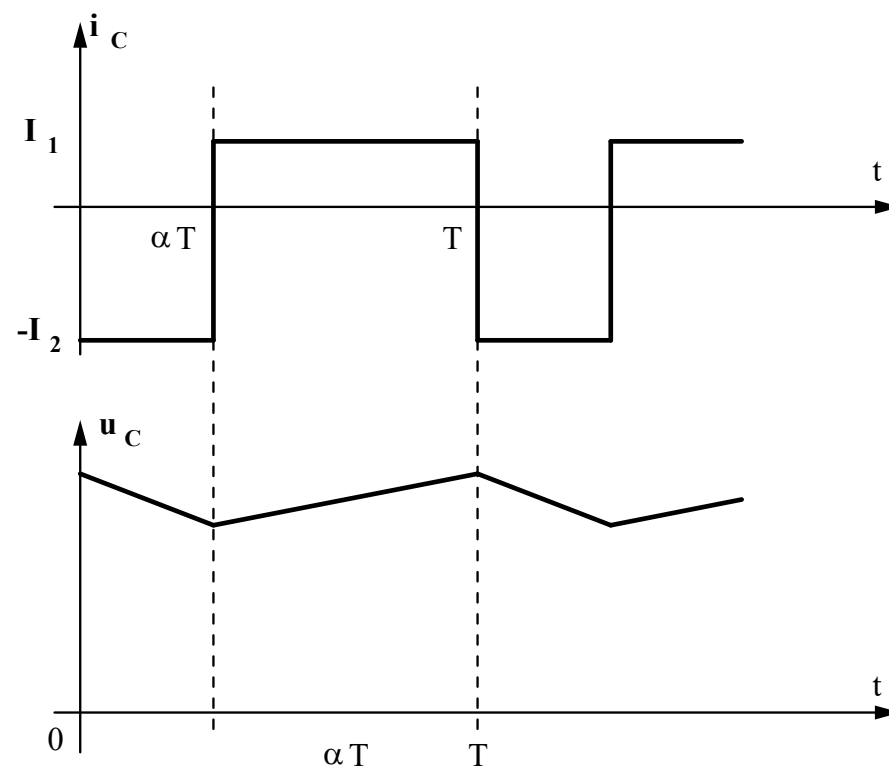
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CUK structure: Chronograms

$$\langle i_C \rangle = 0$$

$$I_2 = I_1 \cdot \frac{1 - \alpha}{\alpha}$$



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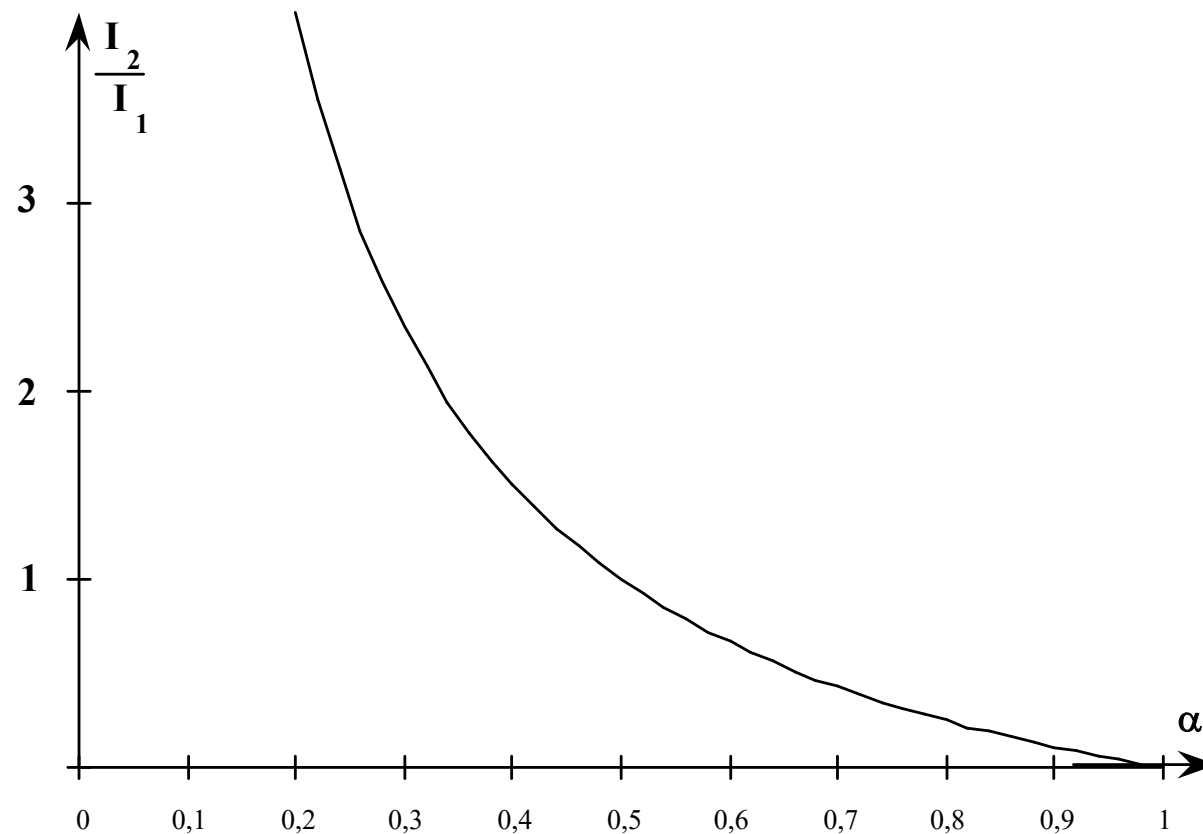
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CUK structure: Transfer characteristic



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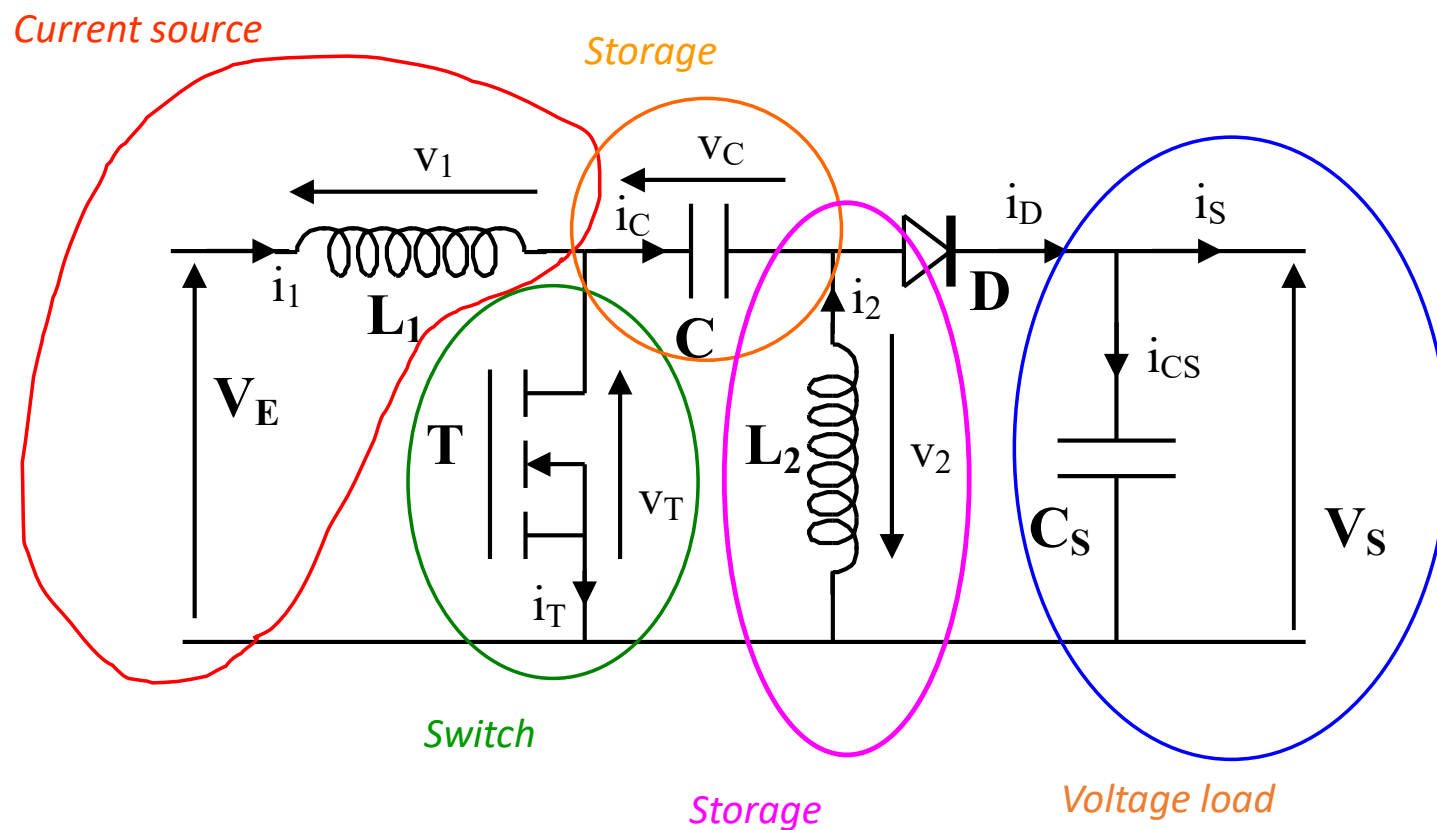
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SEPIC: Single Ended Primary Inductor Convertor



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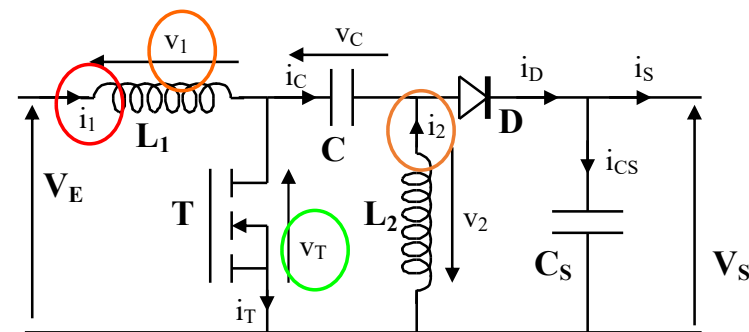
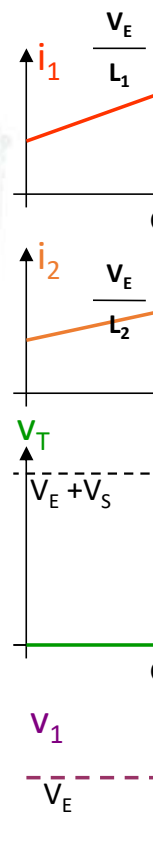
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SEPIC: Continuous mode (voltage transfer)



$$\langle v_1 \rangle = 0 \text{ et } \langle v_2 \rangle = 0 \Rightarrow \langle v_C \rangle = V_E = \text{Constant}$$

$$V_S = V_E \cdot \frac{\alpha}{1 - \alpha}$$

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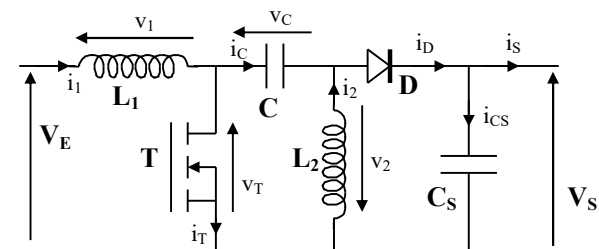
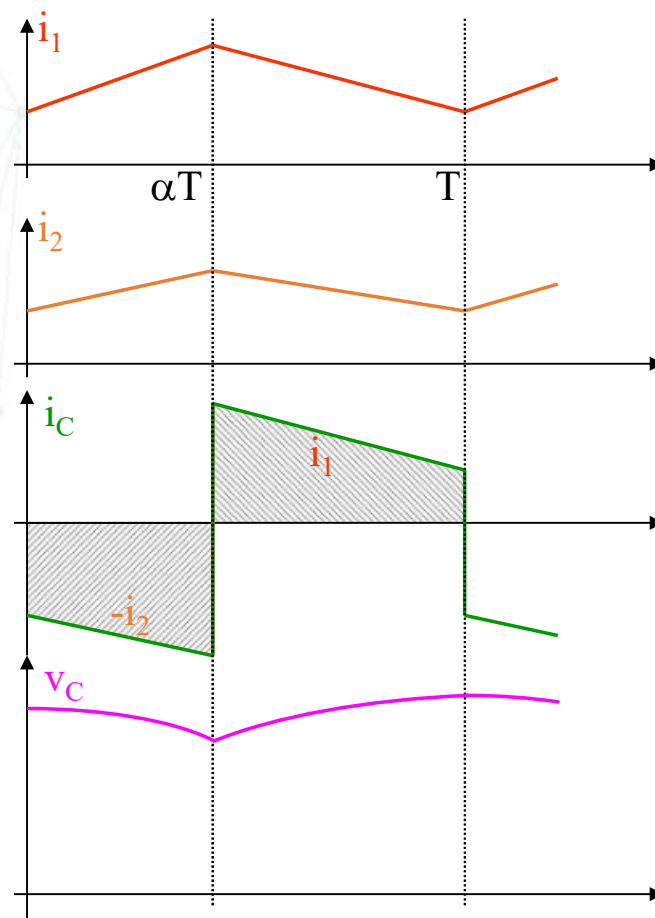
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SEPIC: Continuous mode (coupling capacitor)



$$I_2 = I_S \quad I_S = I_E \cdot \frac{1-\alpha}{\alpha}$$

$$\Delta V_C = \frac{\alpha \cdot I_S}{C \cdot f}$$

$$I_{Ceff} = I_S \cdot \sqrt{\frac{\alpha}{1-\alpha}}$$

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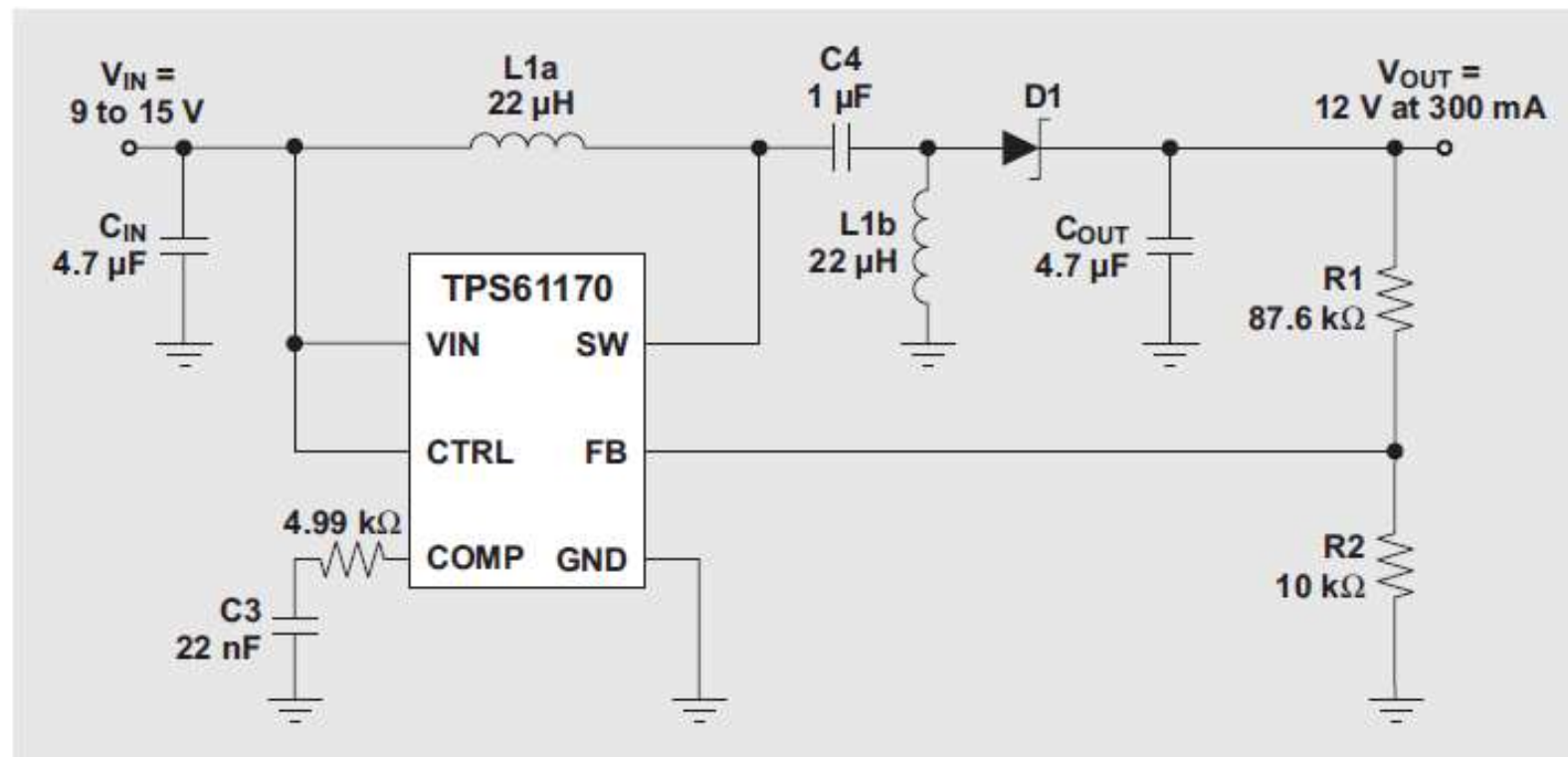
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SEPIC: circuit example
TPS 61170 – Texas Instrument



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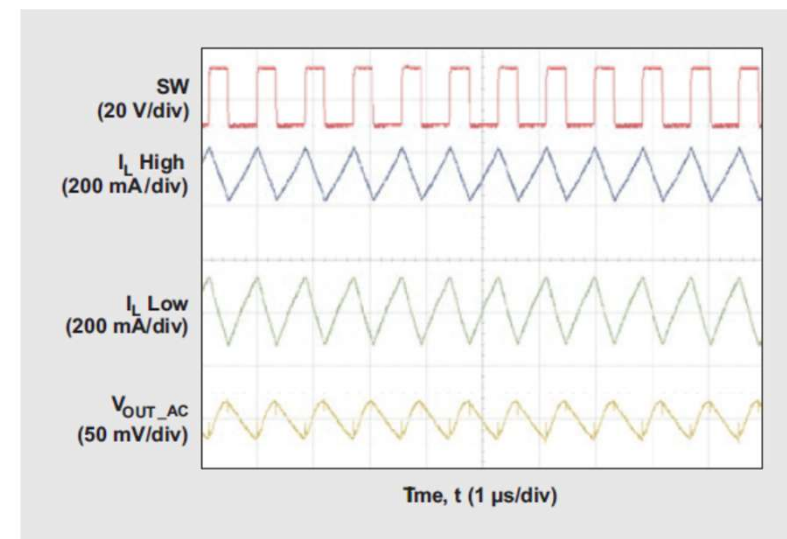
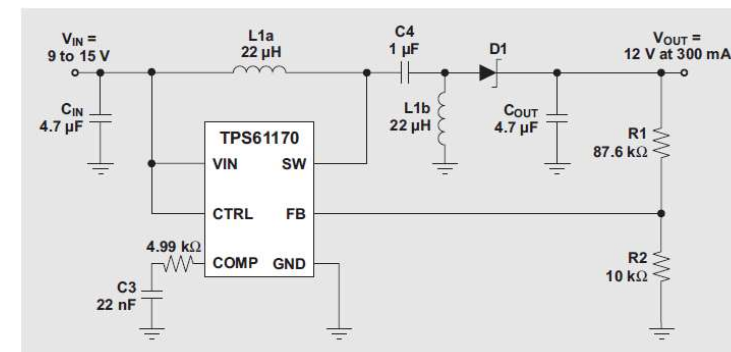
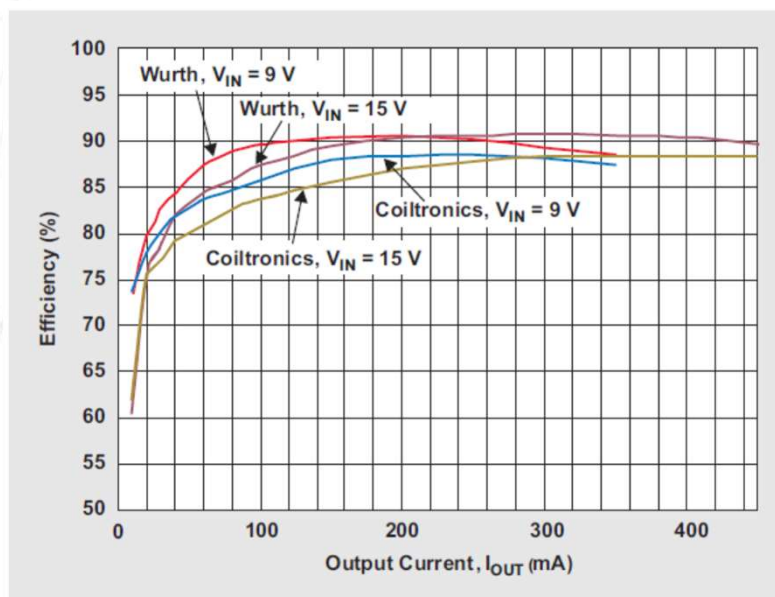
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SEPIC: circuit example
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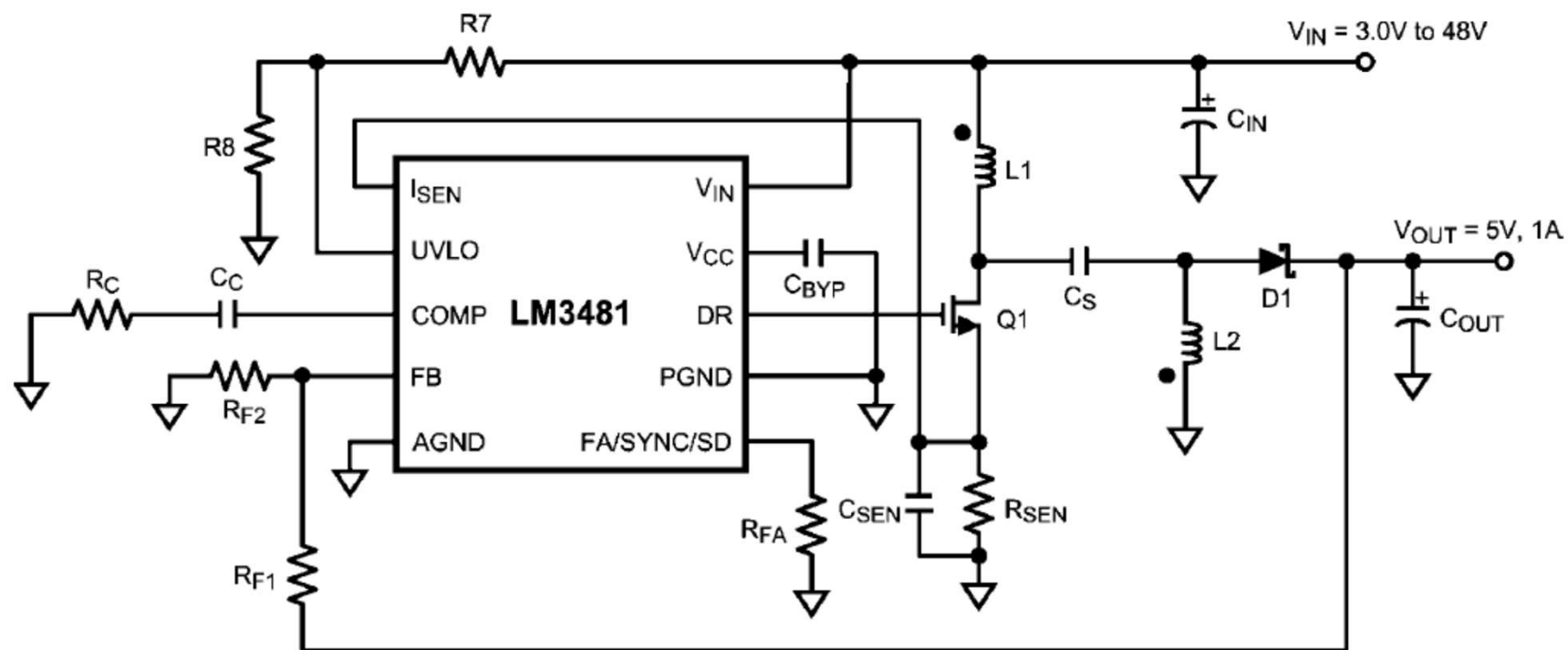
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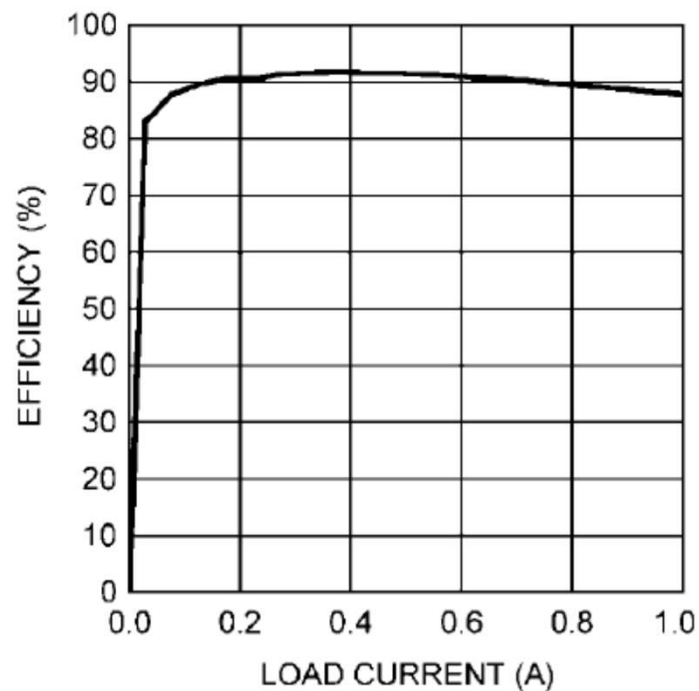
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Efficiency vs. Load Current ($3.3V_{IN}$ and $12V_{OUT}$)



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LM3481: internal schematic

